



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, April 23, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

Flexible Fluidic Actuators

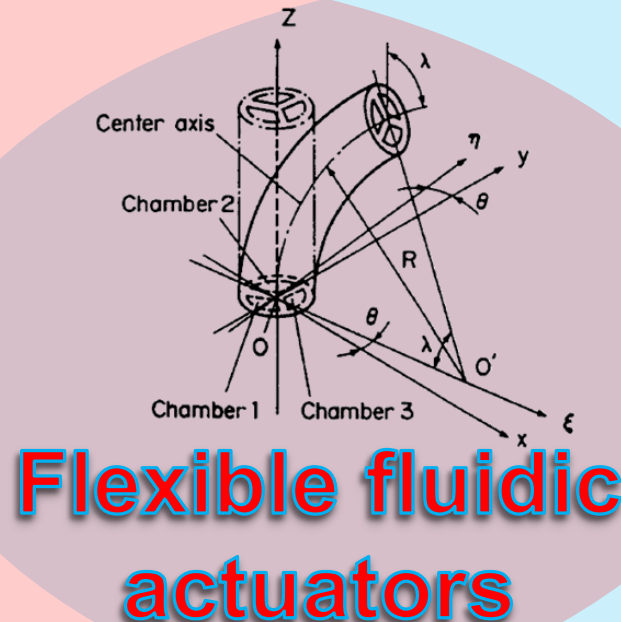
Matteo Cianchetti - matteo.cianchetti@santannapisa.it



Definition

Pneumatic or hydraulic actuators operating in a soft bodied chamber

**Flexible
structure**



**Fluidic
actuation**

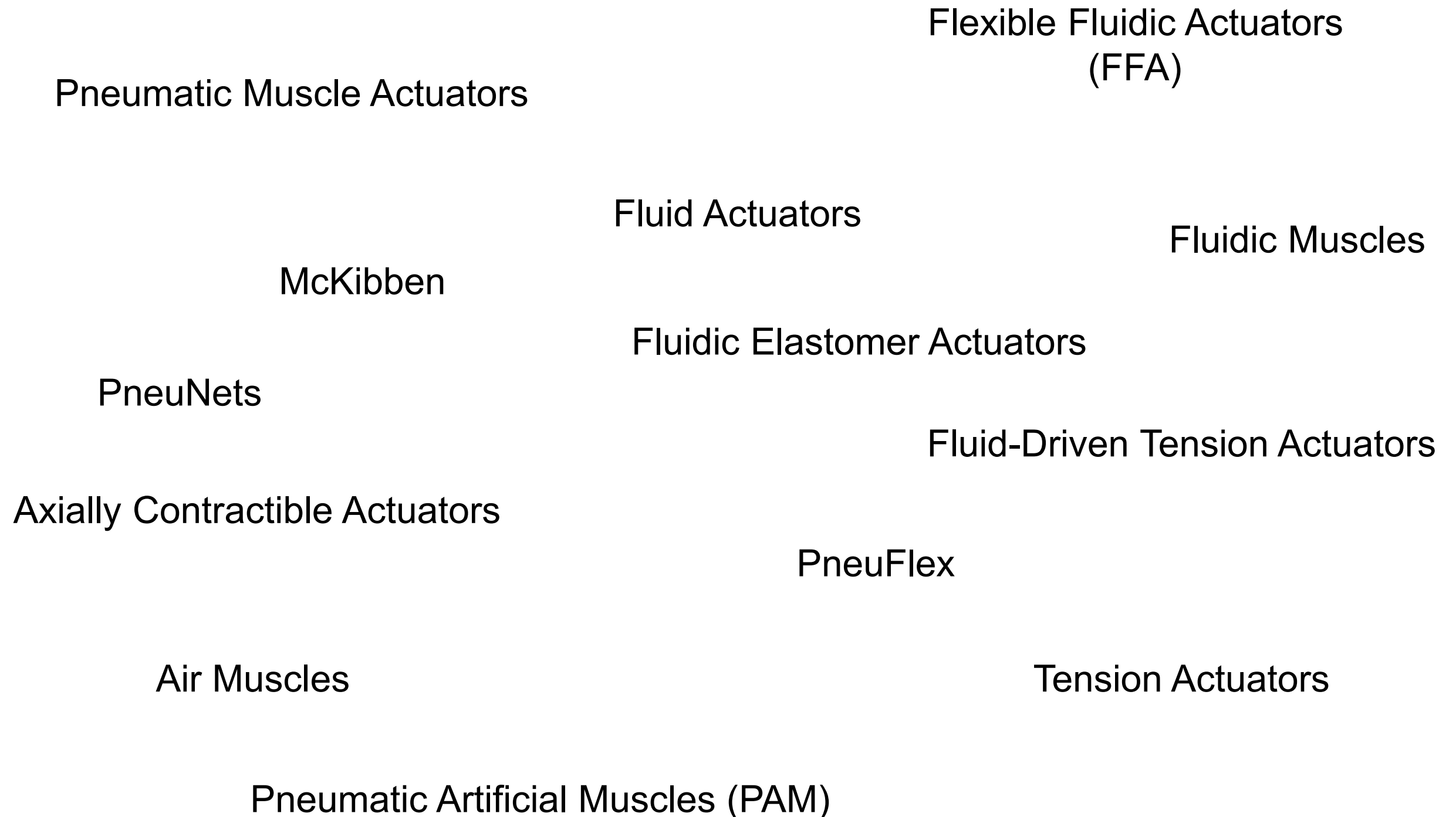


Advantages:

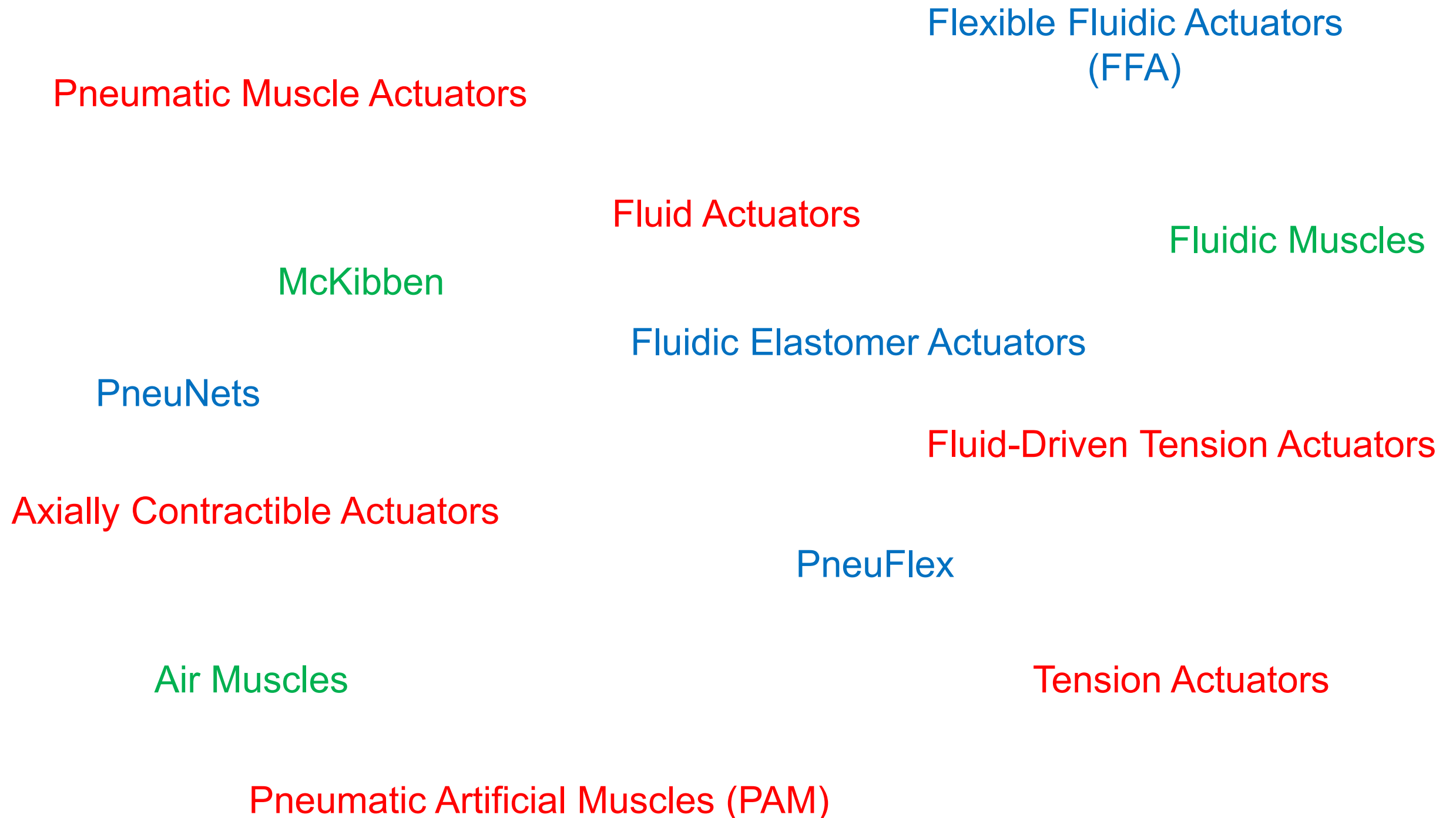
- Compatible with radioactivity and magnetic fields
- No relative motion of rigid parts
- No lubricant, no leakages and no wear particles
- No friction
- Smooth and complex motion
- Mechanically, chemically and electrically safe



Definition

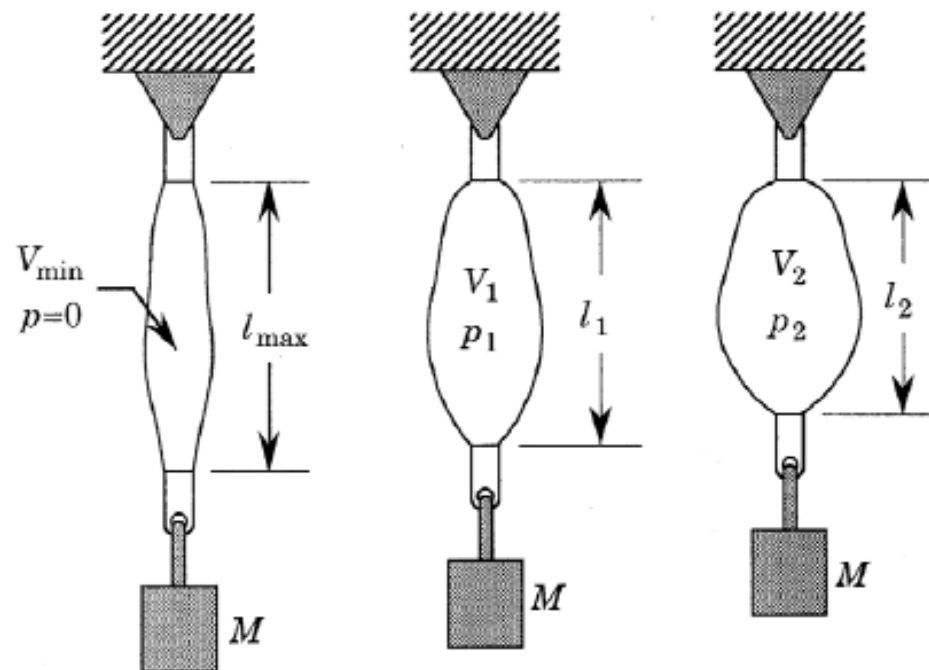


Definition



Flexible Fluidic Actuators

These devices are based on an elastomeric chamber inflated by a pneumatic or hydraulic external source.



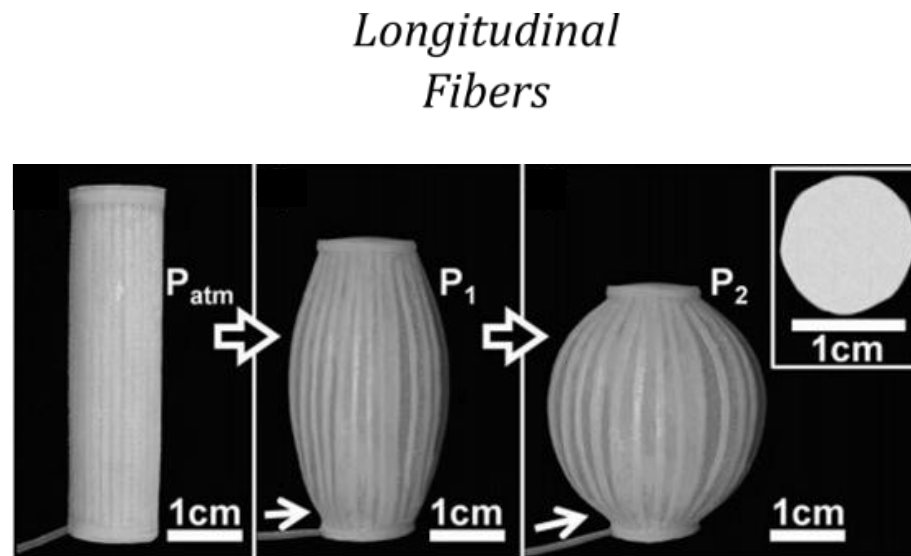
The volumetric deformation of the chamber is usually driven to cause expansion toward specific directions so to result in predefined deformation modalities.



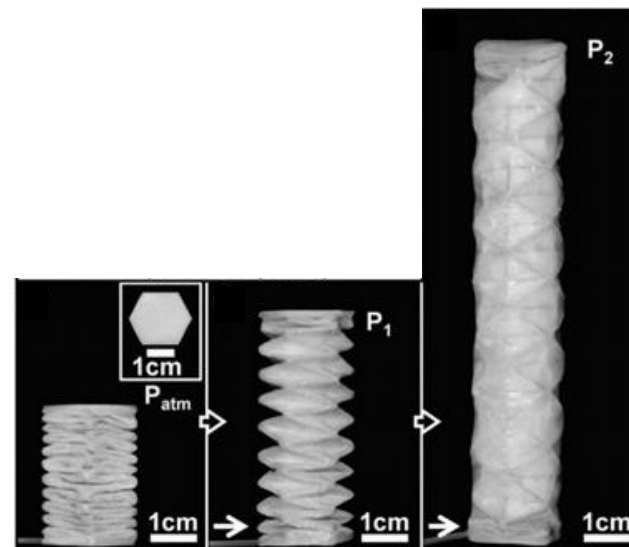
Flexible Fluidic Actuators

Contraction/elongation

Contraction



Elongation



*Corrugated
Membrane*

*Braided
Fibers $\theta_0 < 54.7^\circ$*

**McKibben
actuators**

*Braided
Fibers $\theta_0 > 54.7^\circ$*



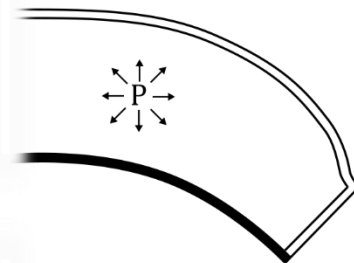
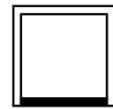
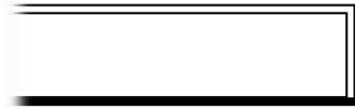
Flexible Fluidic Actuators

Bending

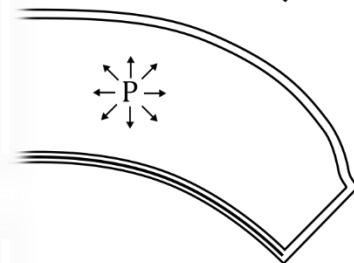
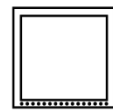
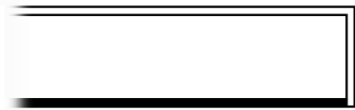
*Longitudinal
Cross Section*

*Transversal
Cross Section*

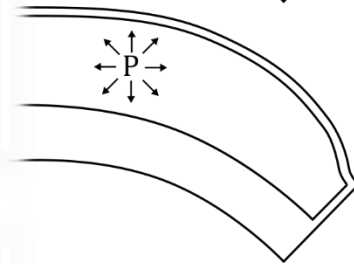
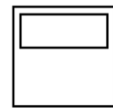
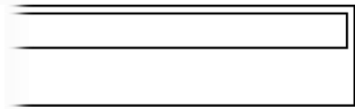
Bending



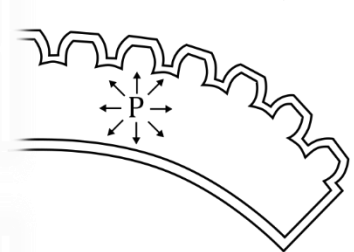
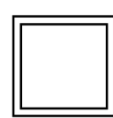
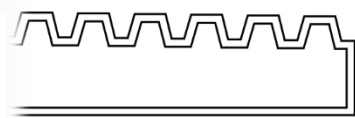
*Inextensible
2D flexible layer*



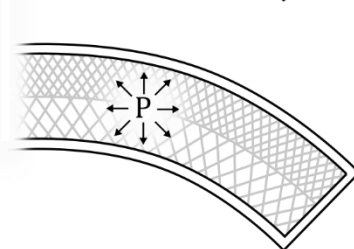
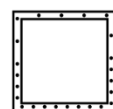
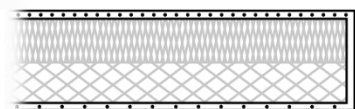
*Inextensible
1D (braided) fibers*



Eccentricity

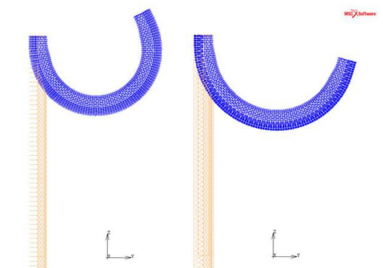
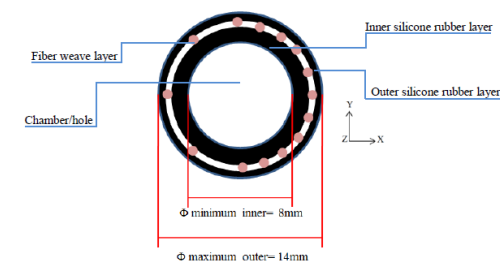
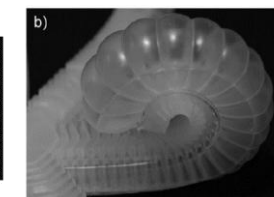
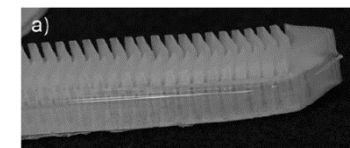
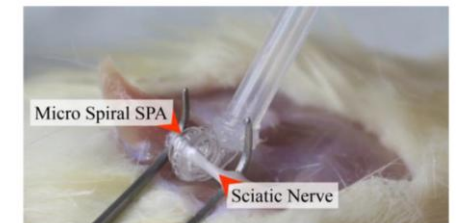
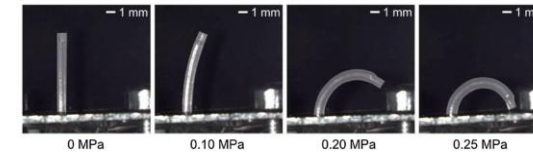
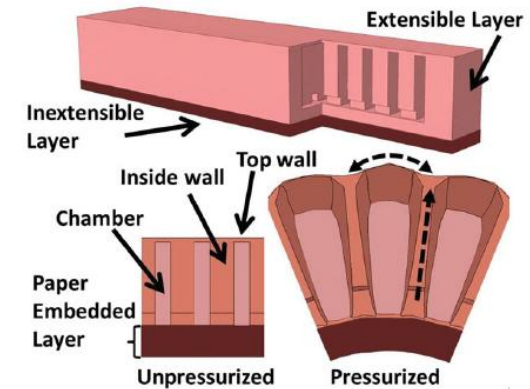


Patterning



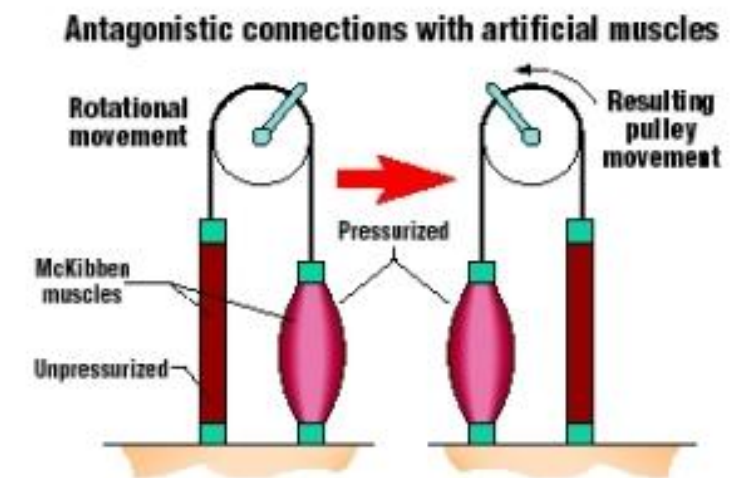
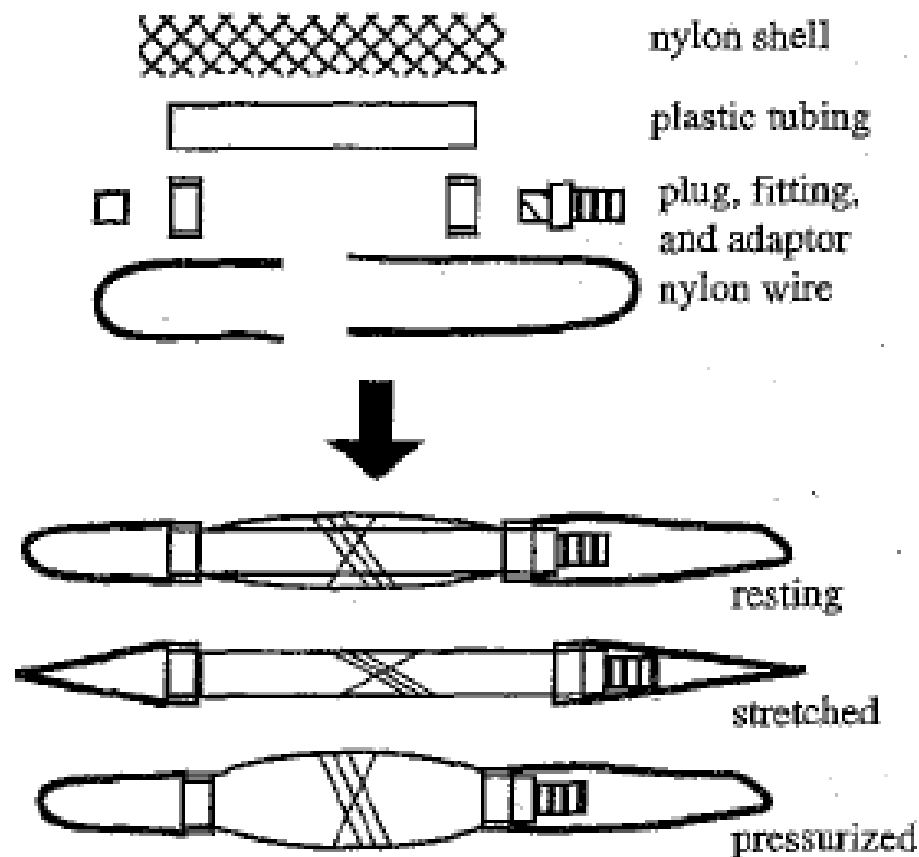
*Asymmetric
braided sleeve*

A Slow Pneu-net (sPN)



McKibben

PAMs were first developed (under the name of McKibben Artificial Muscles) in the 1950s for use in artificial limbs (orthotic appliance for polio patients). The Bridgestone rubber company (Japan) commercialized the idea in the 1980s under the name of Rubbertuators.



History of Artificial Muscles (McKibben)

Although "fluidic actuators" had been around for a long time prior to Joseph Laws McKibben's invention, none had been used previously for prosthetic applications, yet alone robotics.

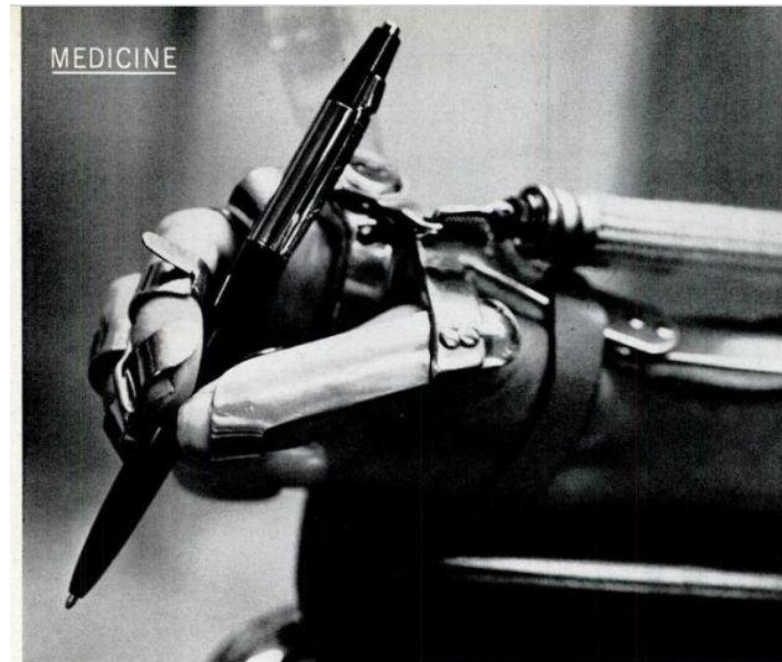
It was McKibben's use that coined the term "Artificial Muscle".

**TIGHTENED BY ARTIFICIAL
MUSCLE LEADING DOWN HER
ARM. FINGERS OF A PARALYZED
GIRL GRASP AND HOLD A PEN**

ARTIFICIAL MUSCLE

No part of man's body is more distinctively human than his hand—and when it becomes paralyzed, few disabilities are more tragic. For years doctors have been looking for a substitute for hand muscles which would enable victims of paralysis to touch fingers to thumb and pick things up. The device above, developed at the Rancho Los Amigos Rehabilitation Center in Downey, Calif., solves the problem.

[Life Magazine 14 March 1960]



TIGHTENED BY ARTIFICIAL MUSCLE LEADING DOWN HER ARM, FINGERS OF A PARALYZED GIRL GRASP AND HOLD A PEN

More Help For Polio Victims

To bring motion to his little daughter's polio-paralyzed hands, Dr. Joseph Laws McKibben, an atomic physicist at Los Alamos, N.M., has developed a new mechanical "muscle" which some day may help thousands of other paralyzed fingers to move, to grasp, even to write.

[The Buckingham Post – Mar 21, 1958]

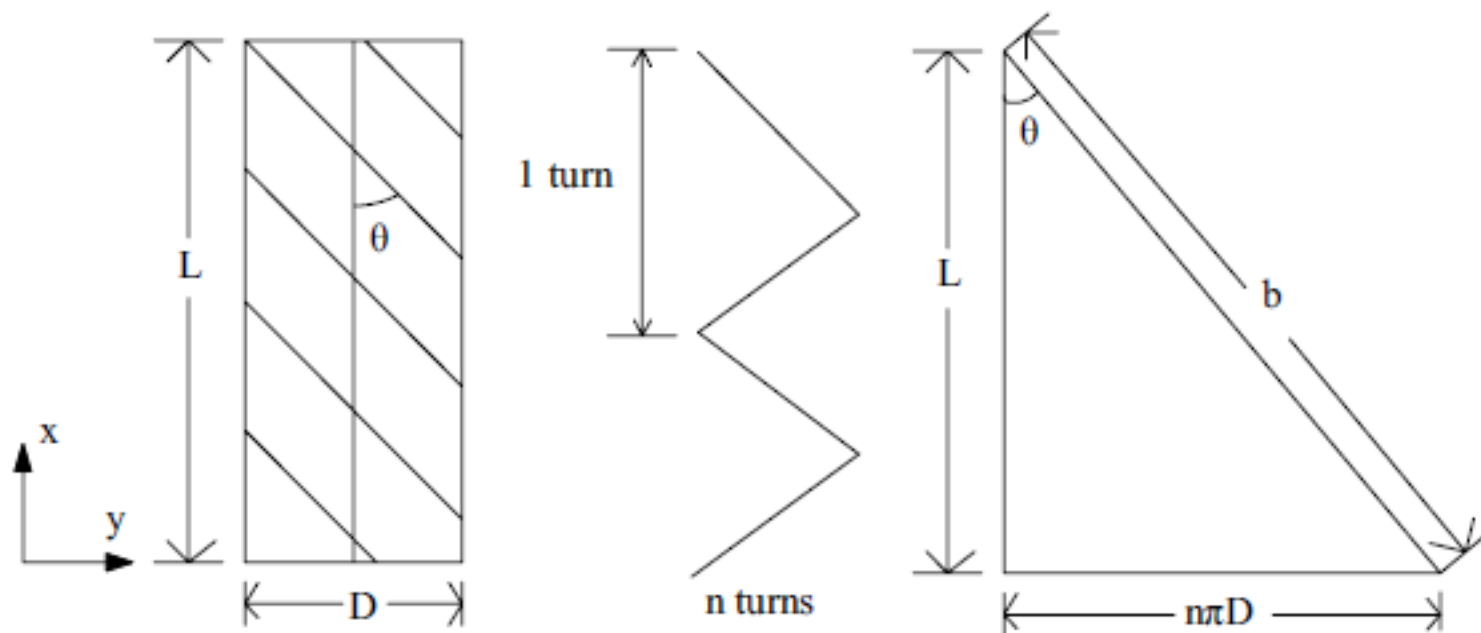


KARAN AND PARENTS leave Rancho Los Amigos Hospital where latest model of muscle was fitted. Unable to walk, she does high school work at home.

McKibben

Geometrical Modeling

- The actuator is cylindrical in shape.
- The threads in the sheath are inextensible and always in contact with the outside diameter of the latex bladder.
- Frictional forces between the tubing and the sheath and between the fibers of the sheath are negligible.
- Latex tubing forces are negligible.



$$L = b \cos(\theta)$$

$$D = \frac{b \sin(\theta)}{n\pi}$$

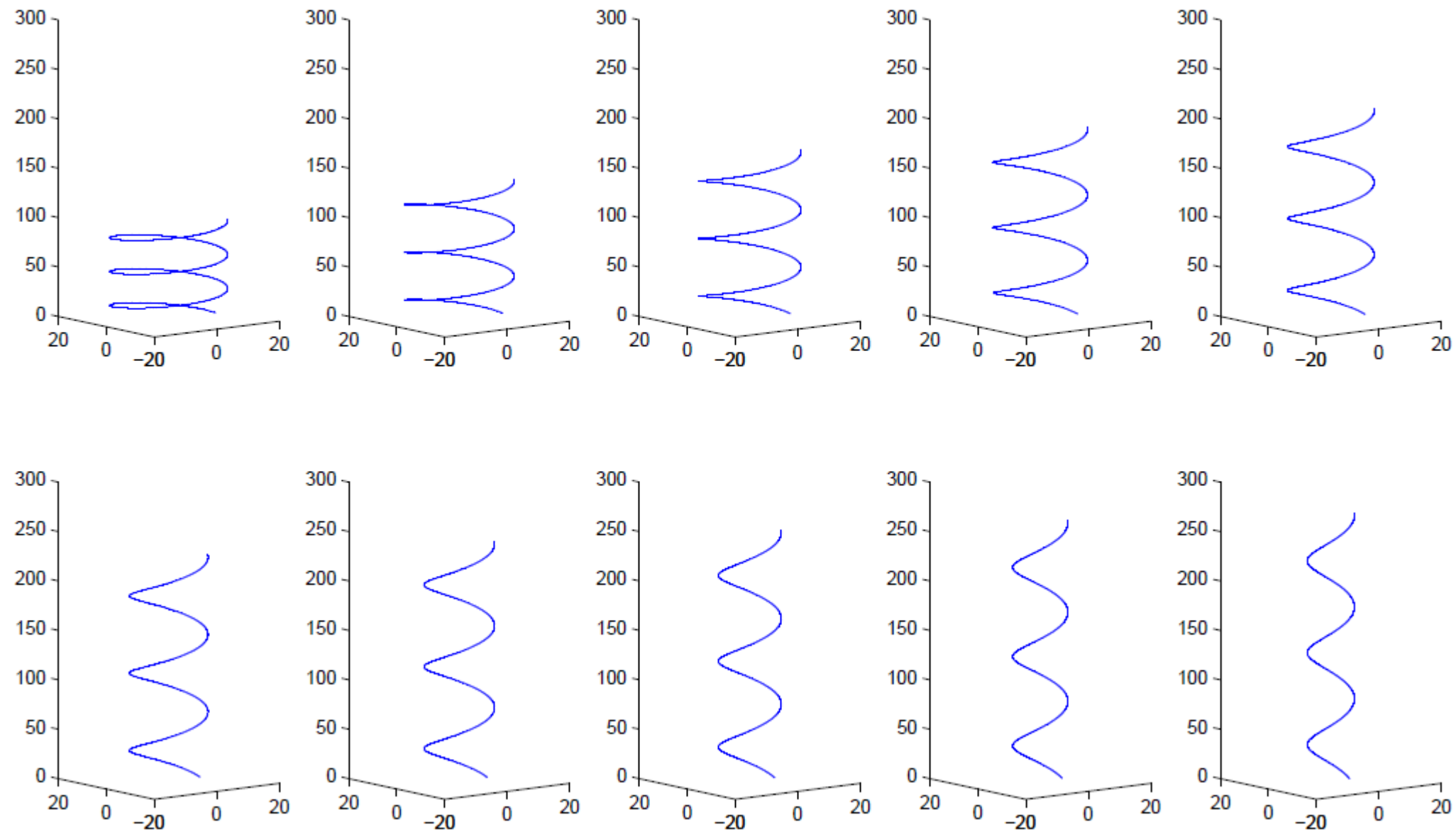
$$b = \sqrt{L^2 + D^2 n^2 \pi^2}$$

$$V = \frac{b^3 \cos(\theta) \sin^2(\theta)}{4n^2 \pi}$$



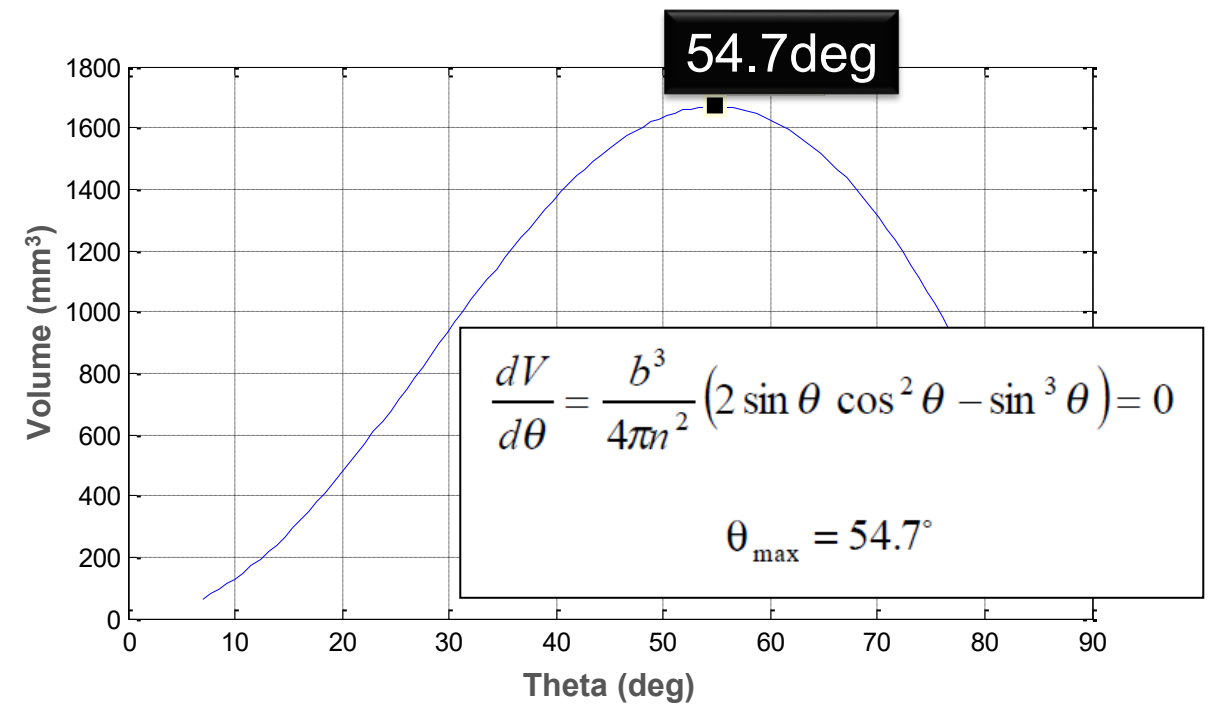
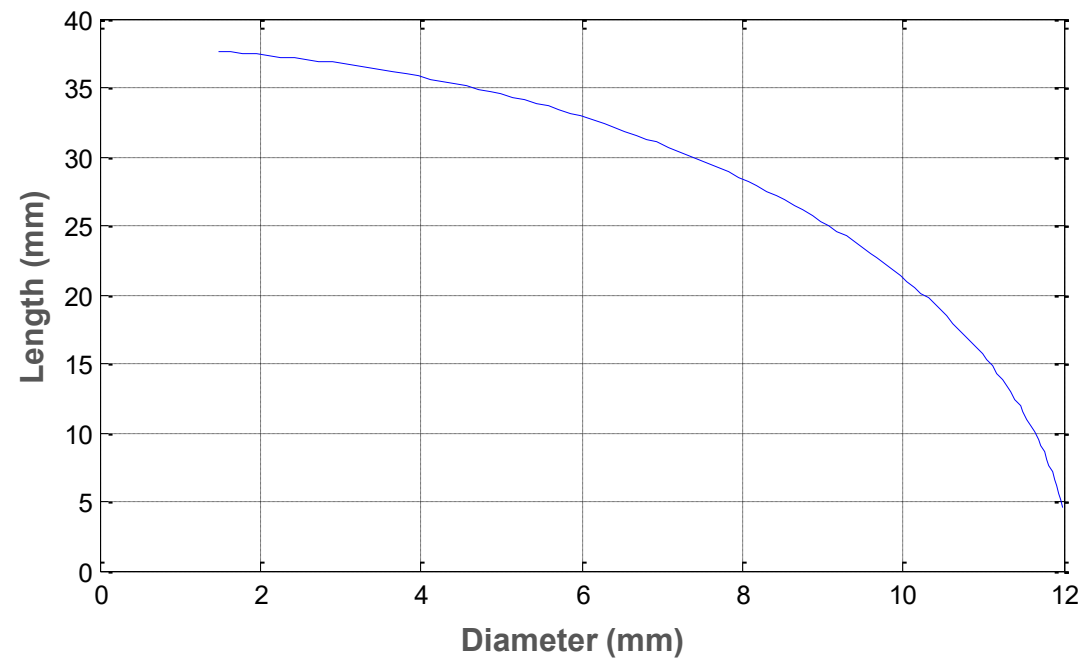
McKibben

Geometrical Modeling



McKibben

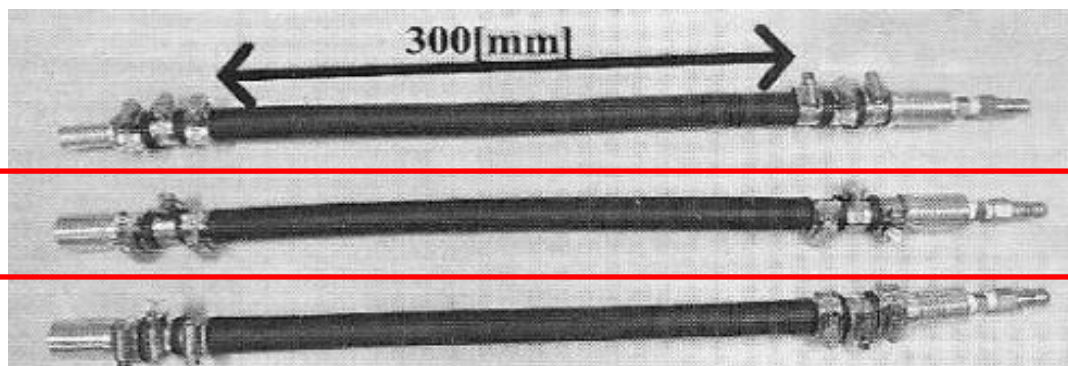
Geometrical Modeling



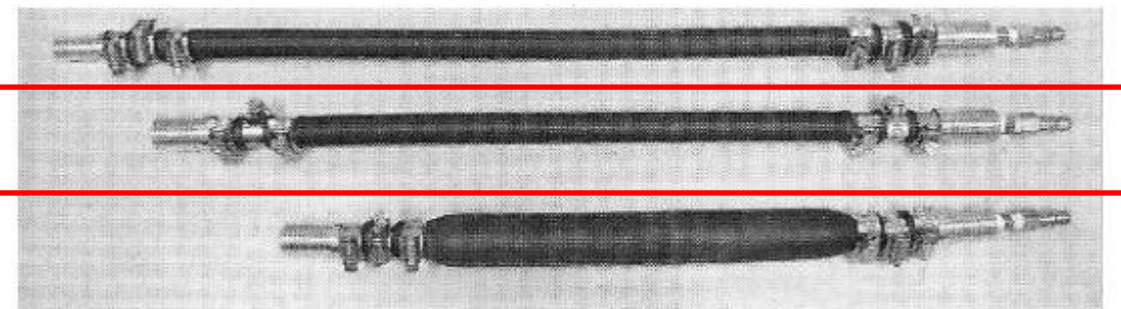
$\theta > 54.7^\circ$

$\theta = 54.7^\circ$

$\theta < 54.7^\circ$



At rest



Inflated



McKibben

Static Modeling

The input work:

$$dW_{\text{in}} = \int_{S_i} (P - P_0) d\mathbf{l}_i \cdot d\mathbf{s}_i = (P - P_0) \int_{S_i} d\mathbf{l}_i \cdot d\mathbf{s}_i = P' dV$$

The output work:

$$dW_{\text{out}} = -F dL$$

Using the “virtual work” argument:

$$dW_{\text{out}} = dW_{\text{in}} \quad F = -P' \frac{dV}{dL} = \frac{P' b^2 (3 \cos^2 \theta - 1)}{4\pi n^2}$$

P : absolute internal pressure

P_0 : environment pressure

P' : relative pressure

S_i : total inner surface

$d\mathbf{s}_i$: area vector

$d\mathbf{l}_i$: the inner surface displacement

dV : volume change

F : axial tension

dL : axial displacement

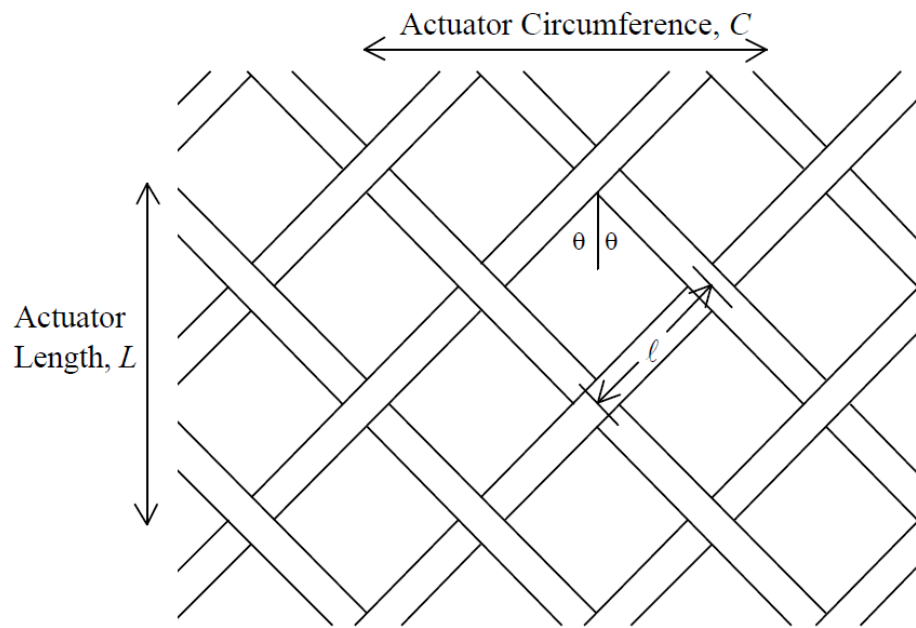
How to determine n ?

How to determine b ?



McKibben

Static Modeling



$$L = 2A\ell \cos\theta$$

A: Number of lengthwise trapezoids

$$b = 2A\ell$$

$$L = b \cos\theta$$

$$C = 2B\ell \sin\theta$$

B: Number of circumferential trapezoids

$$D = \frac{2B\ell}{\pi} \sin\theta$$

$$n = \frac{A}{B}$$

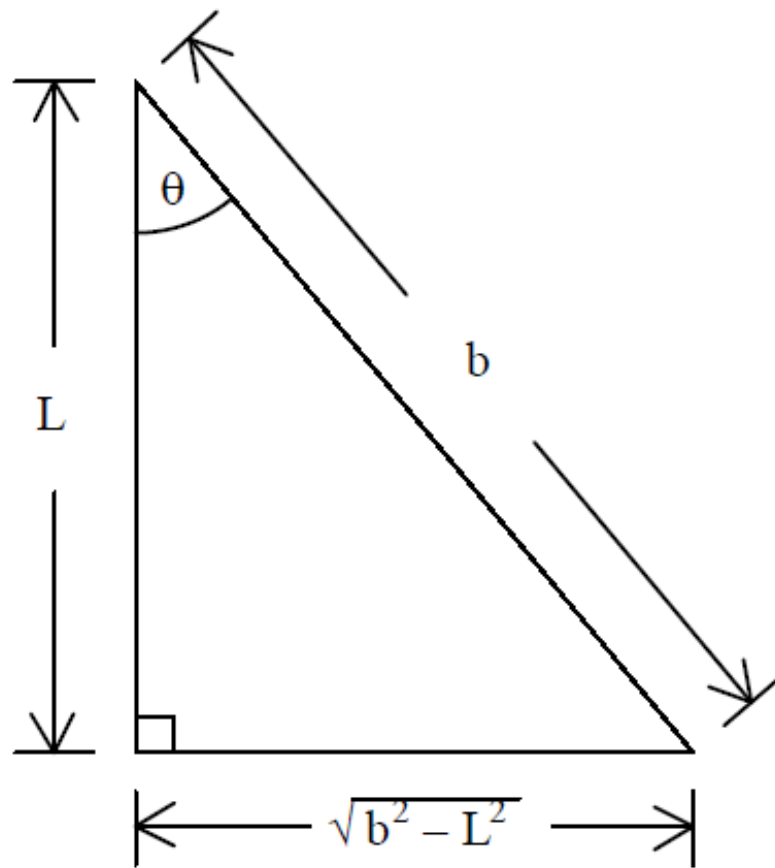
$$D = \frac{b \sin\theta}{n\pi}$$



McKibben

Static Modeling

To make the model useful for control purposes, the dependency on theta has to be removed (hard to measure)



$$\sin\theta = \frac{\sqrt{b^2 - L^2}}{b}$$

$$\cos\theta = \frac{L}{b}$$

$$F = \frac{P_g b^2 (3 \cos^2 \theta - 1)}{4\pi n^2} = \frac{P_g b^2}{4\pi n^2} \left(\frac{3L^2}{b^2} - 1 \right) =$$

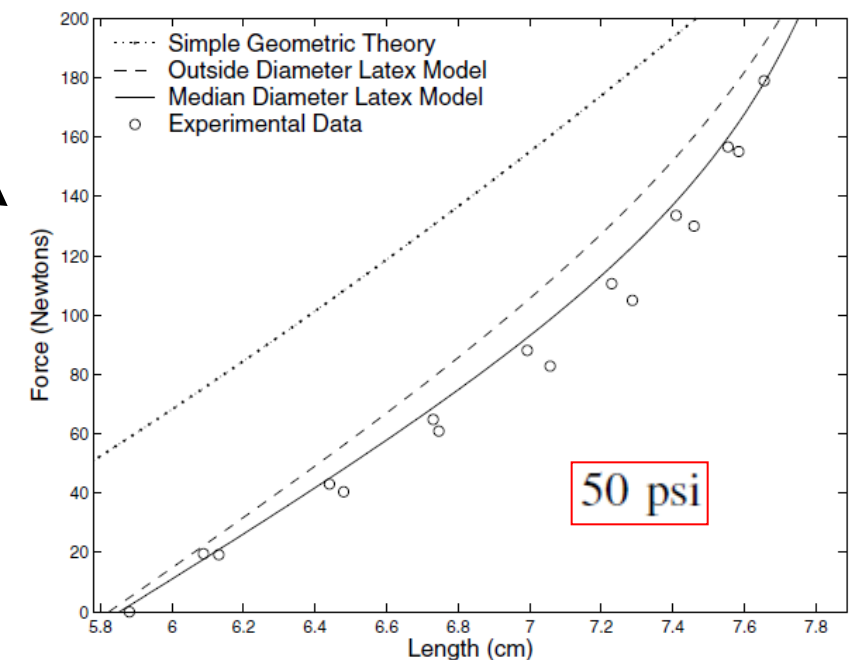
$$= \frac{P_g B^2 l^2}{\pi} \left(\frac{3L^2}{4A^2 l^2} - 1 \right)$$



McKibben

Static Modeling

- A factor to account for internal friction: Tondu and Lopez, 2000
- A model to account stretch in sheath threads: Davis et al., 2003
- A model of the end-effects due to the non-cylindrical shape of the McKibben actuator at either end: Tsagarakis and Caldwell, 2000
- A model for latex bladder forces: Delson et al., 2005



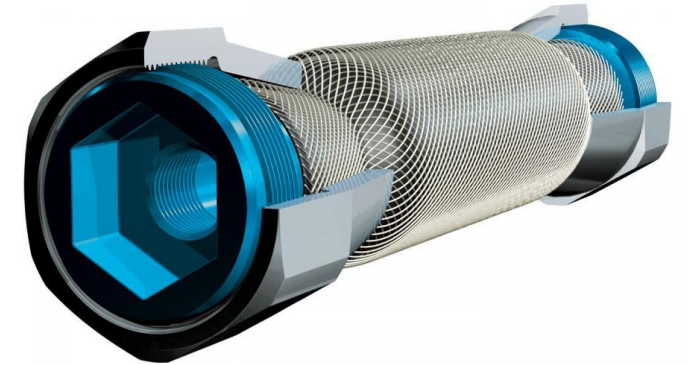
McKibben

Examples



'Use with a Mitsubishi PA10 Arm',
26t 2ix

Shadow - Air Muscles

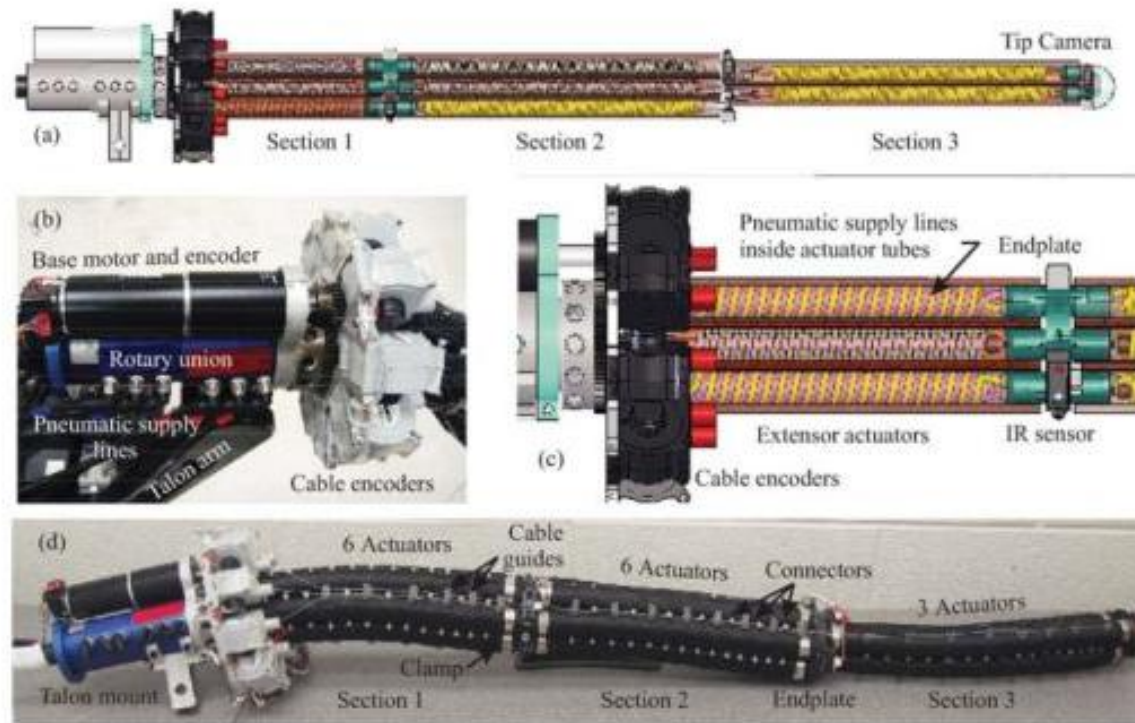


FESTO - Fluidic Muscles

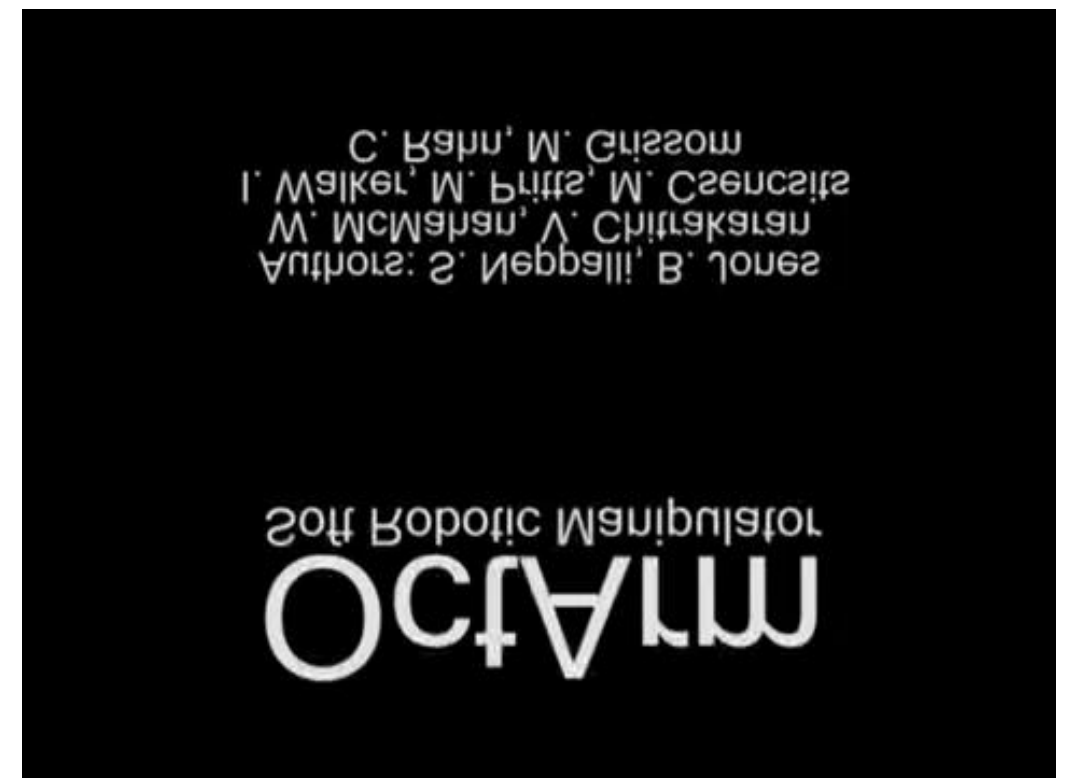


McKibben

Examples

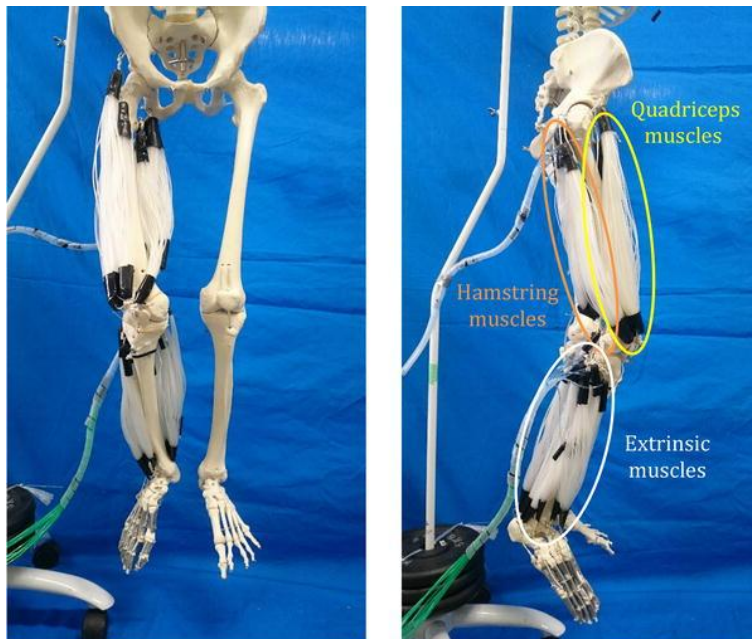


- biologically-inspired maneuvering (bending in different directions while holding a basketball, expanding and contracting, and rotating through 360 degrees with the help of a motor attached to the base)
- the entire length of its arm can be used to grasp objects of different shapes and sizes



McKibben

Examples



Multifilament muscles based on micro-McKibben actuators



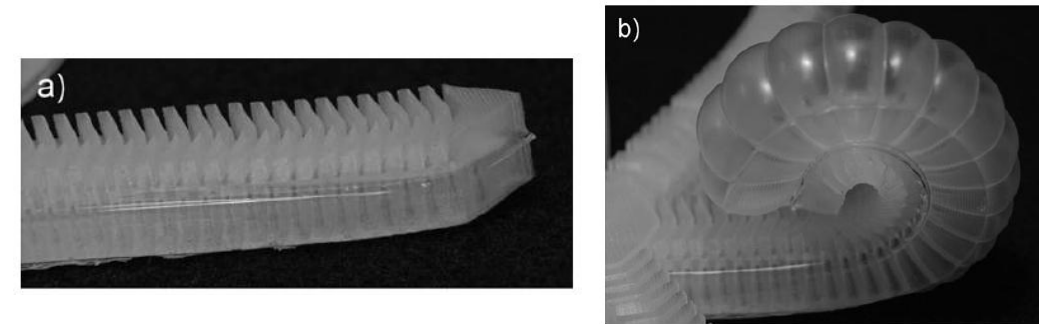
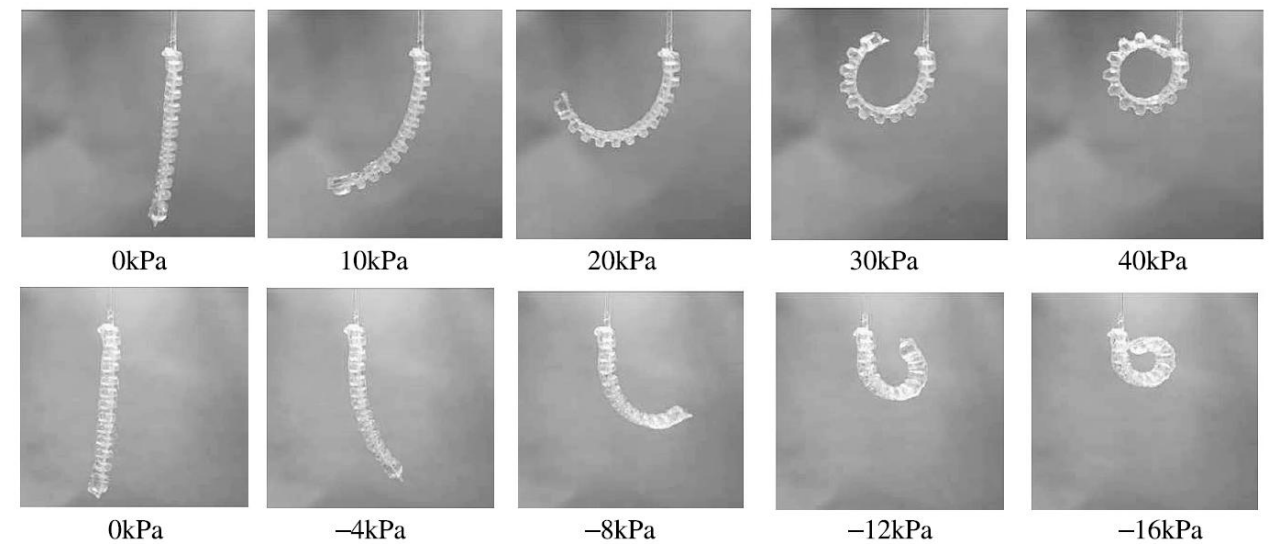
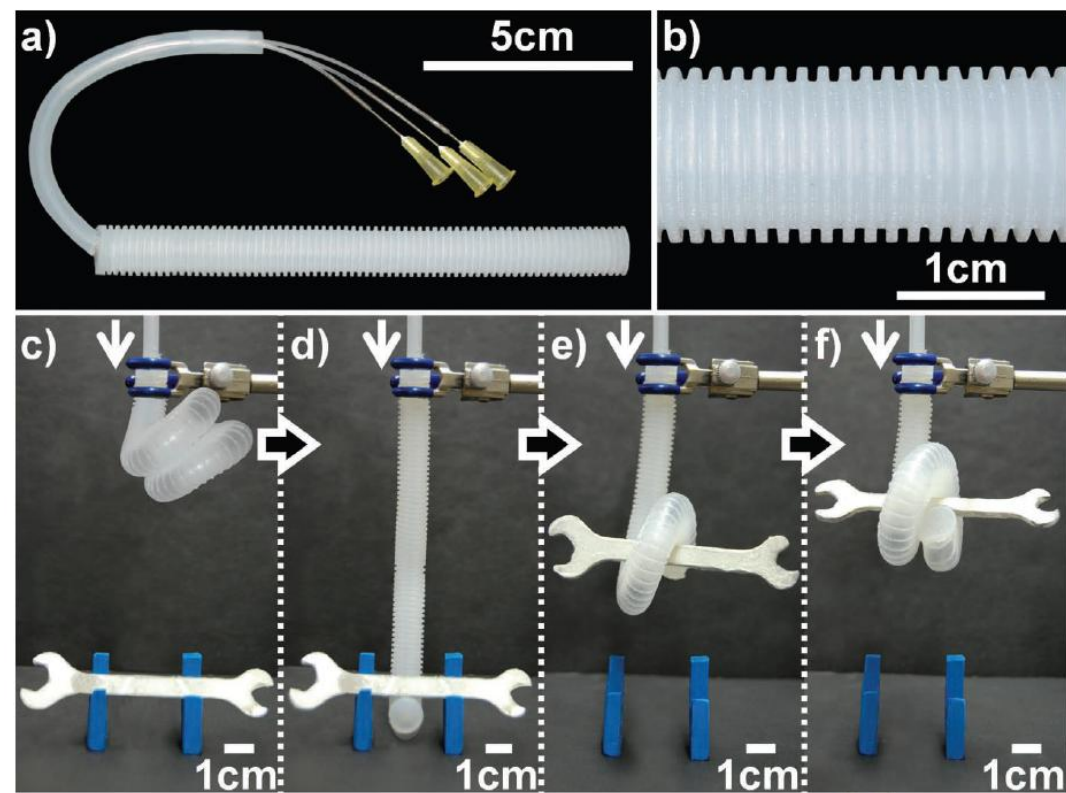
Kurumaya, S. et al. «Musculoskeletal lower-limb robot driven by multifilament muscles» ROBOMECH Journal, 2016, 3, 18



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Internal / external patterning



Wakimoto, S.; Suzumori, K. & Ogura, K. Miniature Pneumatic Curling Rubber Actuator Generating Bidirectional Motion with One Air-Supply Tube Advanced Robotics, 2011, 25, 1311-1330

Ilievski, F.; Mazzeo, A. D.; Shepherd, R. F.; Chen, X. & Whitesides, G. M. Soft Robotics for Chemists Angewandte Chemie International Edition, WILEY-VCH Verlag, 2011, 50, 1890-1895

Martinez, R. V.; Branch, J. L.; Fish, C. R.; Jin, L.; Shepherd, R. F.; Nunes, R. M. D.; Suo, Z. & Whitesides, G. M. Robotic Tentacles with Three-Dimensional Mobility Based on Flexible Elastomers Advanced Materials, WILEY-VCH Verlag, 2013, 25, 205-212



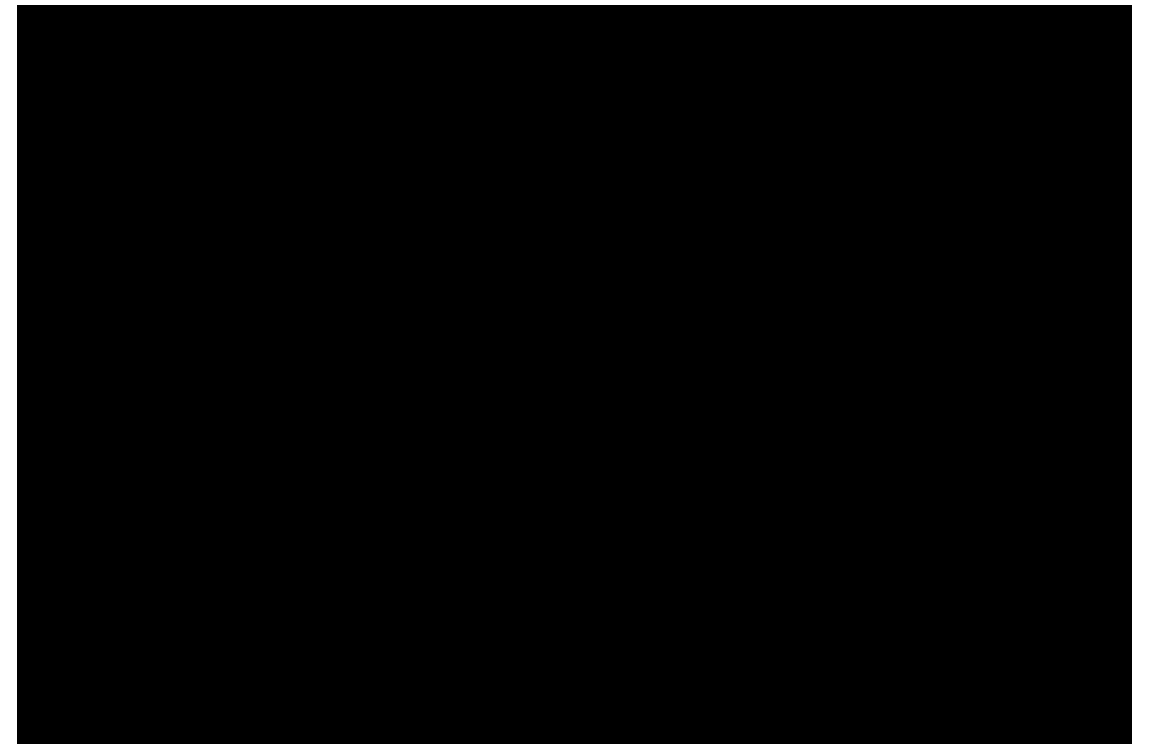
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Internal / external patterning



Shepherd, R. F.; Ilievski, F.; Choi, W.; Morin, S. A.; Stokes, A. A.; Mazzeo, A. D.; Chen, X.; Wang, M. & Whitesides, G. M. Multigait soft robot Proceedings of the National Academy of Sciences, 2011, 108, 20400-20403



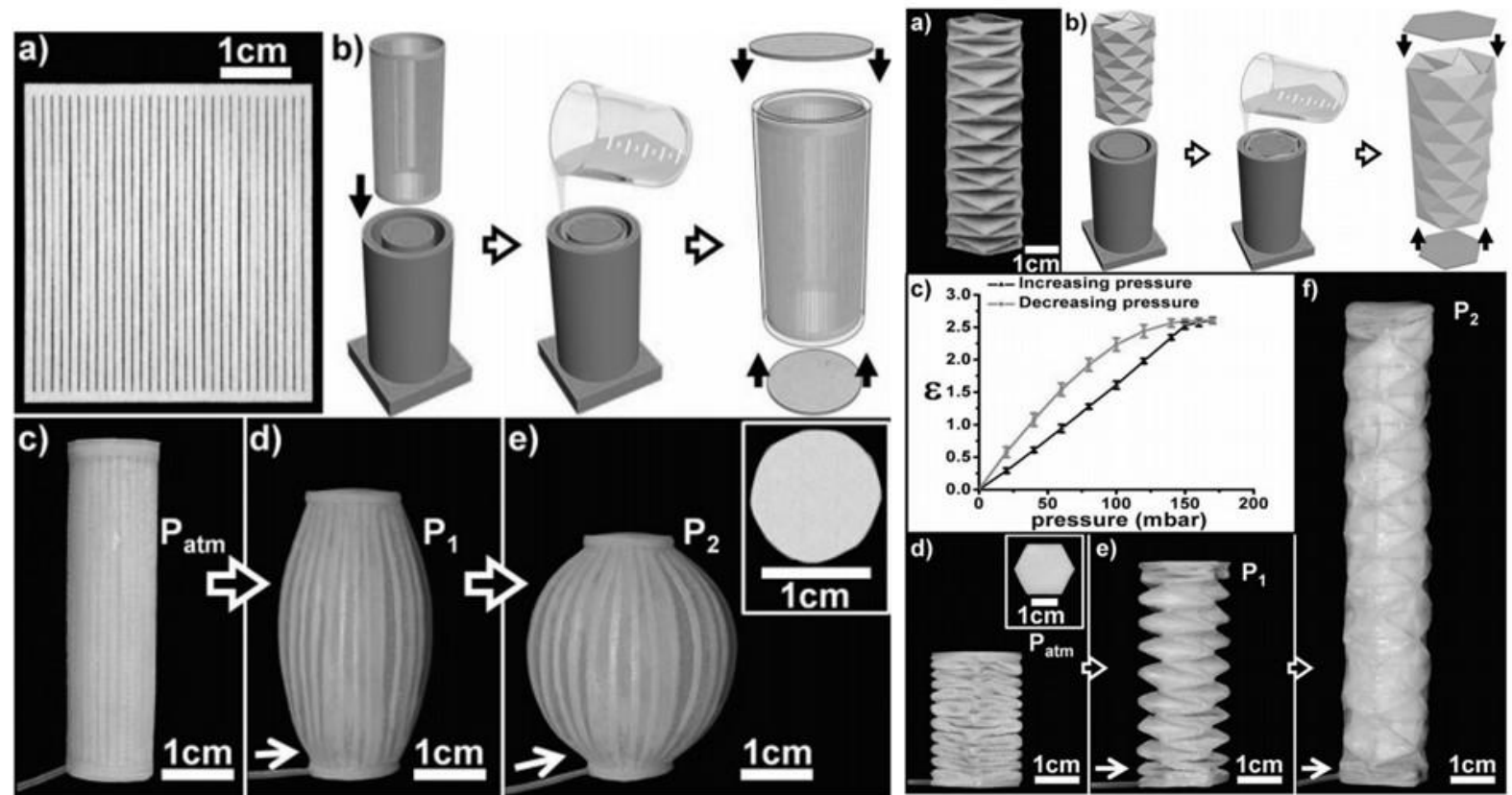
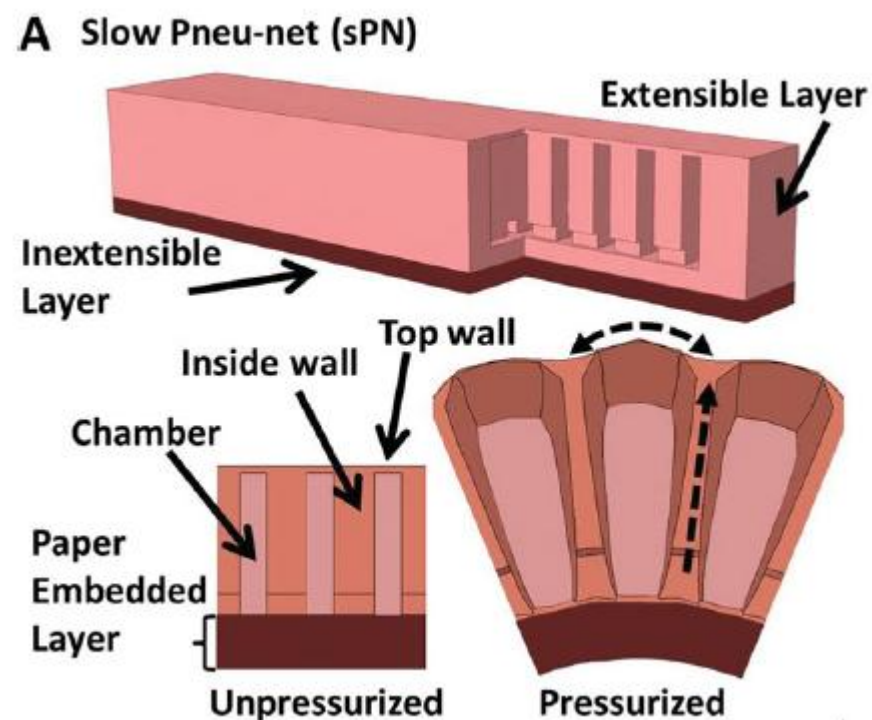
Ilievski, F.; Mazzeo, A. D.; Shepherd, R. F.; Chen, X. & Whitesides, G. M. Soft Robotics for Chemists Angewandte Chemie International Edition, WILEY-VCH Verlag, 2011, 50, 1890-1895



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Different materials



Martinez, R. V.; Fish, C. R.; Chen, X. & Whitesides, G. M. Elastomeric Origami: Programmable Paper-Elastomer Composites as Pneumatic Actuators *Advanced Functional Materials*, WILEY-VCH Verlag, 2012, 22, 1376-1384

Mosadegh, B.; Polygerinos, P.; Keplinger, C.; Wennstedt, S.; Shepherd, R. F.; Gupta, U.; Shim, J.; Bertoldi, K.; Walsh, C. J. & Whitesides, G. M. Pneumatic Networks for Soft Robotics that Actuate Rapidly *Advanced Functional Materials*, 2014, 24, 2163-2170



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Different materials



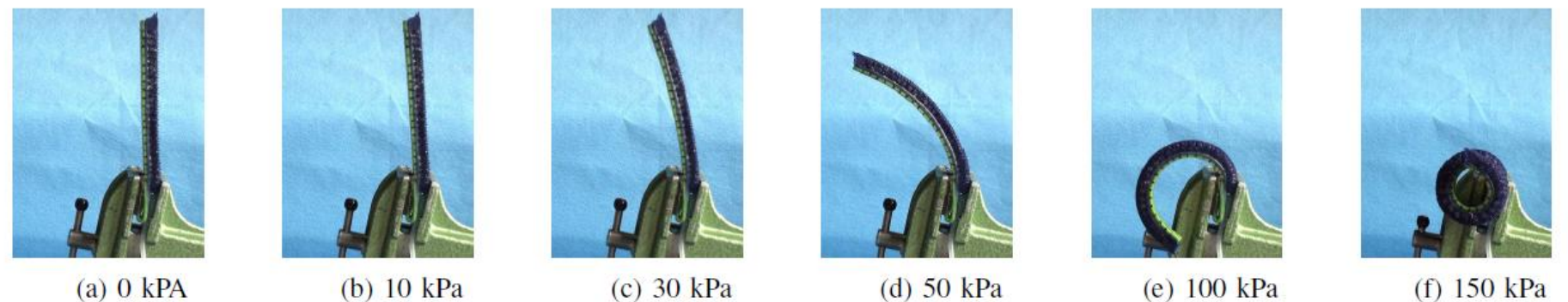
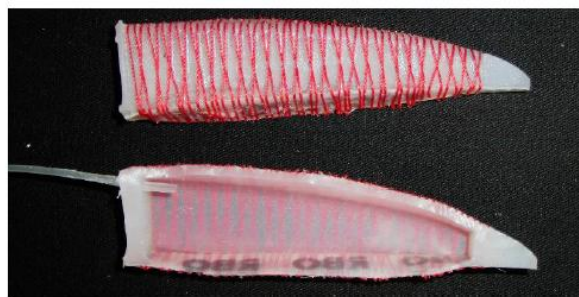
P. Polygerinos et al., Soft robotic glove for combined assistance and at-home rehabilitation. Robot. Auton. Syst. 73, 135–143 (2015).



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Crossed yarns



Deimel, R. & Brock, O. A Compliant Hand Based on a Novel Pneumatic Actuator Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2013, 01-07

Deimel, R. & Brock, O. A Novel Type of Compliant, Underactuated Robotic Hand for Dexterous Grasping Proceedings of Robotics: Science and Systems, 2014



Flexible Fluidic Actuators

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Crossed yarns

Summary

The Video demonstrates 31 different grasp postures taken from the Feix grasp list.

Deimel, R. & Brock, O. A Compliant Hand Based on a Novel Pneumatic Actuator Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2013, 01-07

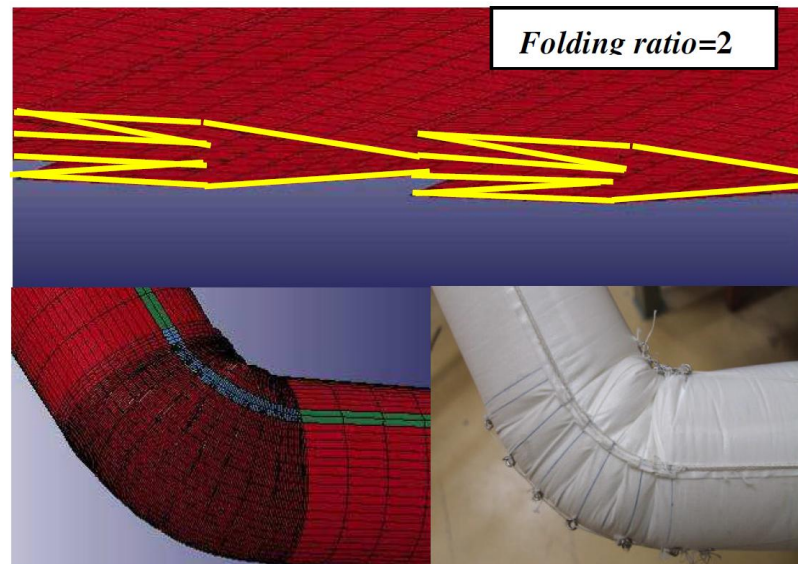
Deimel, R. & Brock, O. A Novel Type of Compliant, Underactuated Robotic Hand for Dexterous Grasping Proceedings of Robotics: Science and Systems, 2014



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Inextensible chambers



S. Voisembert, A. Riwan, N. Mechbal, Numerical evaluation of a new robotic manipulator based on inflatable joints, CASE 2012, pp. 544–549.



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Inextensible chambers



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Inextensible chambers

“The origin of *Big Hero 6* reflects my passion for Marvel’s comics – says Don Hall (*movie director*) – ...I am interested in human-robot interaction, in **soft robotics** research and in Japanese culture. As I heard about an **inflatable** vinyl robotic arm that was able to perform simple actions, such as cleaning teeth of a human and infinite other applications, I realized the idea of a healthcare assistant robot”

https://www.youtube.com/watch?v=6PMJRe_B2QQ

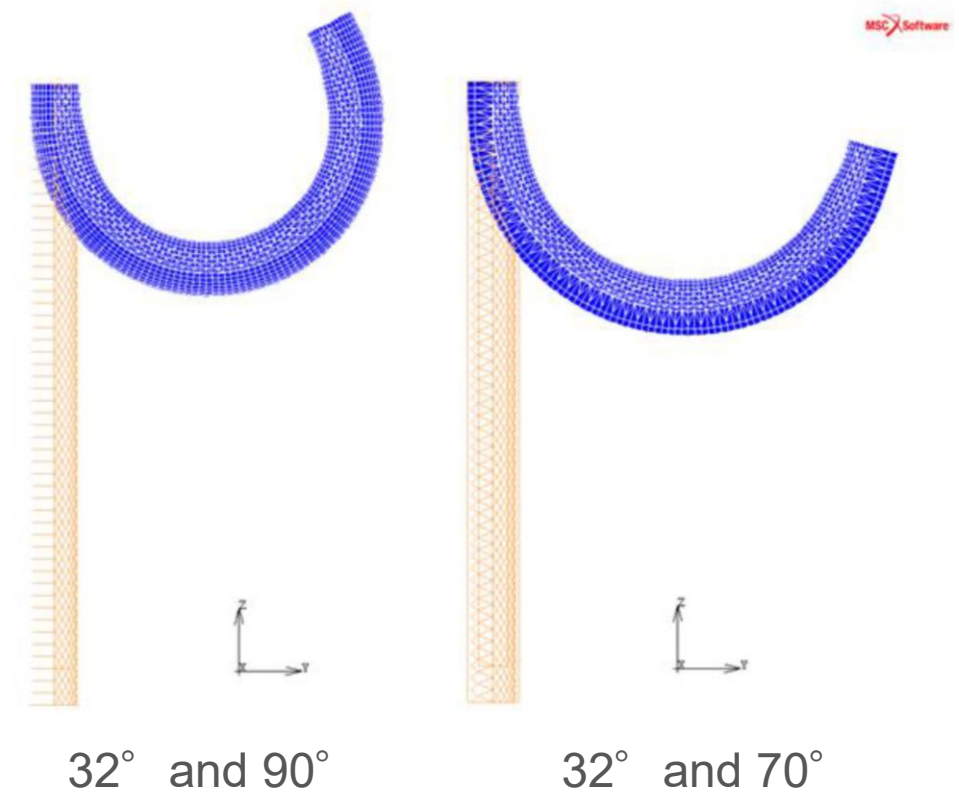
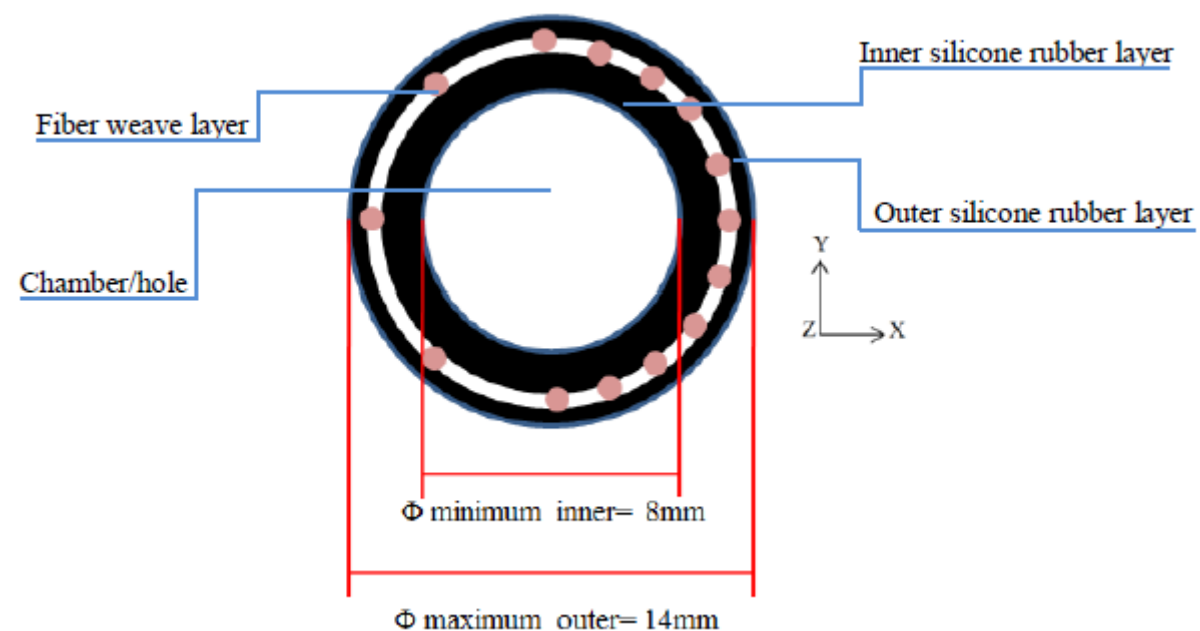
Fictional Baymax in the Movie "Big Hero 6" Embodies Research on Soft Robotics



Flexible Fluidic Actuators

All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Different braided structures



Nordin, I. N. A. M.; Faudzi, A. A. M.; Razif, M. R. M.; Natarajan, E.; Wakimoto, S. & Suzumori, K. Simulations of Two Patterns Fiber Weaves Reinforced in Rubber Actuator Journal Teknologi, 2014, 69

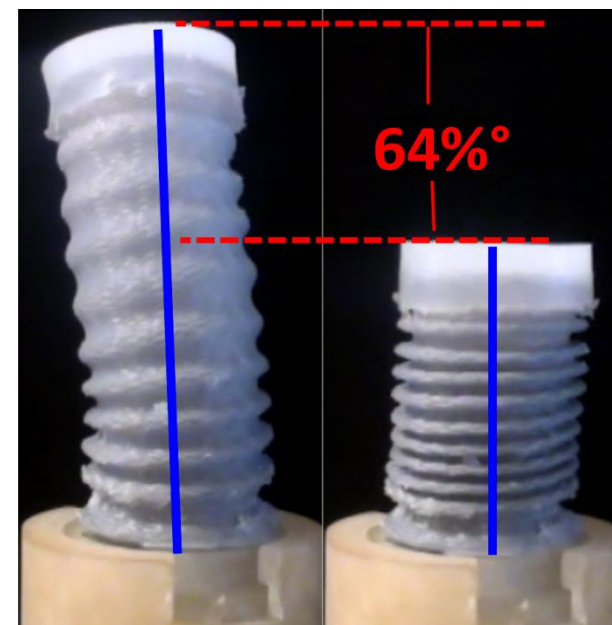
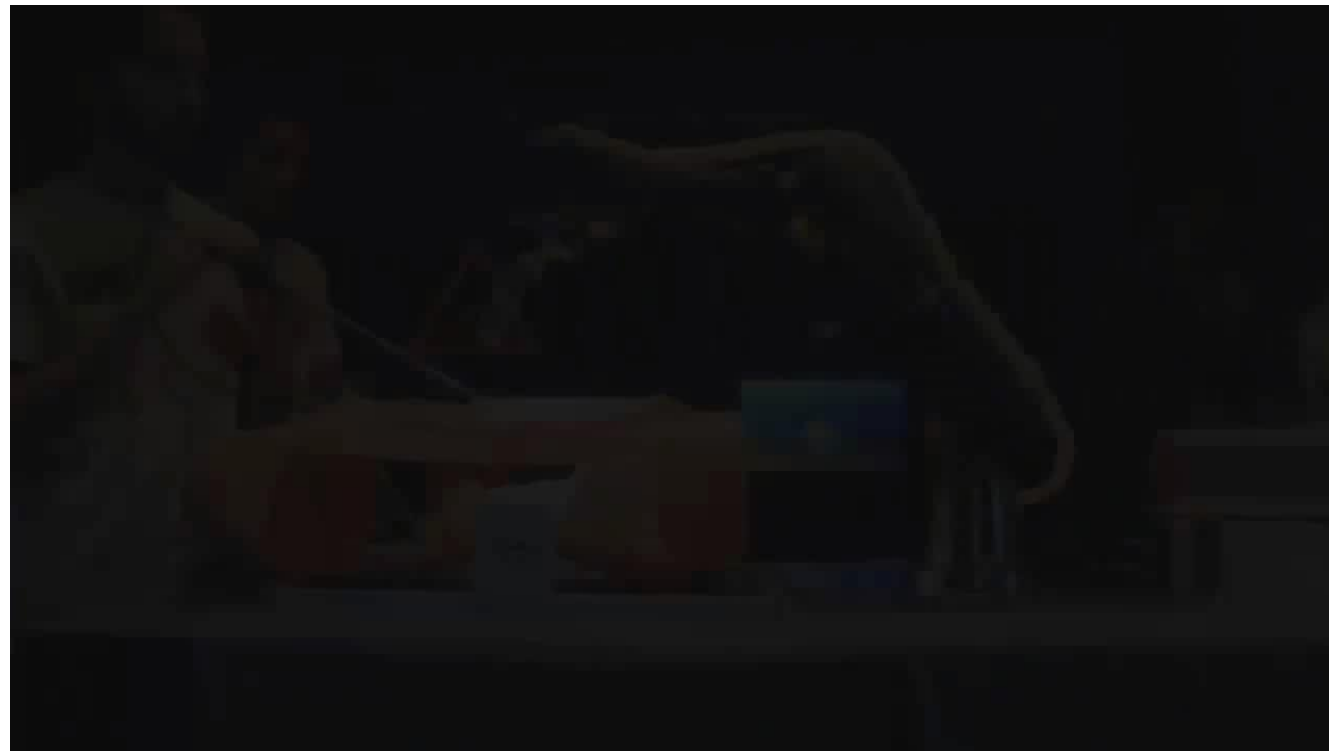


Flexible Fluidic Actuators



All the other fluidic actuators based on elastomers which use other constraint methods to obtain deformation.

Different braided structures



Cianchetti M, Ranzani T, Gerboni G, De Falco I, Laschi C, Menciassi A (2013) "STIFF-FLOP Surgical Manipulator: mechanical design and experimental characterization of the single module", Conf Proc IEEE on Intelligent and Robotic Systems – IROS 2013, 3567-3581



The STIFF-FLOP modular arm



Overall

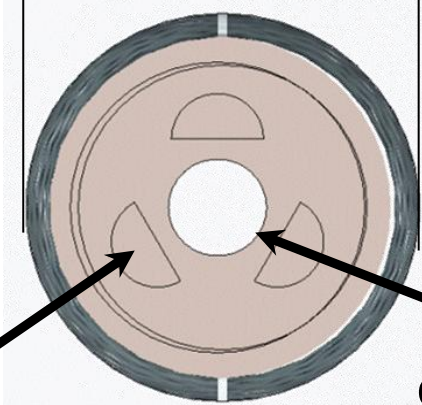
Fluidic chamber

Stiffening chamber

30 mm

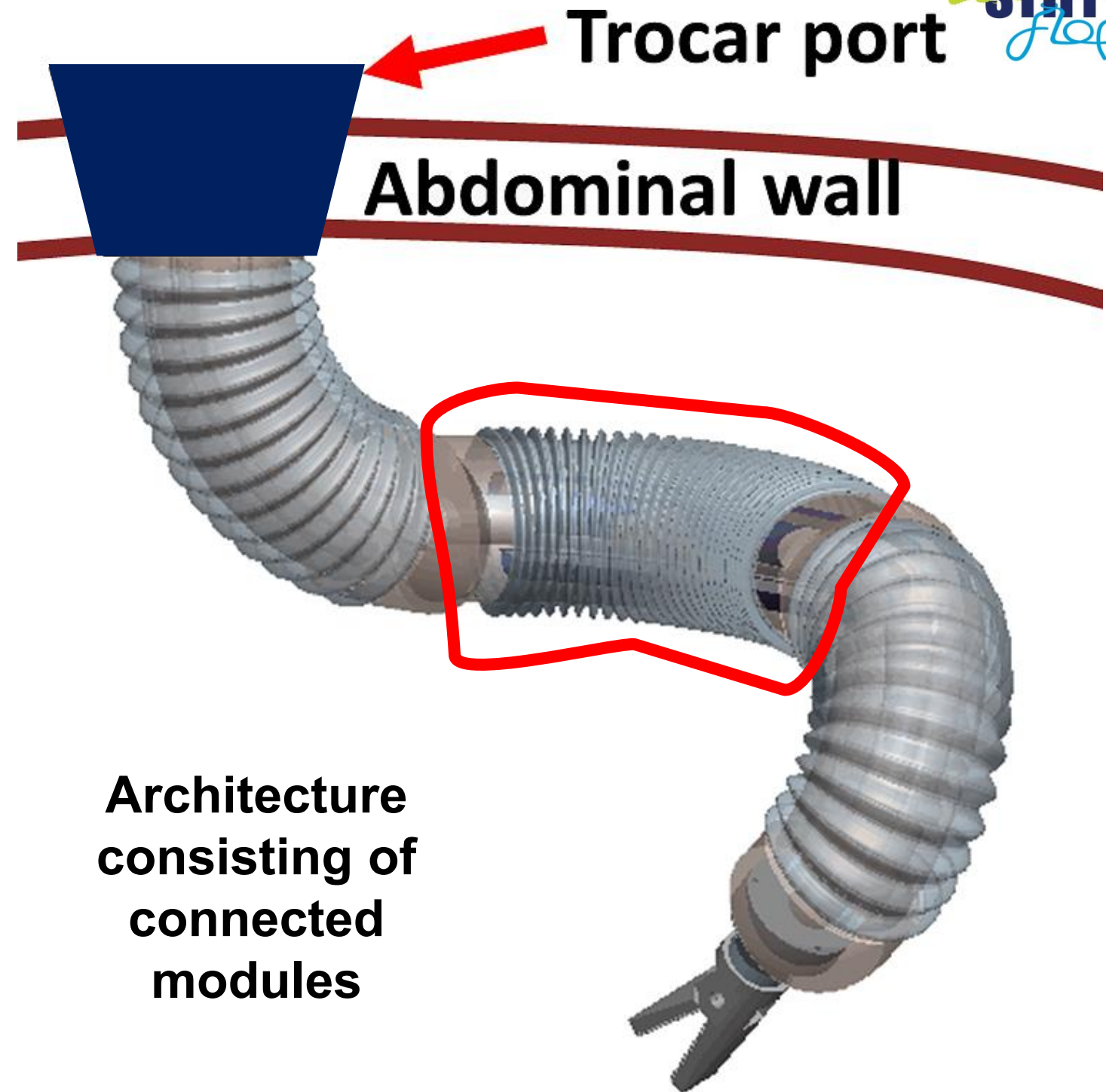
50 mm

D= 35 mm



d= 8 mm

d= 8 mm



Architecture consisting of connected modules

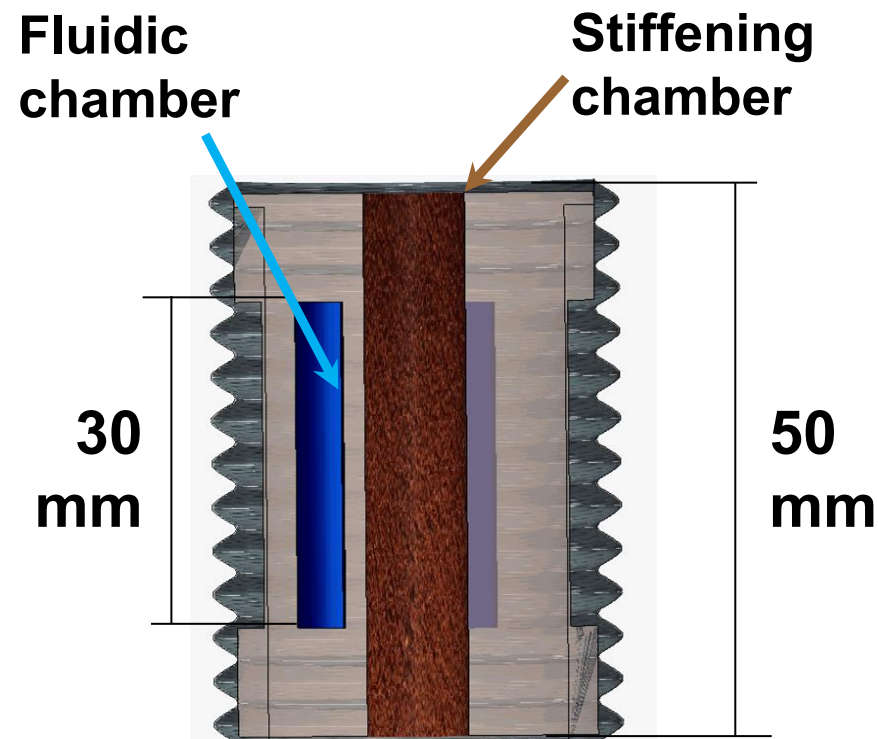
Cianchetti M, Ranzani T, Gerboni G, De Falco I, Laschi C, Menciassi A (2013) "STIFF-FLOP Surgical Manipulator: mechanical design and experimental characterization of the single module", Conf Proc IEEE on Intelligent and Robotic Systems – IROS 2013, 3567-3581



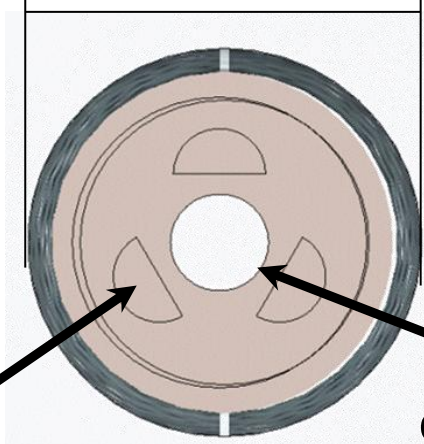
The STIFF-FLOP modular arm



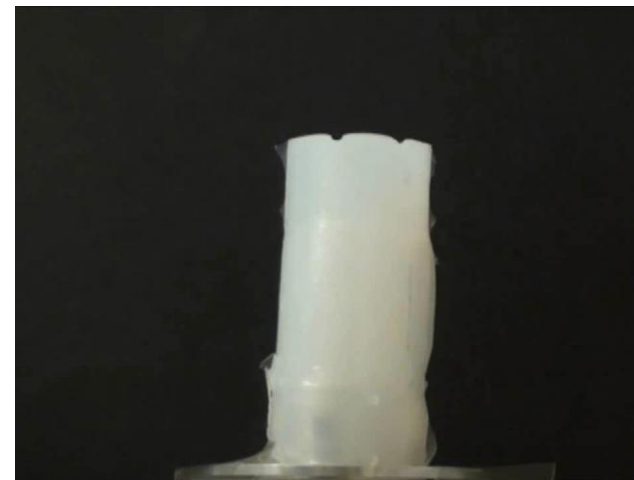
Flexible fluidic actuation



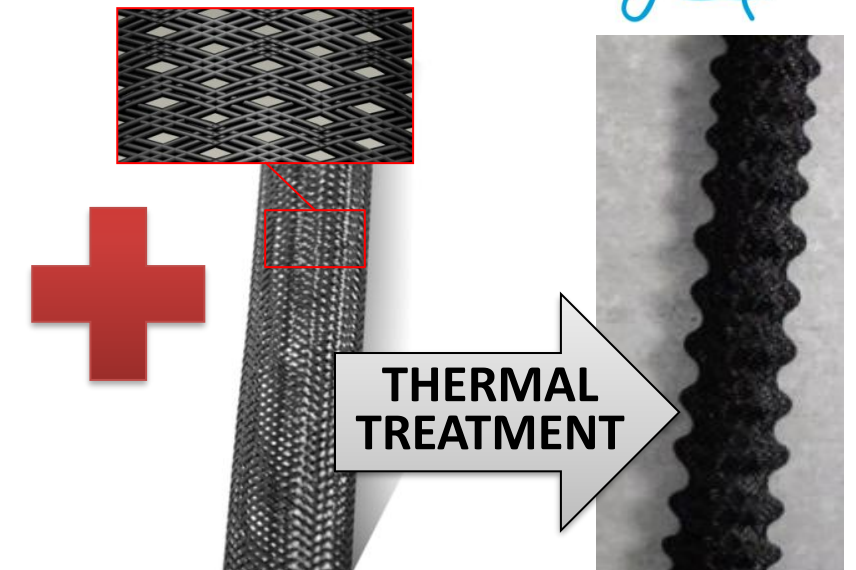
D= 35 mm



d= 8 mm



Flexible silicone actuator with three embedded fluidic chambers



External braided sheath limiting chambers lateral expansion



- easy to fabricate
- stable
- withstands very high strains without damage
- not susceptible to loading hysteresis
- cheap

Cianchetti M, Ranzani T, Gerboni G, De Falco I, Laschi C, Menciassi A (2013) "STIFF-FLOP Surgical Manipulator: mechanical design and experimental characterization of the single module", Conf Proc IEEE on Intelligent and Robotic Systems – IROS 2013, 3567-3581



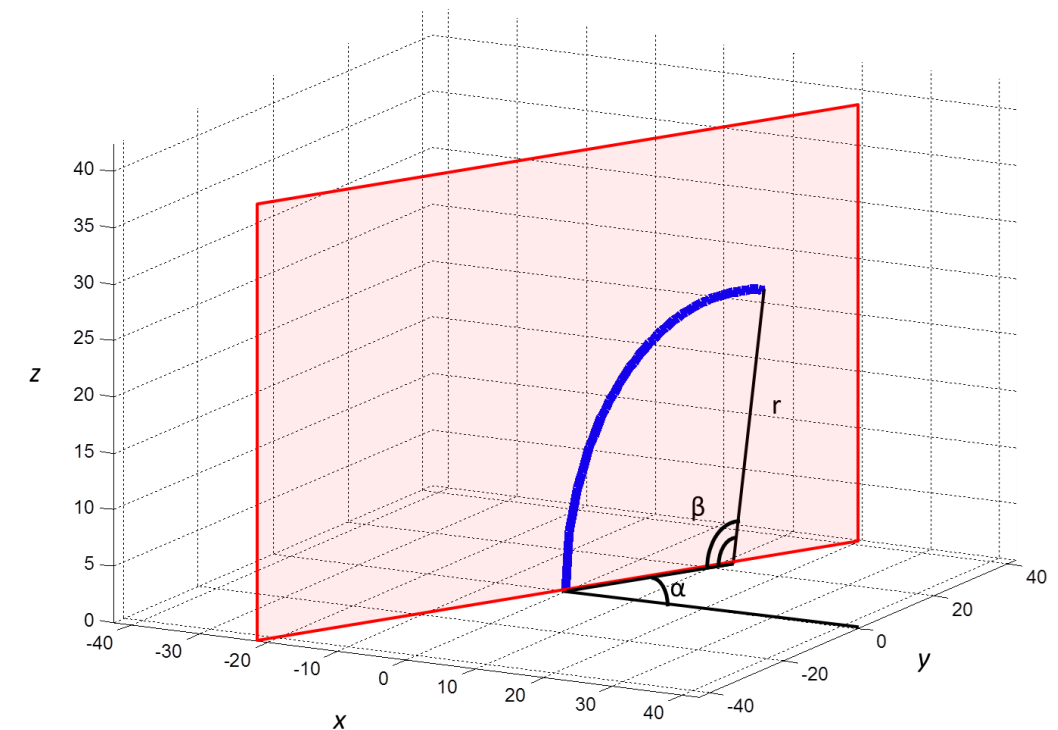
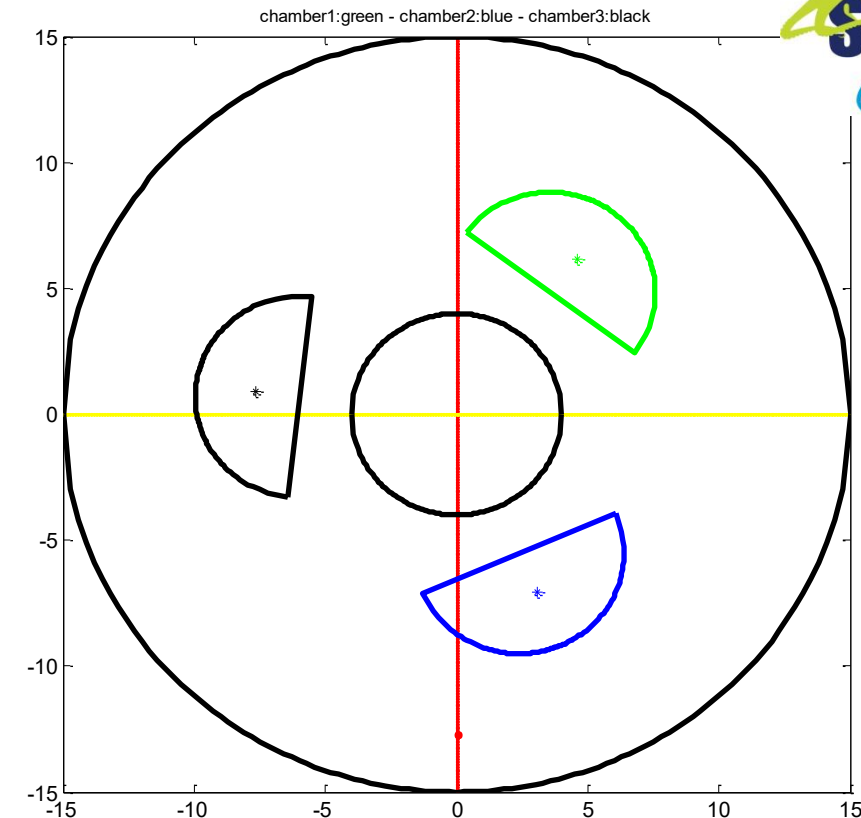
The STIFF-FLOP modular arm



Modelling – Analytic (Eulero-Bernulli)

Assumptions:

- The module can be described as a constant curvature section of a circle
- The module undergoes to small deformations (**STRONG**).
- The module is supposed to be fixed at the origin.
- The rest position, when no chamber is inflated, corresponds to the module being aligned along the vertical axis, with a rest length.
- No external forces are applied to the module.



The STIFF-FLOP modular arm



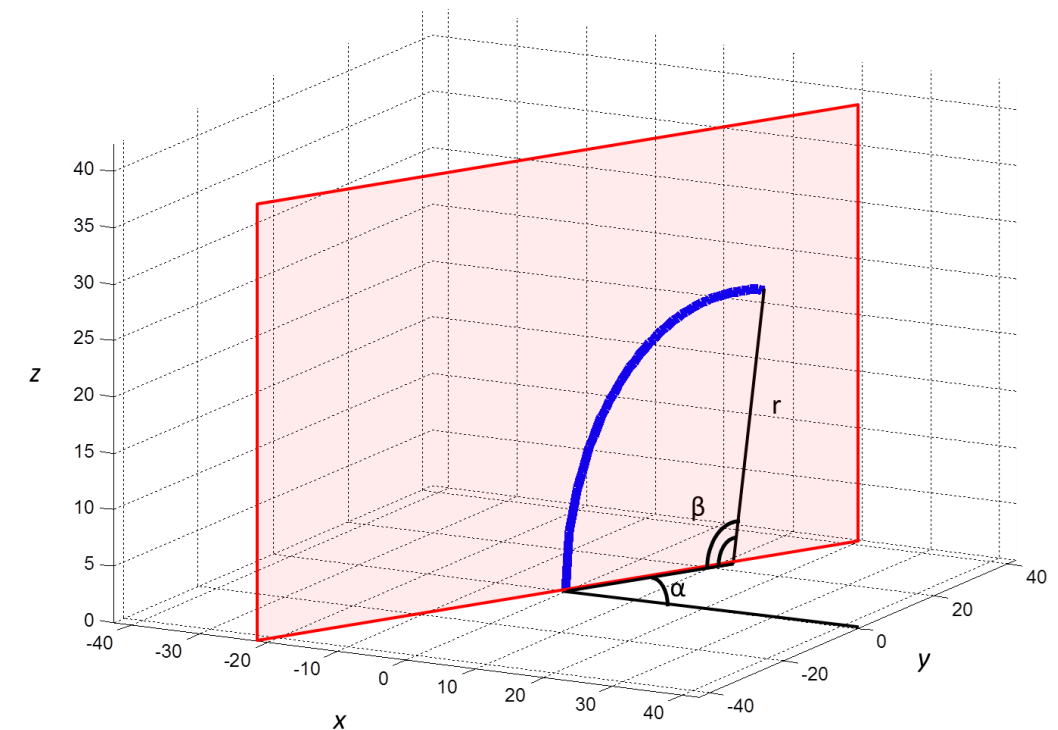
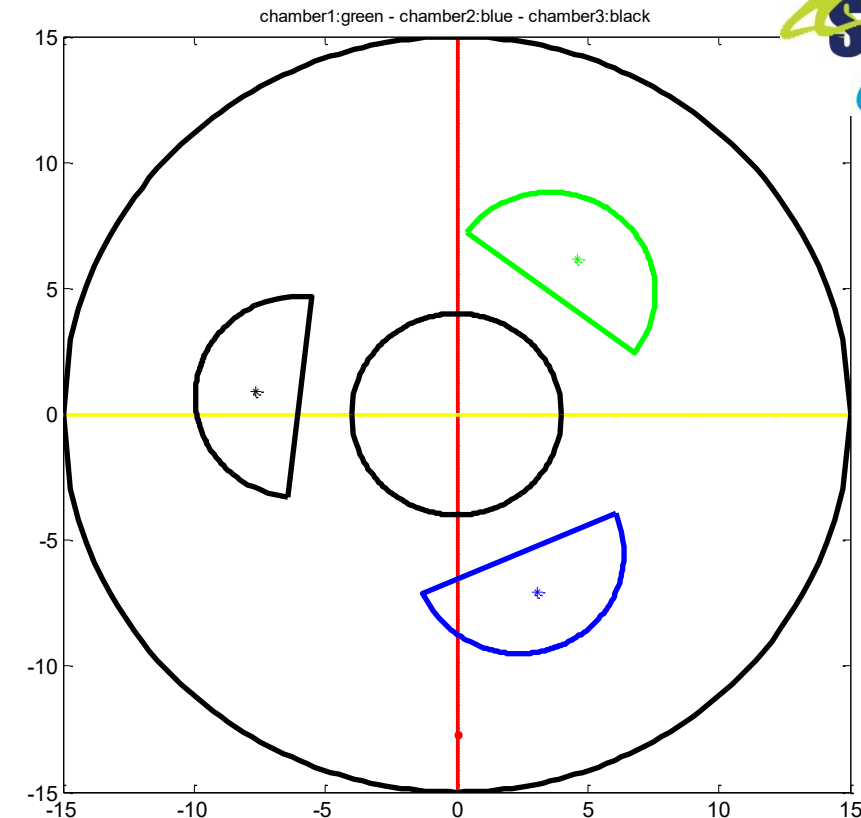
Modelling – Analytic (Eulero-Bernulli)

$$\tan \alpha = \left(\frac{\sqrt{3}(2P_1 - P_2 - P_3)}{3(P_2 - P_3)} \right)$$

$$r = \frac{EI}{M} = \frac{3EI}{A_p d \sum_{i=1}^3 (P_i \sin \alpha_i)}$$

$$L = \frac{A_p L_0}{3A_0 E} \sum_{i=1}^3 P_i + L_0$$

$$\beta = \frac{L}{r}$$

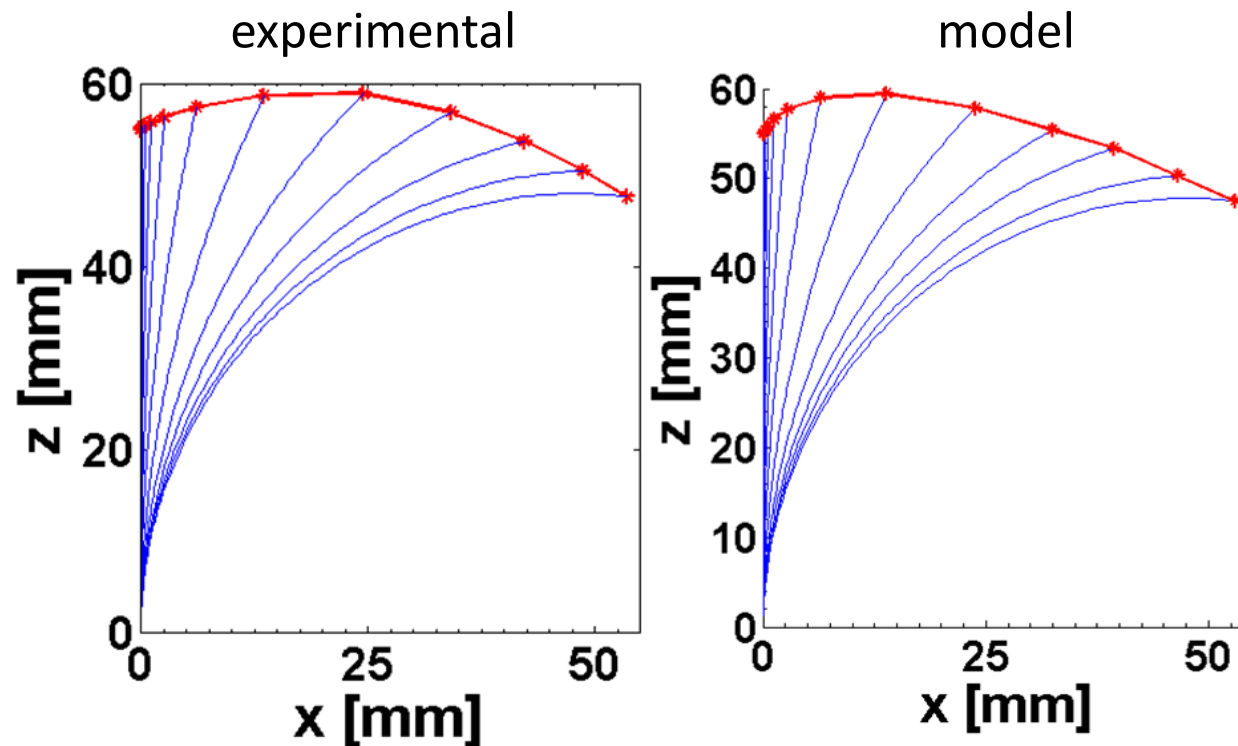


The STIFF-FLOP modular arm

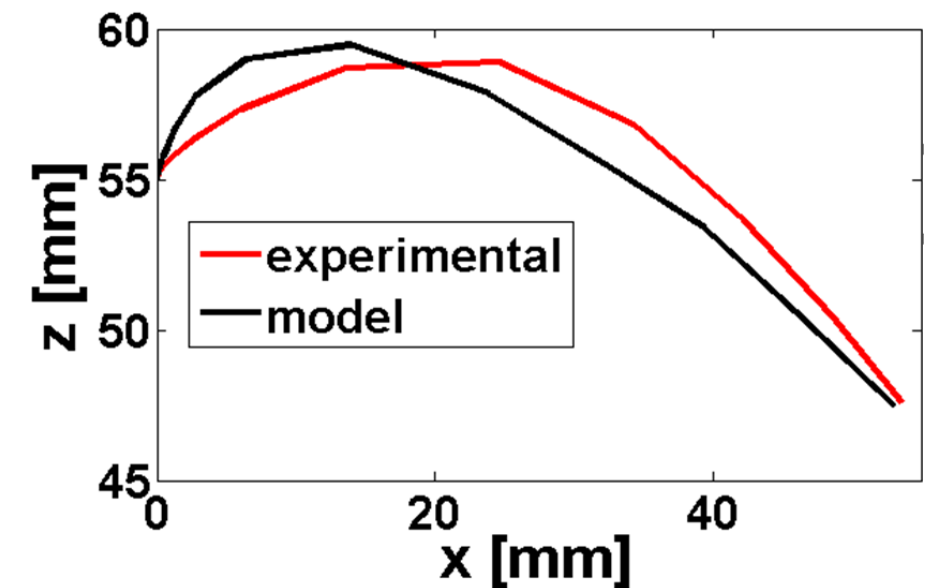
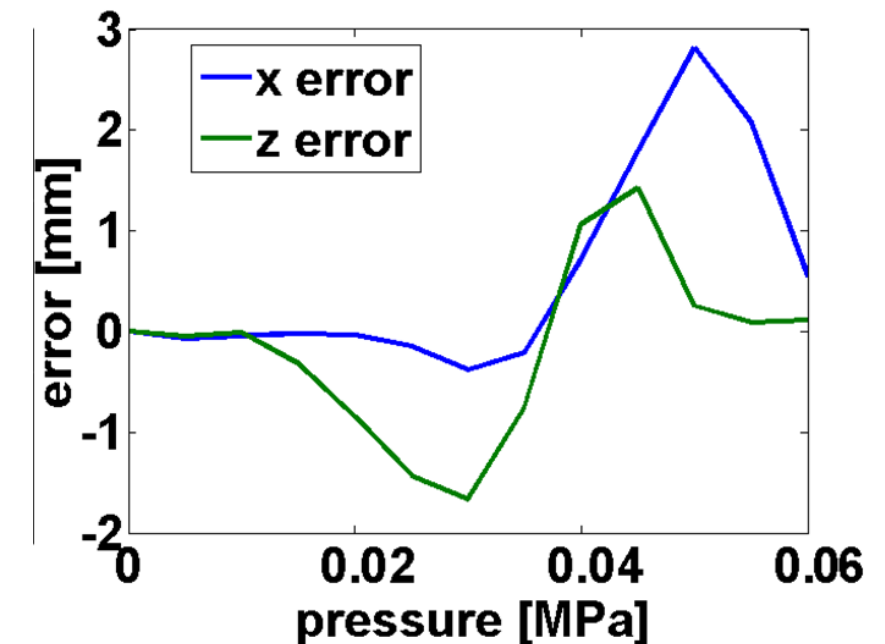


Modelling – Analytic (Eulero-Bernulli)

Results



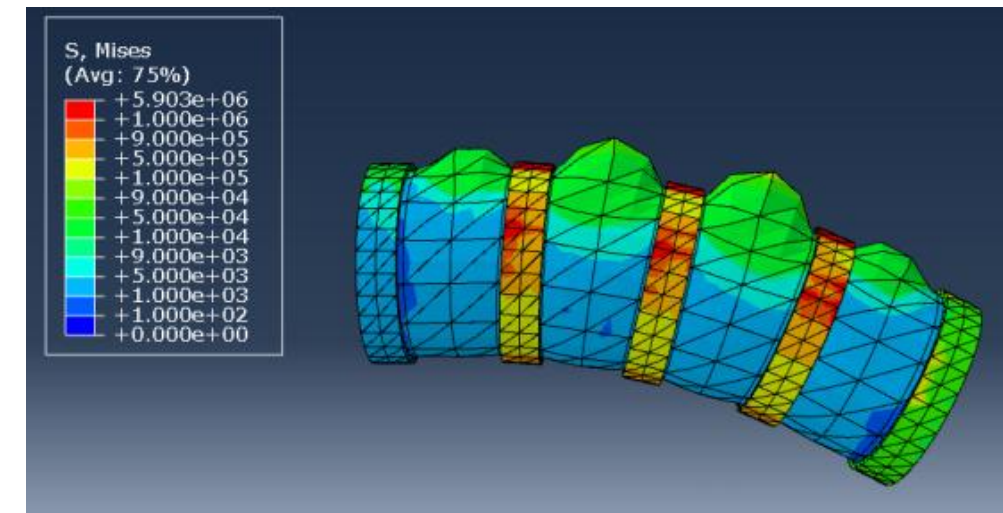
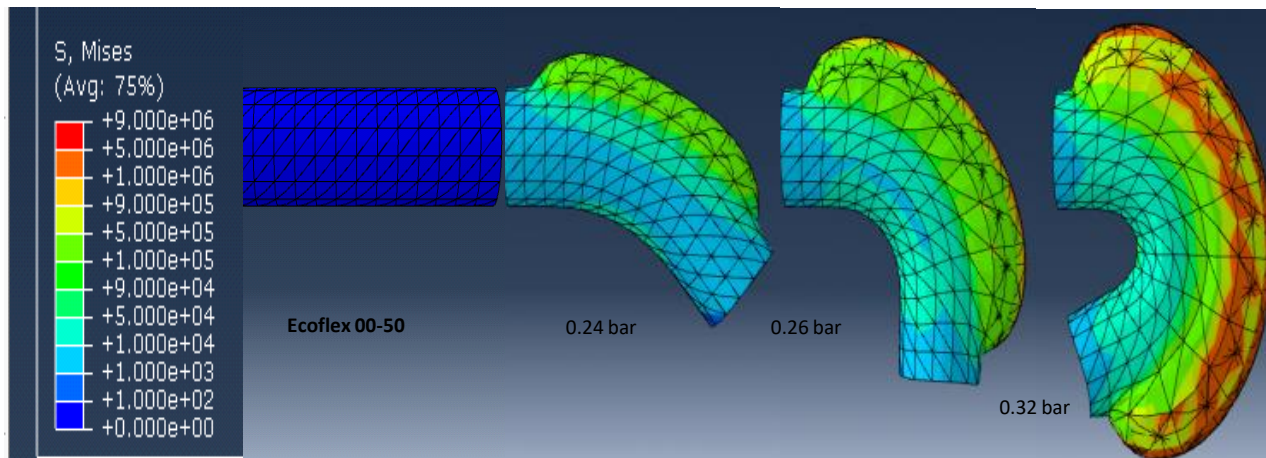
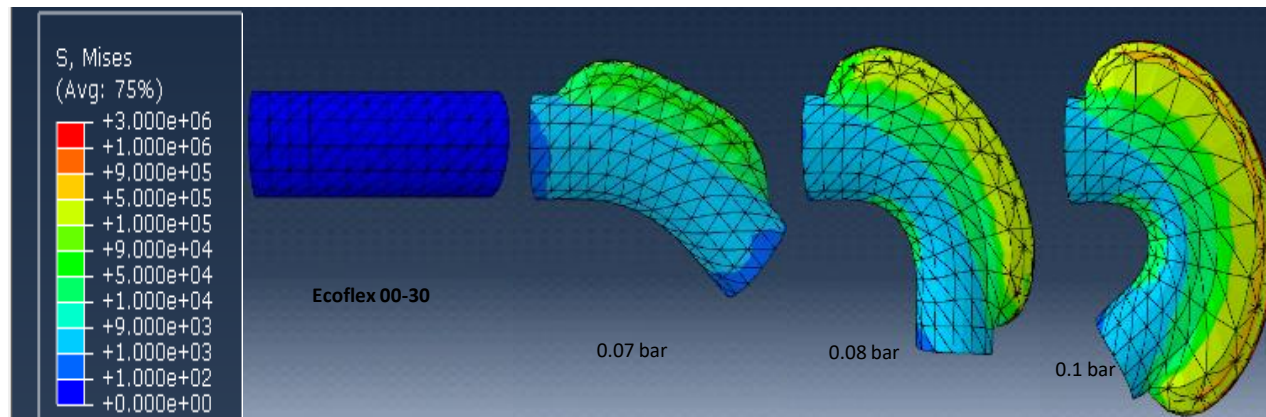
- Not precise
- Fast
- Useful for a general understanding



The STIFF-FLOP modular arm



Modelling – FEM



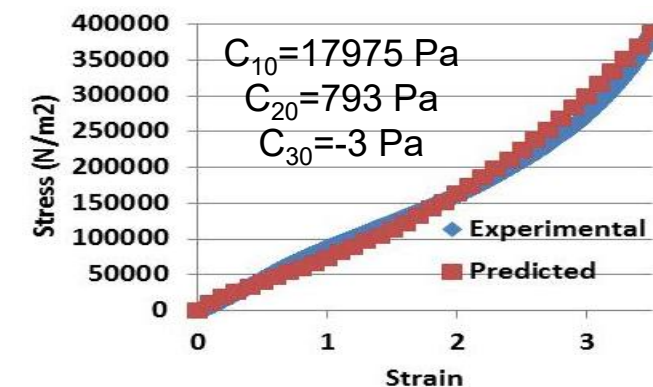
Energy Equation for hyperelastic material:

$$U = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2 + C_{30}(\bar{I}_1 - 3)^3$$

Where for incompressible elastomers

$$\bar{I}_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \quad \lambda_1 \lambda_2 \lambda_3 = 1$$

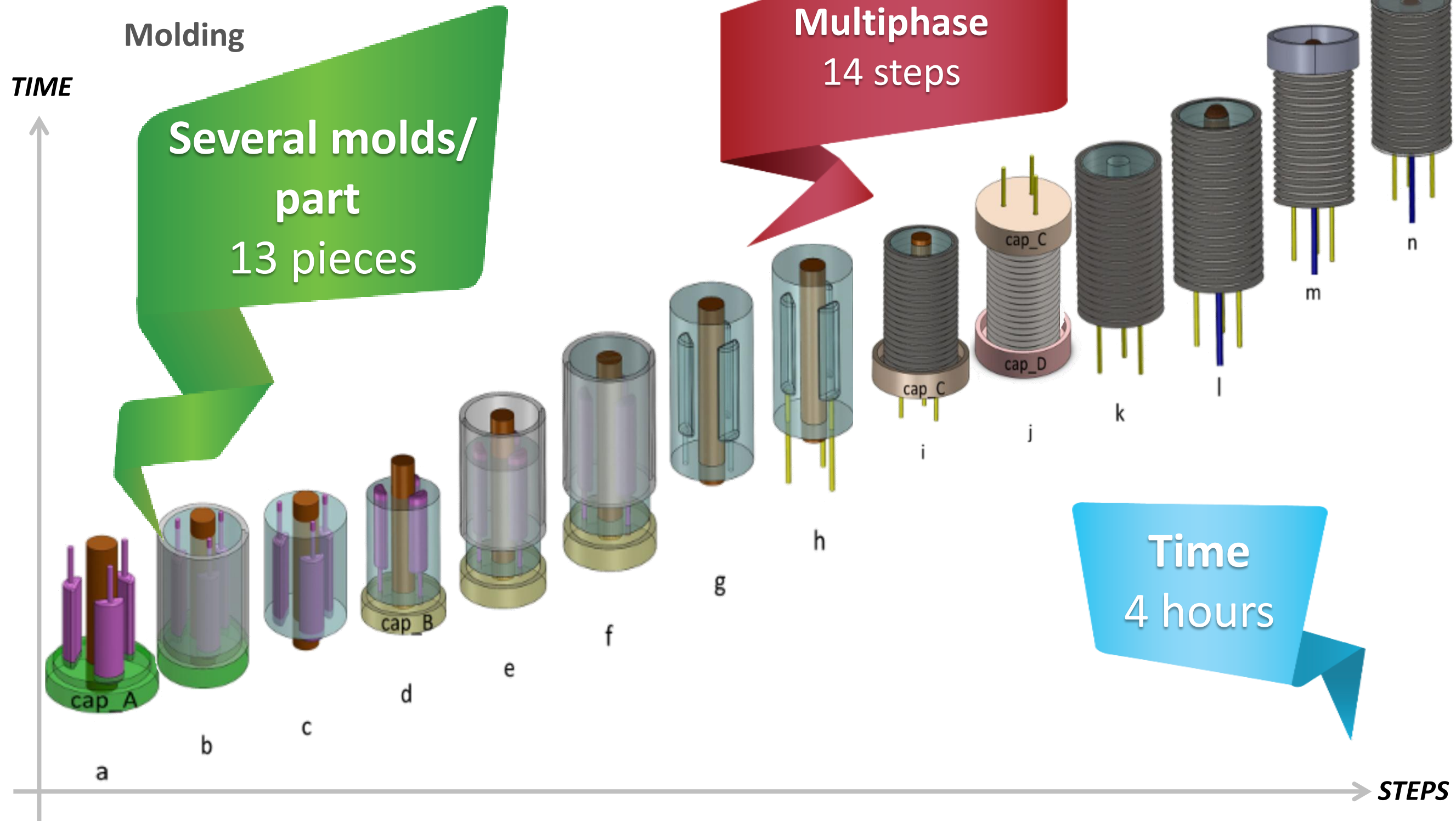
The model fits the stress-strain loading curve but cannot model hysteresis and rate effects



- Slow
- Precise (depending on material properties knowledge) in describing non-linear and complex interactions



Module fabrication



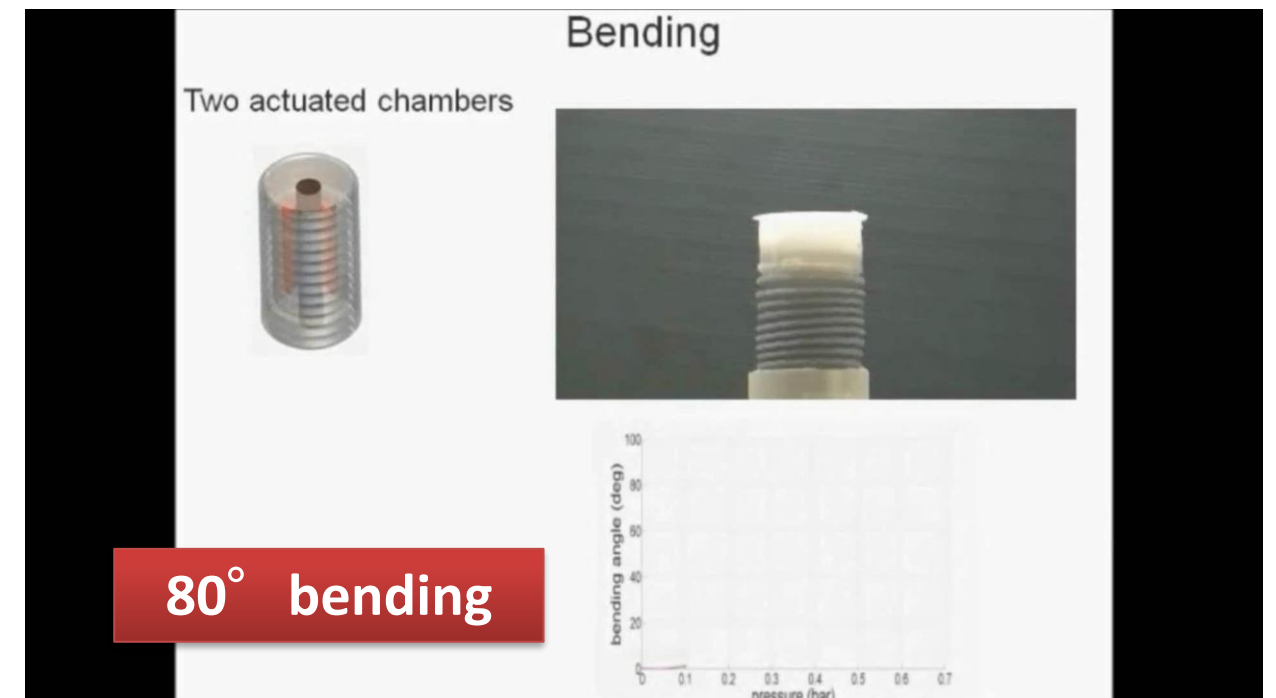
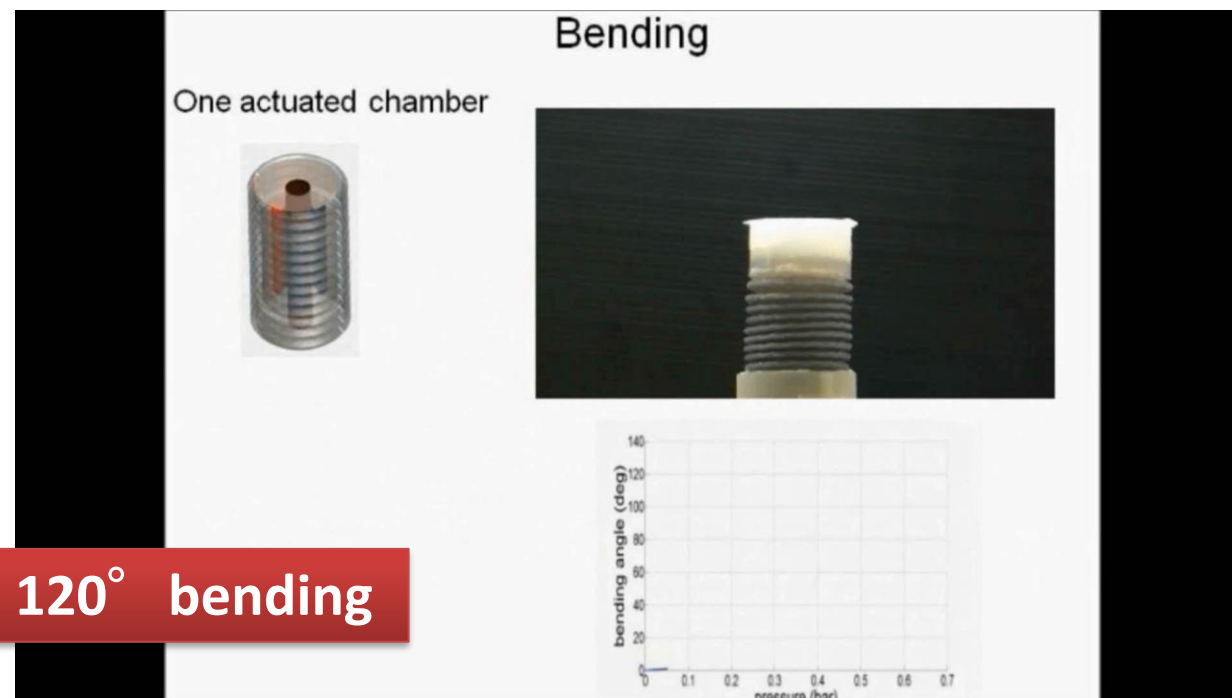
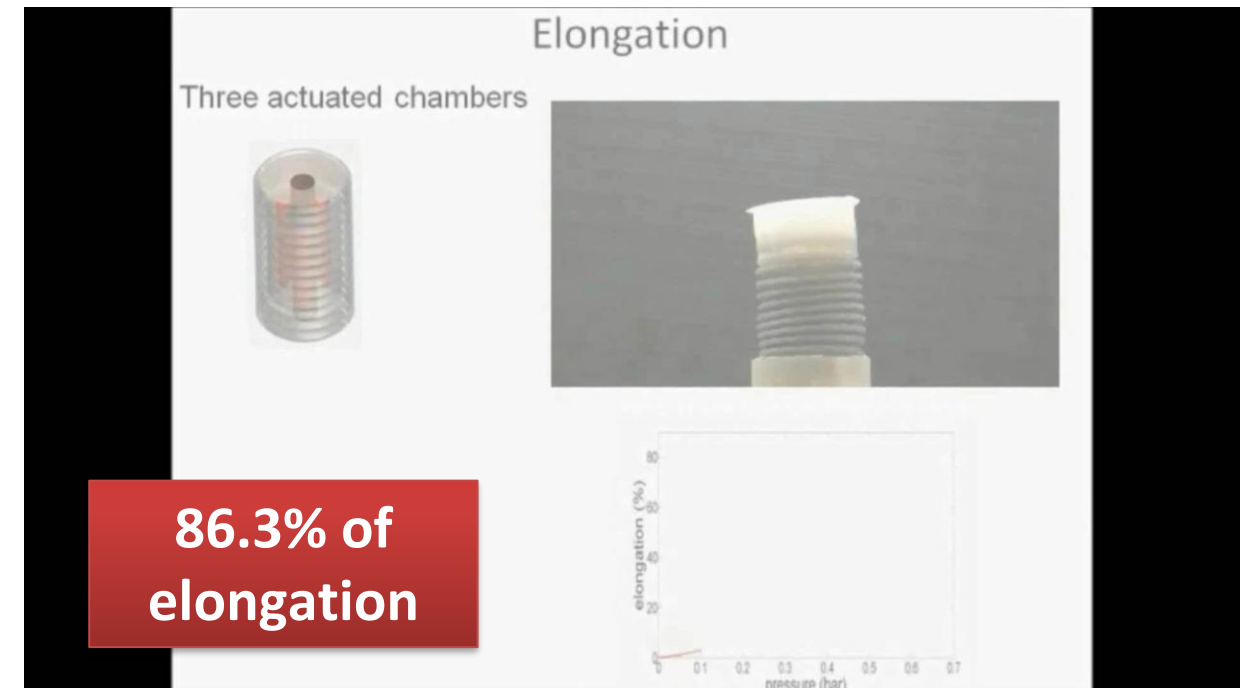
De Falco et al. "Design and Fabrication of an Elastomeric Unit for Soft Modular Robots in Minimally Invasive Surgery" J. Vis. Exp., e53118, 2015.



Module performance



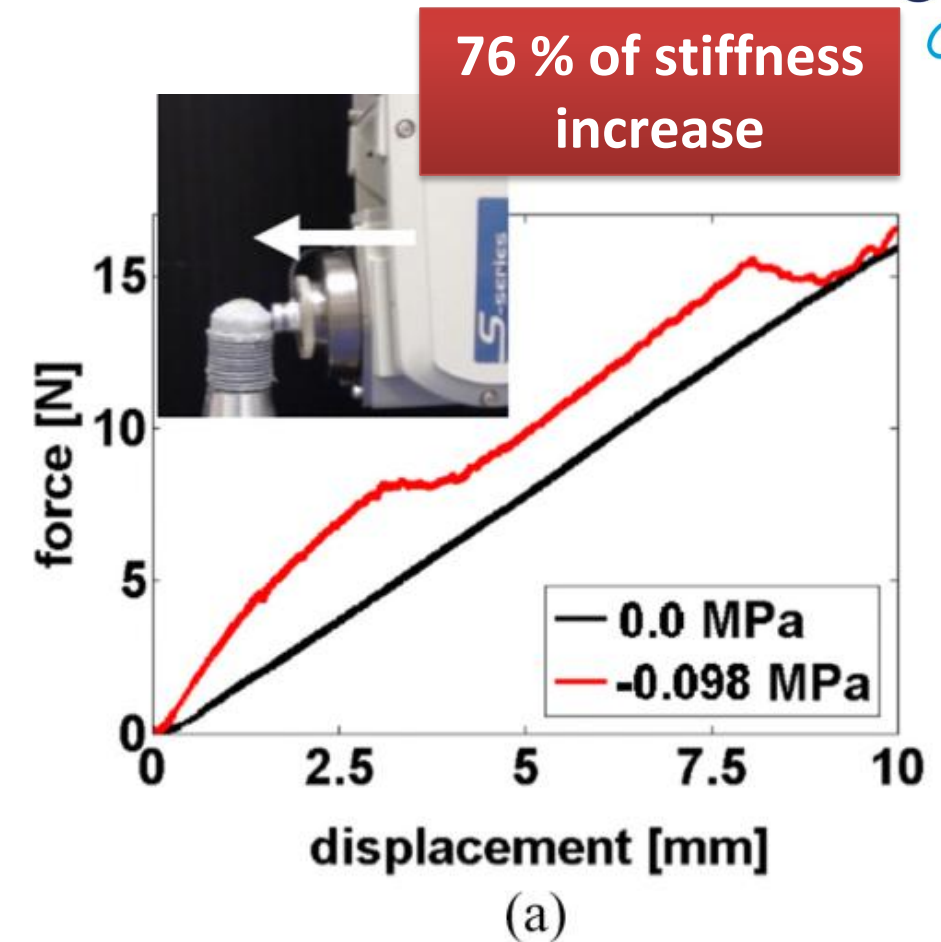
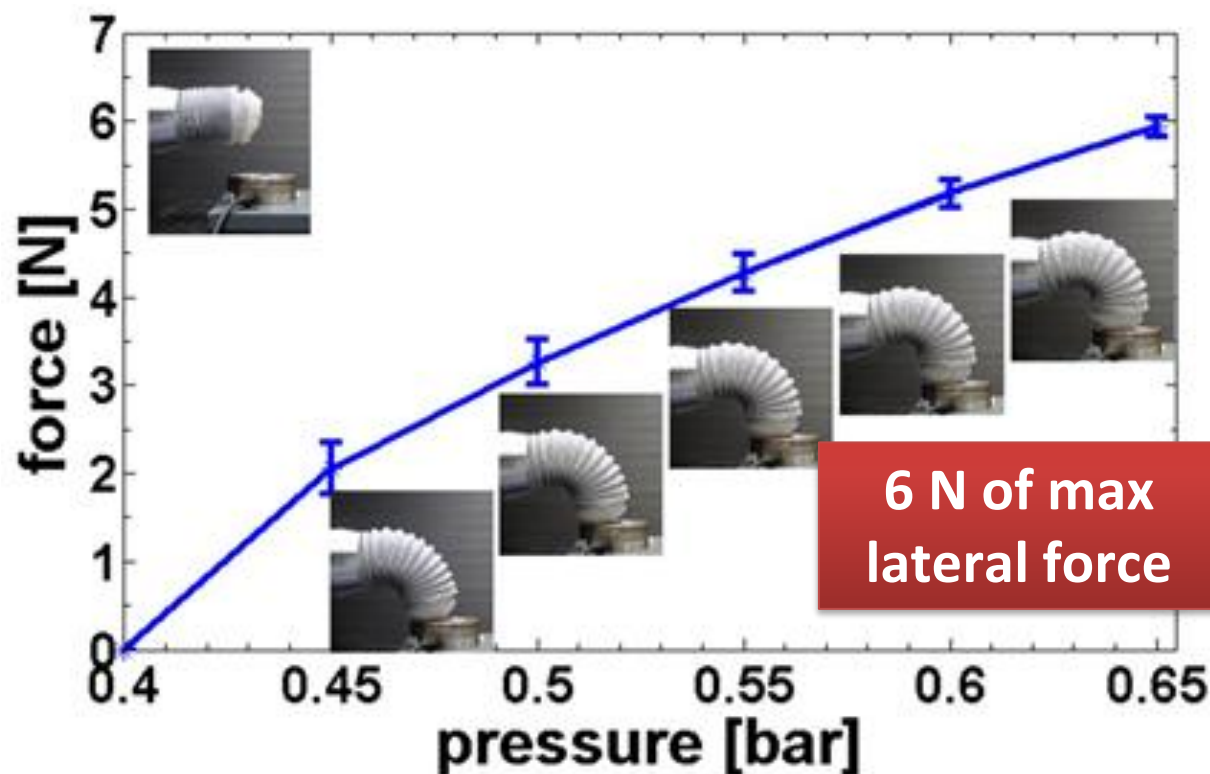
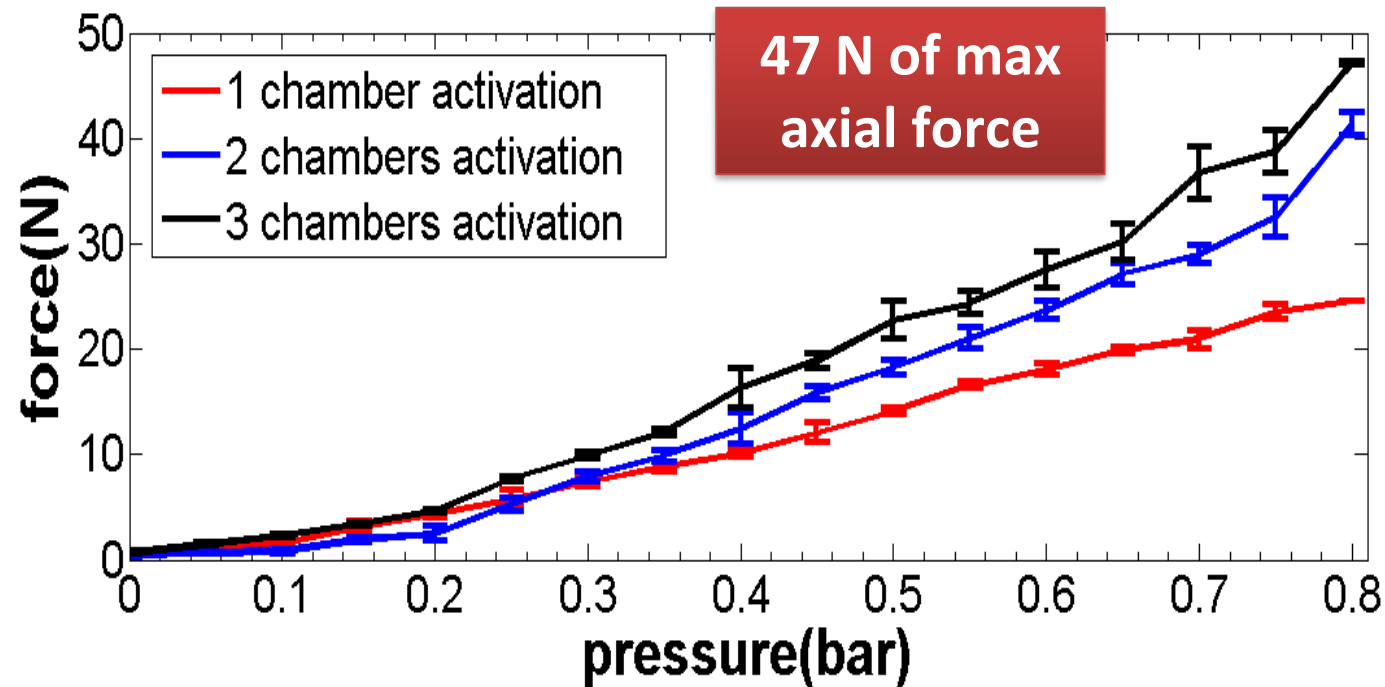
Due to the absence of rigid structures, the module can be deformed for passing through single port laparoscopy incision



Ranzani T, Cianchetti M, Gerboni G, De Falco I, Menciassi A (2016) "A Soft Modular Manipulator for Minimally Invasive Surgery: Design and Characterization of a Single Module", IEEE Transactions on Robotics, 32(1), 187-200.



Module performance



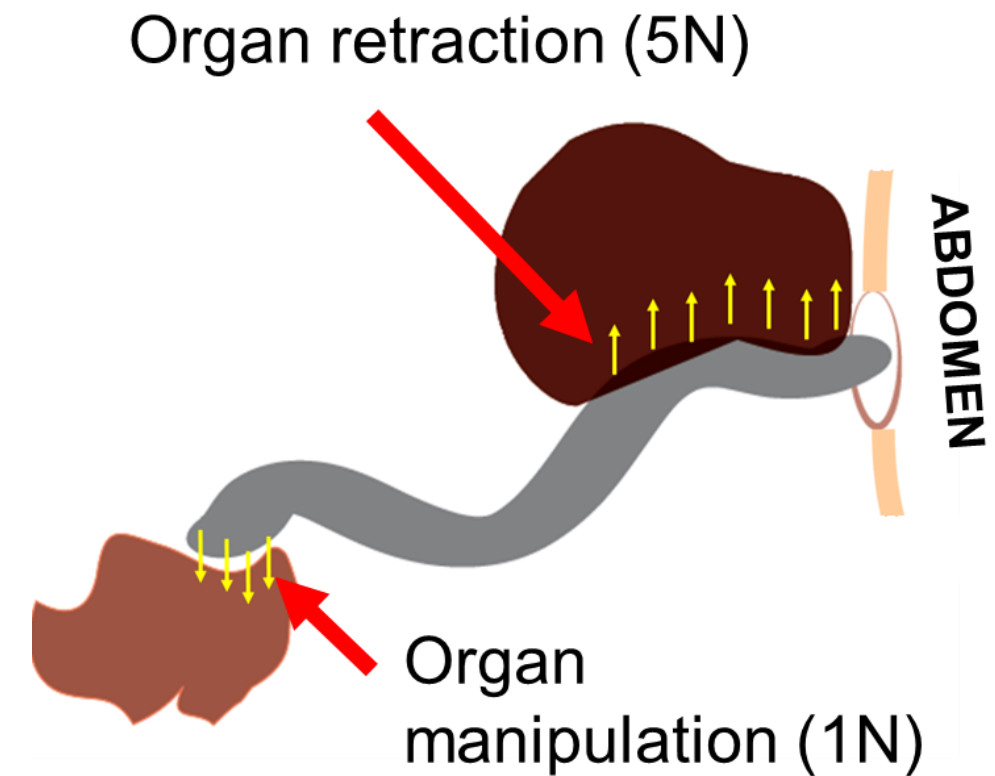
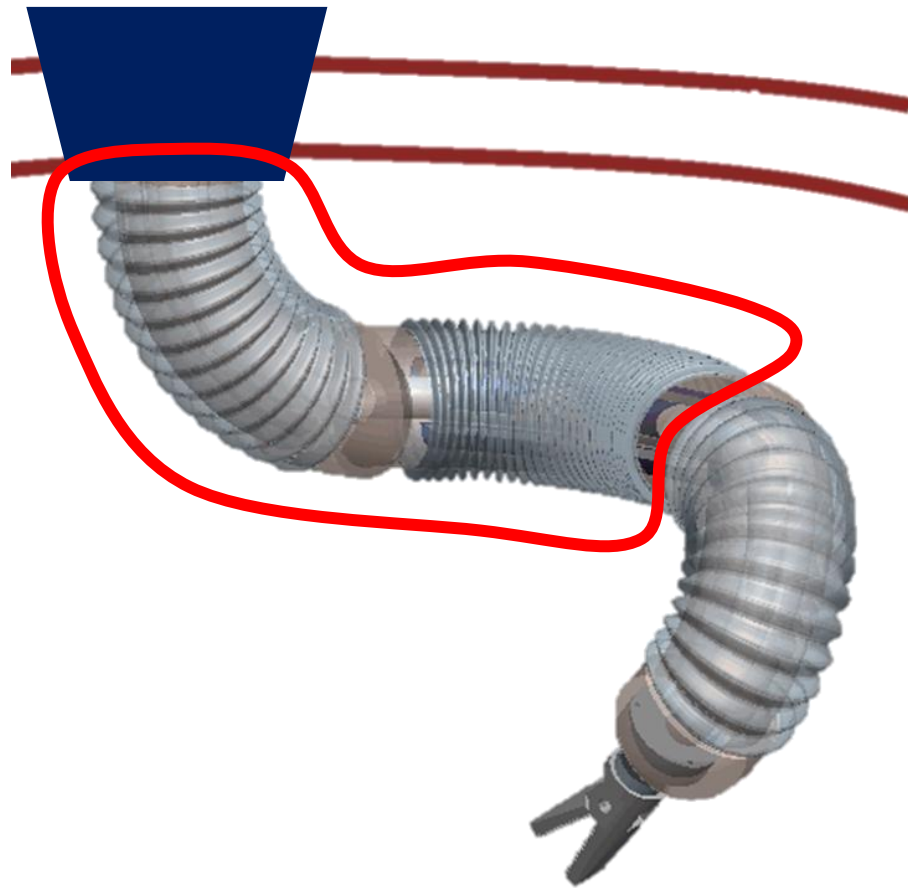
Test type		Elastic constant increase (%)
Axial tests	No chamber inflation	75.70
	90° bending (0.04 MPa pressure in one chamber)	32.52
Lateral tests	Elongation (0.045 MPa pressure in the three chambers)	73.14
	Elongation (0.065 MPa pressure in the three chambers)	44.66

¹ internal pressure set to -0.098 MPa.

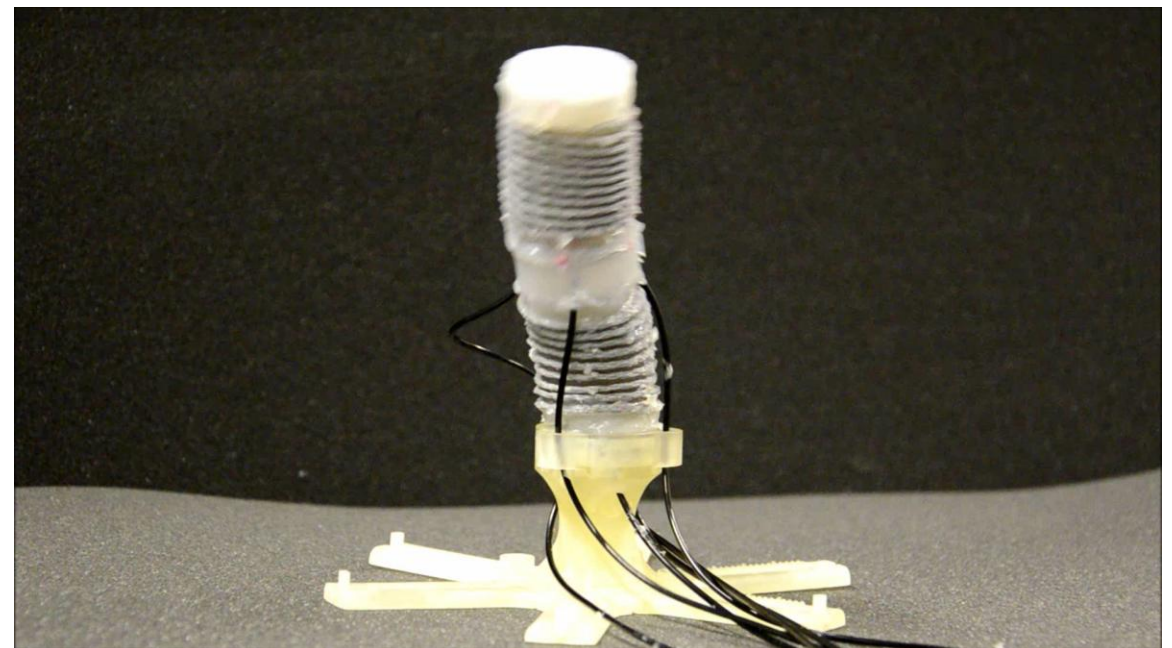
Ranzani T, Cianchetti M, Gerboni G, De Falco I, Menciassi A (2016) "A Soft Modular Manipulator for Minimally Invasive Surgery: Design and Characterization of a Single Module", IEEE Transactions on Robotics, 32(1), 187-200.



The 2-module manipulator



- Modular structure
- Each module can be independently stiffened and actuated

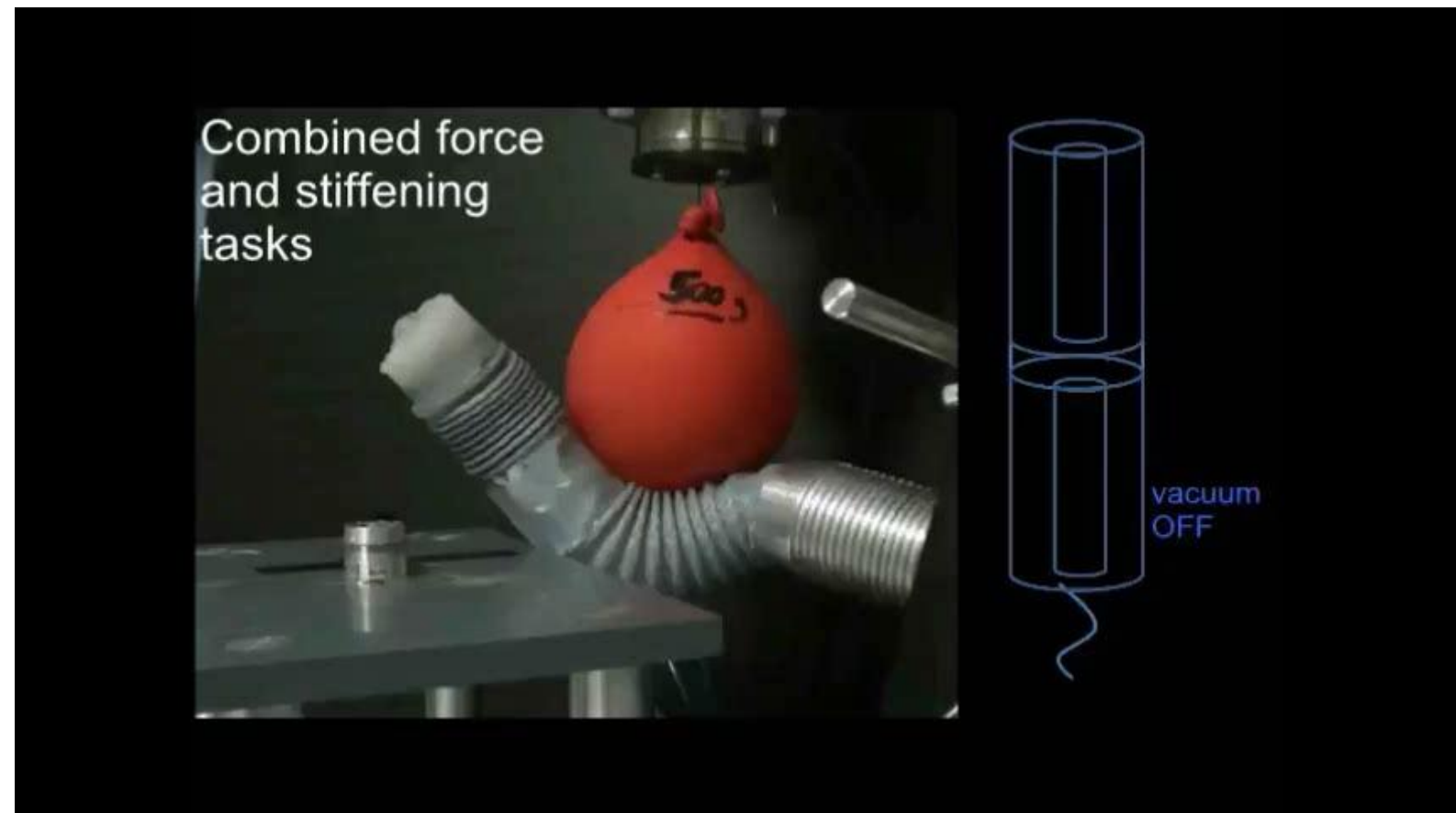
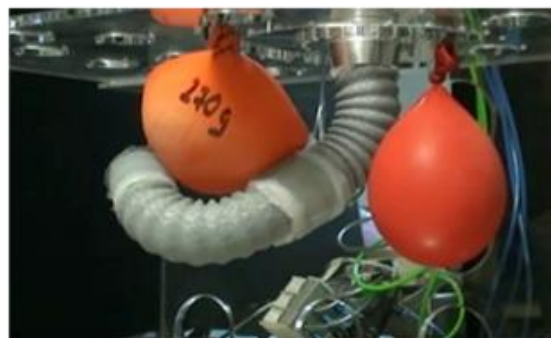


The STIFF-FLOP arm: new surgical scenario



Traditional surgical tasks imply the use of multiple specialized instruments such as graspers, retractors, vision system, dissectors etc. which perform different tasks producing different forces...

... It is possible to exploit the modular structure of the manipulator to carry out **different tasks with different sections of the manipulator**



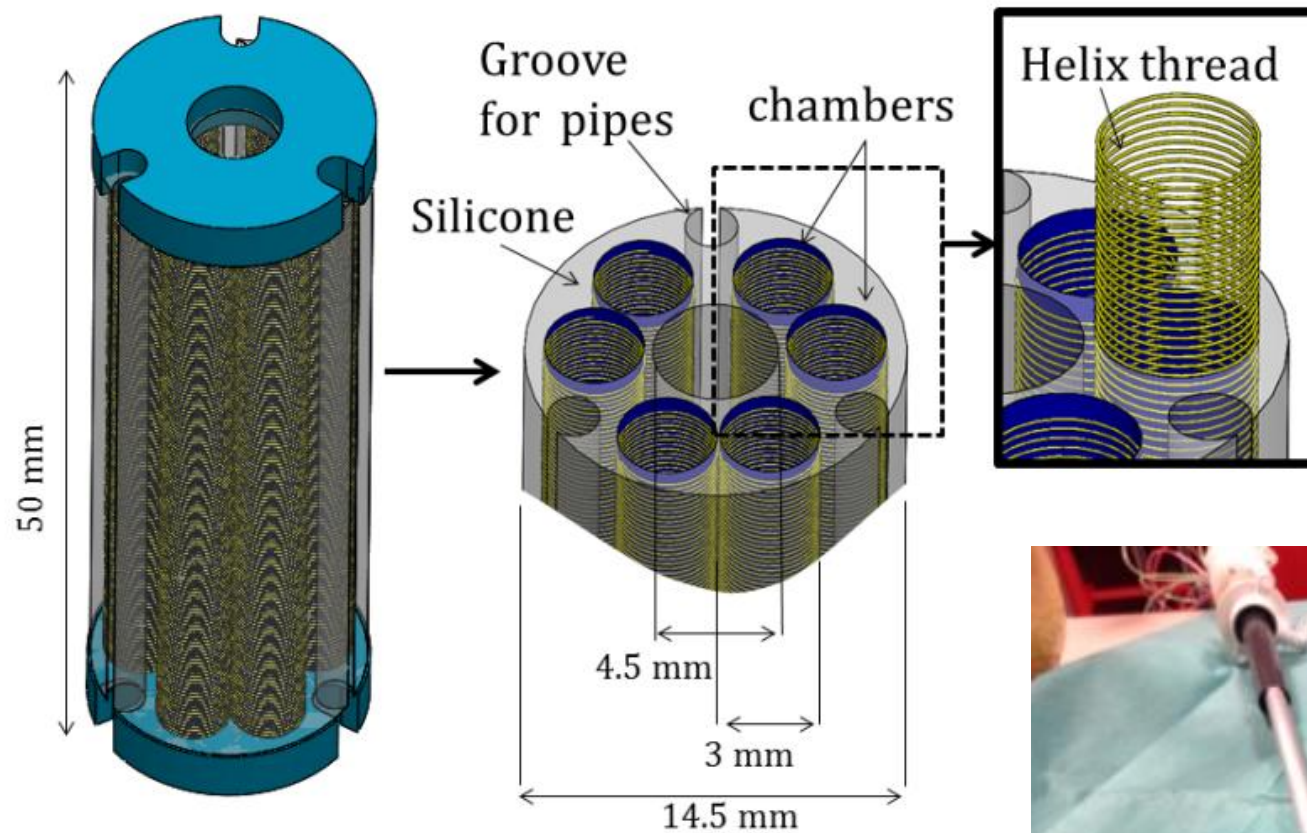
Ranzani T, Gerboni G, Cianchetti M, Menciassi A (2015) "A bioinspired soft manipulator for minimally invasive surgery" Bioinspiration & Biomimetics, 10, 03500



Miniaturized version of the manipulator



2-module manipulator



Diameter: 14.3 mm
Length: 55 mm / 120 mm
pressure chambers: 6
Pipes for actuation: 3 per module



Miniaturized version of the manipulator



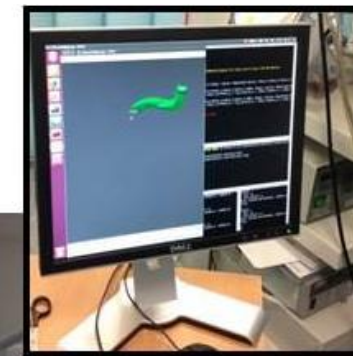
G. Gerboni, M. Brancadoro, A. Diodato, J. Fras, H. A. Wurdemann, K. Althoefer, M. Cianchetti, A. Menciassi. "Flexible and high dexterous 2- module soft robotic manipulator for minimally invasive surgery", submitted to IEEE International Conference on Robotics and Automation (ICRA). 2016.



Miniaturized version of the manipulator



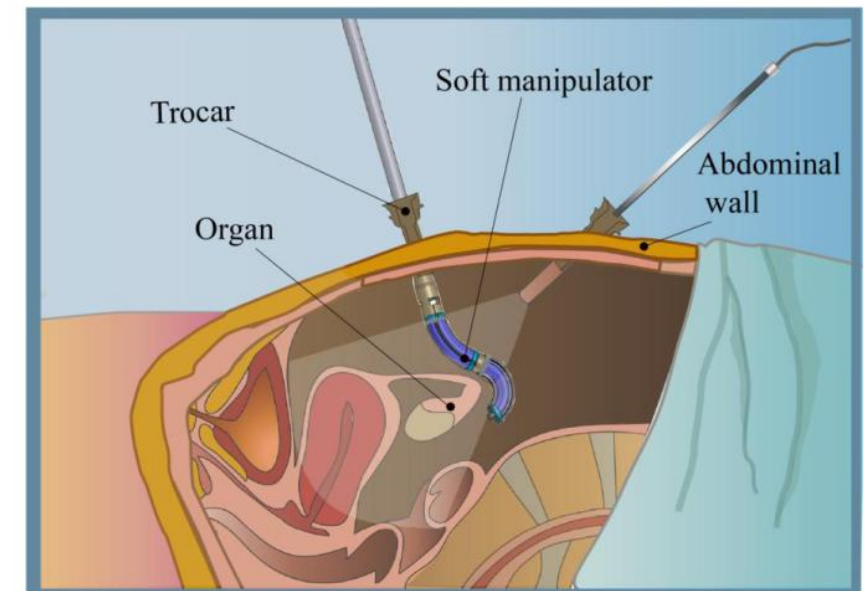
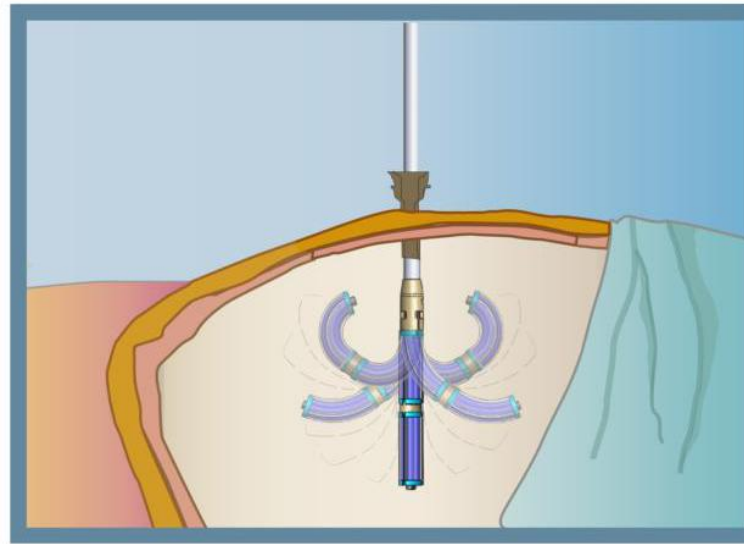
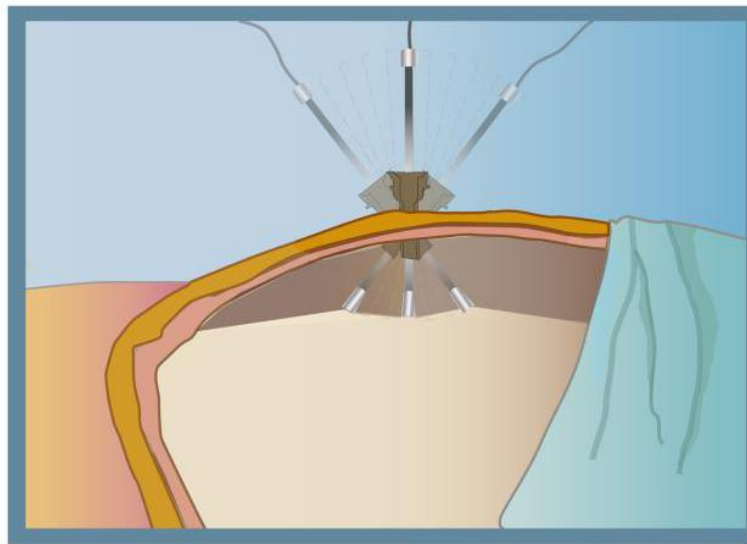
Cadaver tests



Stiff-Flop: Advantages

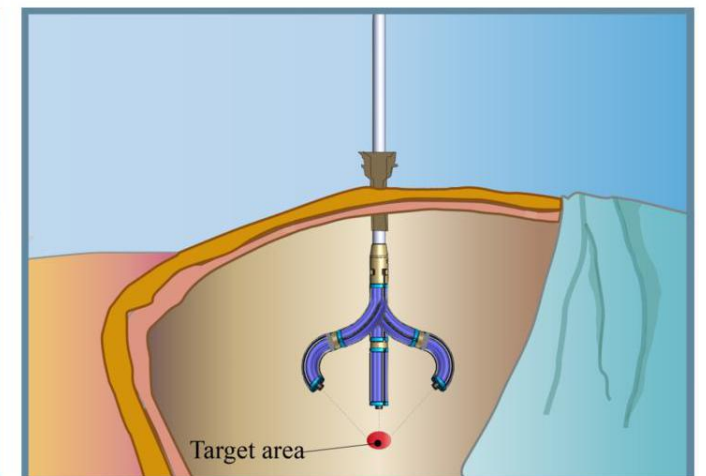
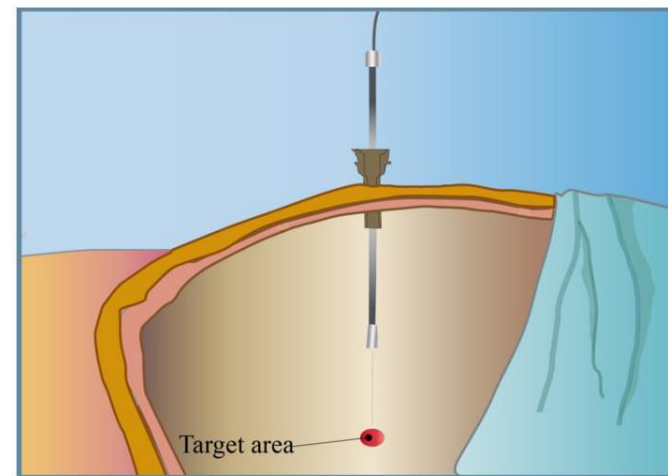


- The improvement of **manoeuvrability** and dexterity of the surgical instrument tip
- The achievement of **safe navigation** around abdominal organs/tissues



- Increase of the **field of view**

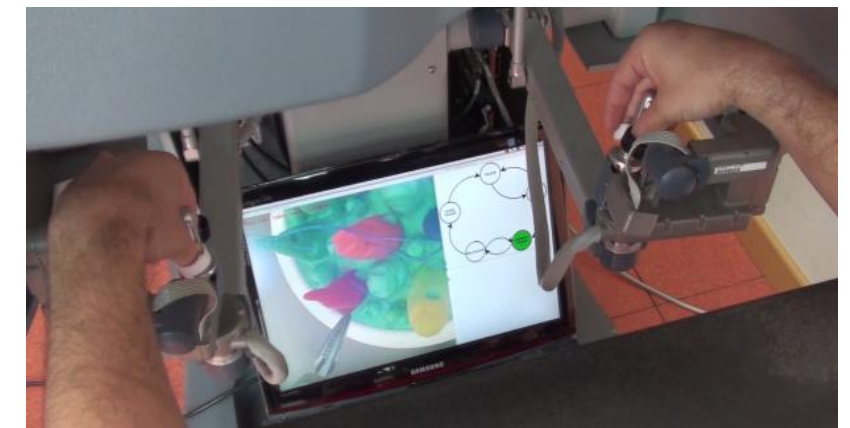
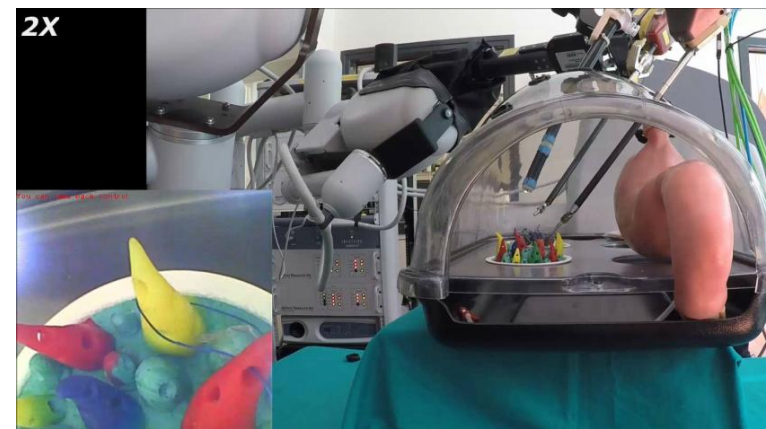
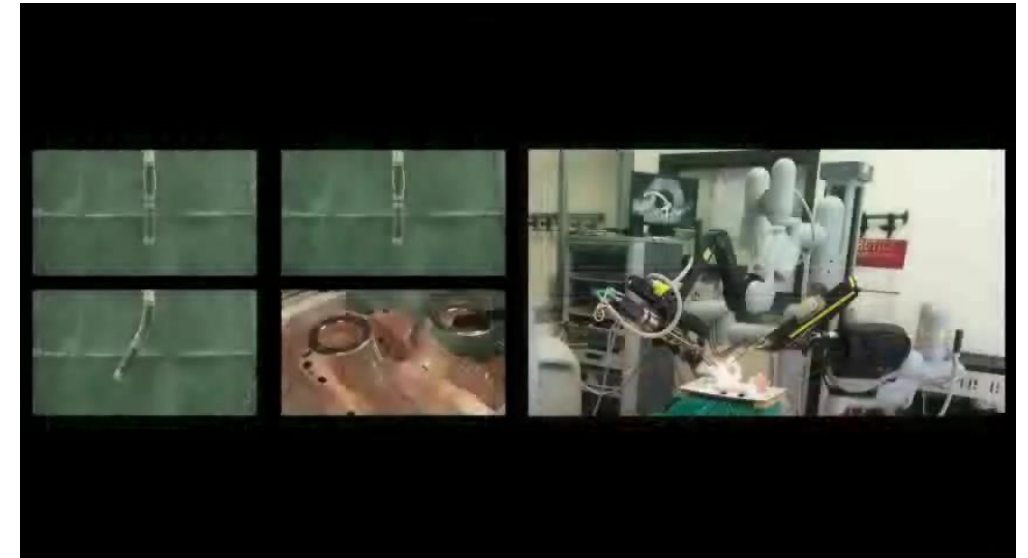
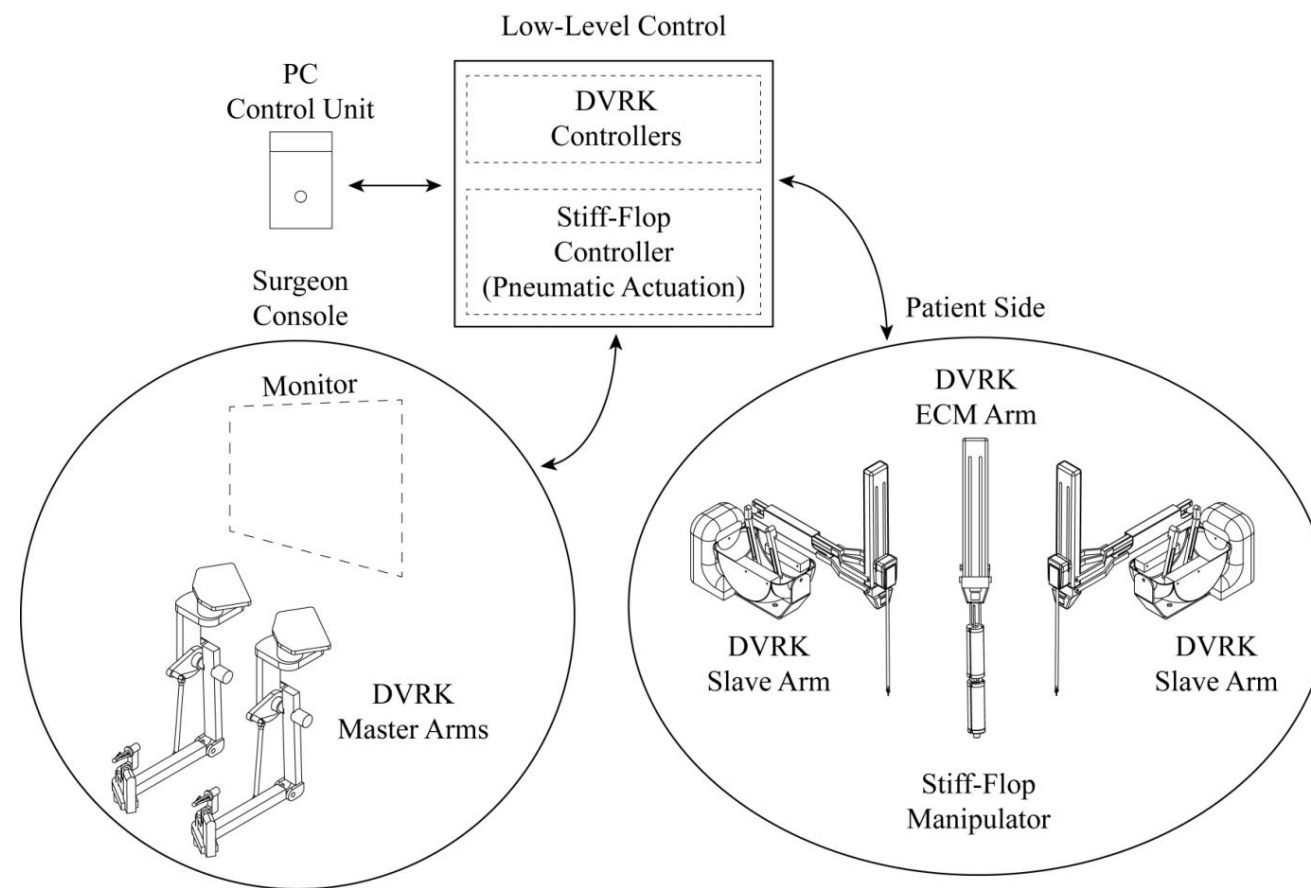
- It allows the surgeon to inspect the target from **different perspectives**, enriching his/her perception and understanding of the on-going surgical procedure



Abidi H, et al. (2018) "Highly Dexterous 2-module Soft Robot for Intra-organ Navigation in Minimally Invasive Surgery", *The International Journal of Medical Robotics and Computer Assisted Surgery*, e1875



Stiff-Flop and DVRK integration



Hardware Integration: HW interfaces to connect Stiff-Flop to ECM arm

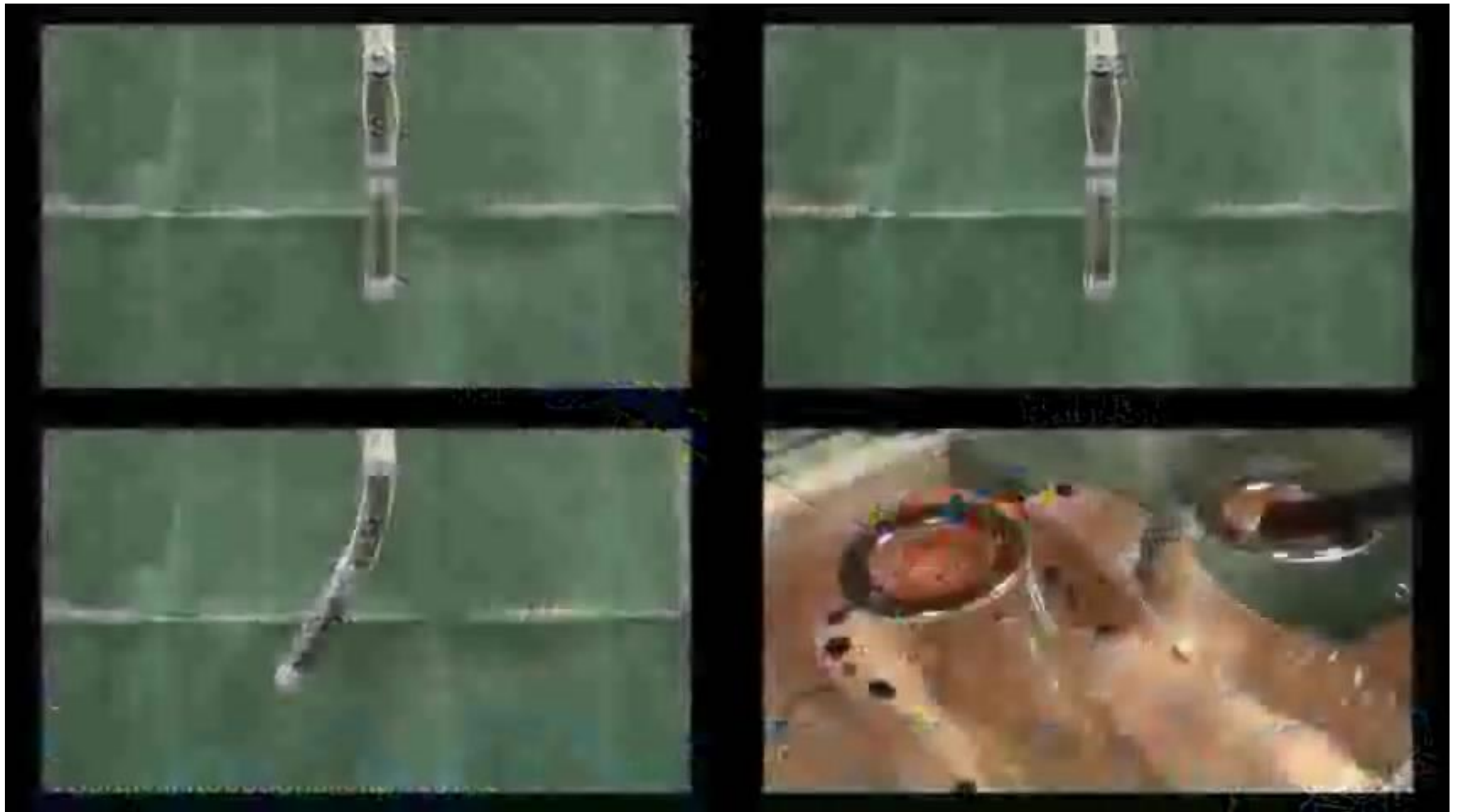
Software Integration:

- State Machine Diagram
- Stiff-Flop control

Diodato A, Brancadoro M, De Rossi G, Abidi H, Dall'Alba D, Muradore R, Ciuti G, Fiorini P, Menciassi A, Cianchetti M (2018) "Soft Robotic Manipulator for Improving Dexterity in Minimally Invasive Surgery", Surgical Innovation, 25(1), 69-76.



Stiff-Flop and DVRK integration

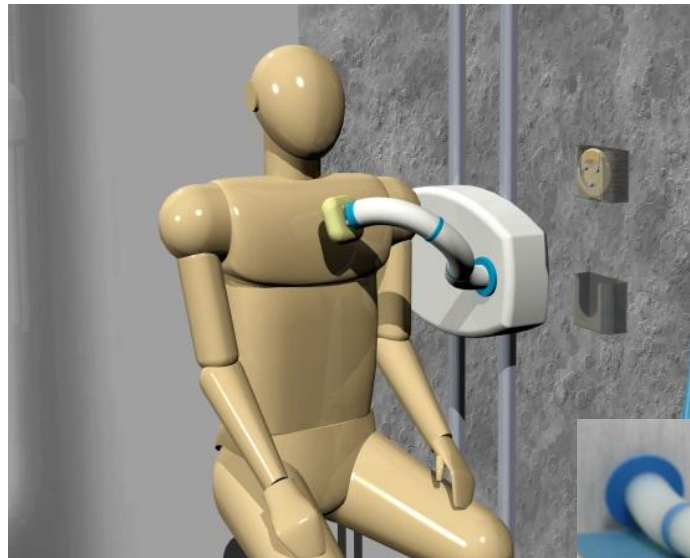


Diodato A, Brancadoro M, De Rossi G, Abidi H, Dall'Alba D, Muradore R, Ciuti G, Fiorini P, Menciassi A, Cianchetti M (2018) "Soft Robotic Manipulator for Improving Dexterity in Minimally Invasive Surgery", *Surgical Innovation*, 25(1), 69-76.



EU funded project

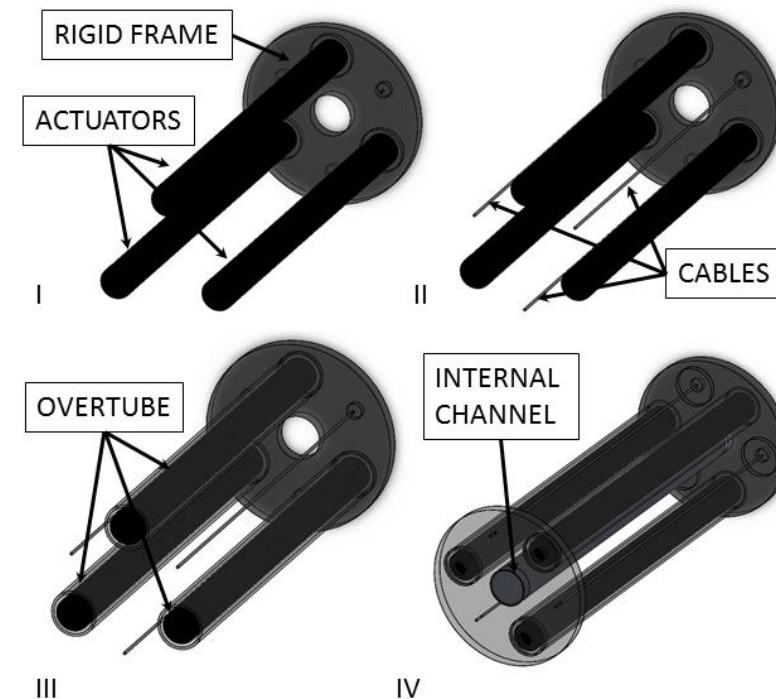
The development and integration of an innovative, modular, ICT-supported service robotics system that supports and enhances frail older adults' motion and force abilities and assists them in successfully, safely and independently completing bathing tasks.



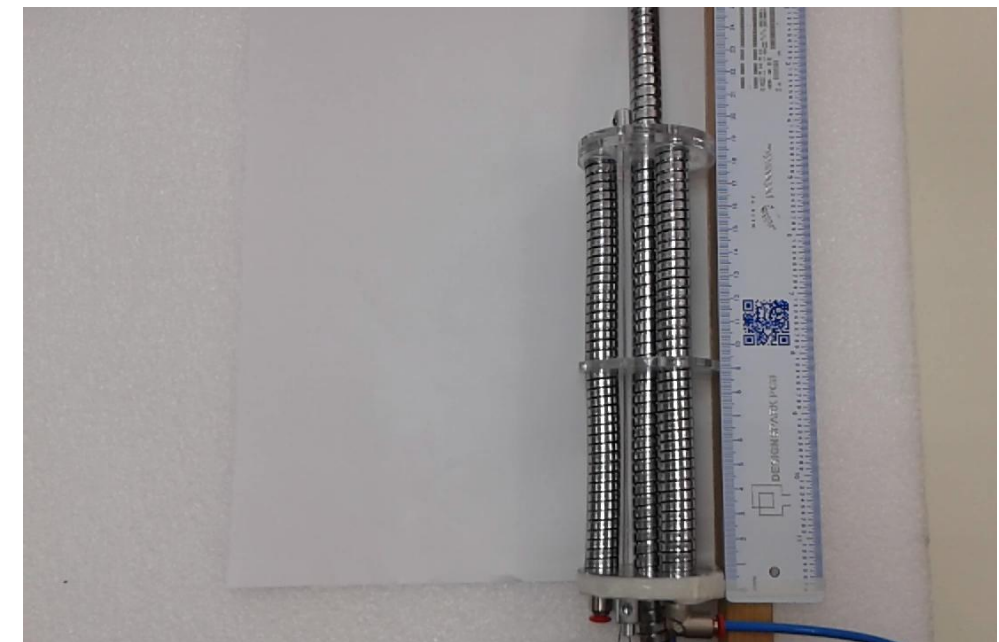
Soft /
flexible
materials

Tendons

Flexible
fluidic
actuators



elongation



bending

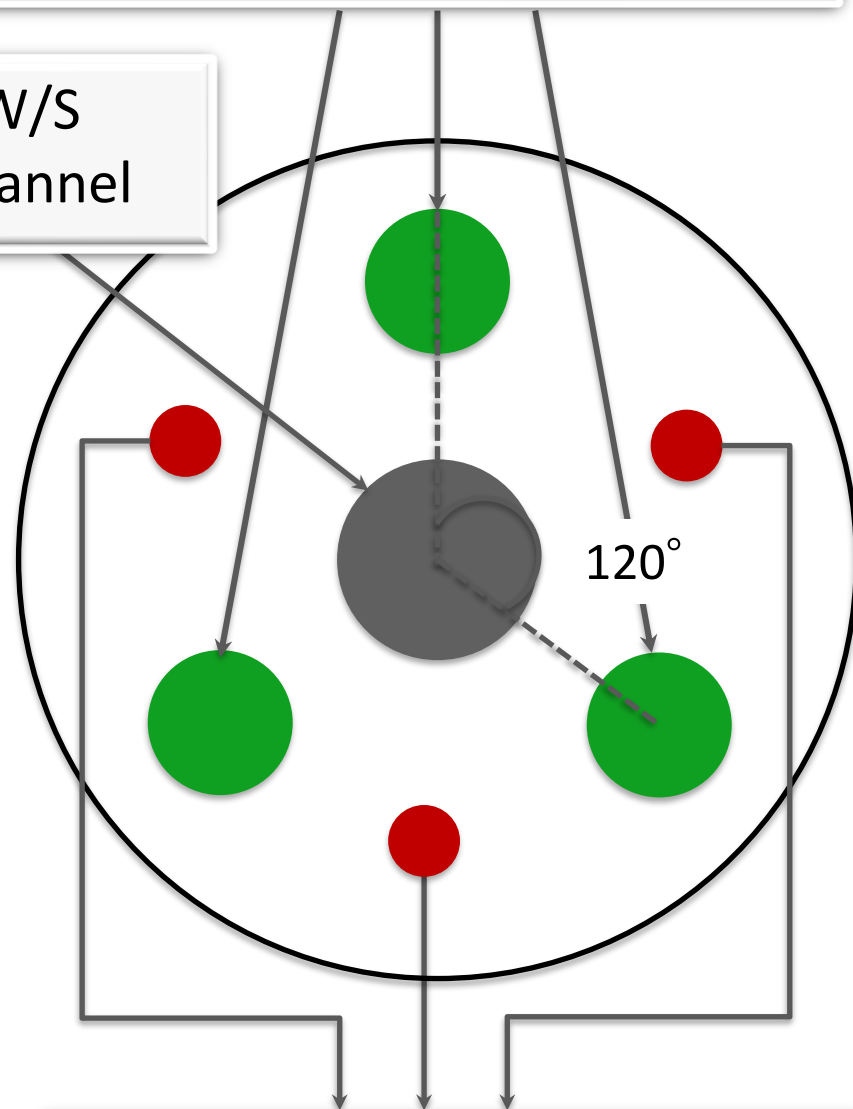


Design of the single module

Section view

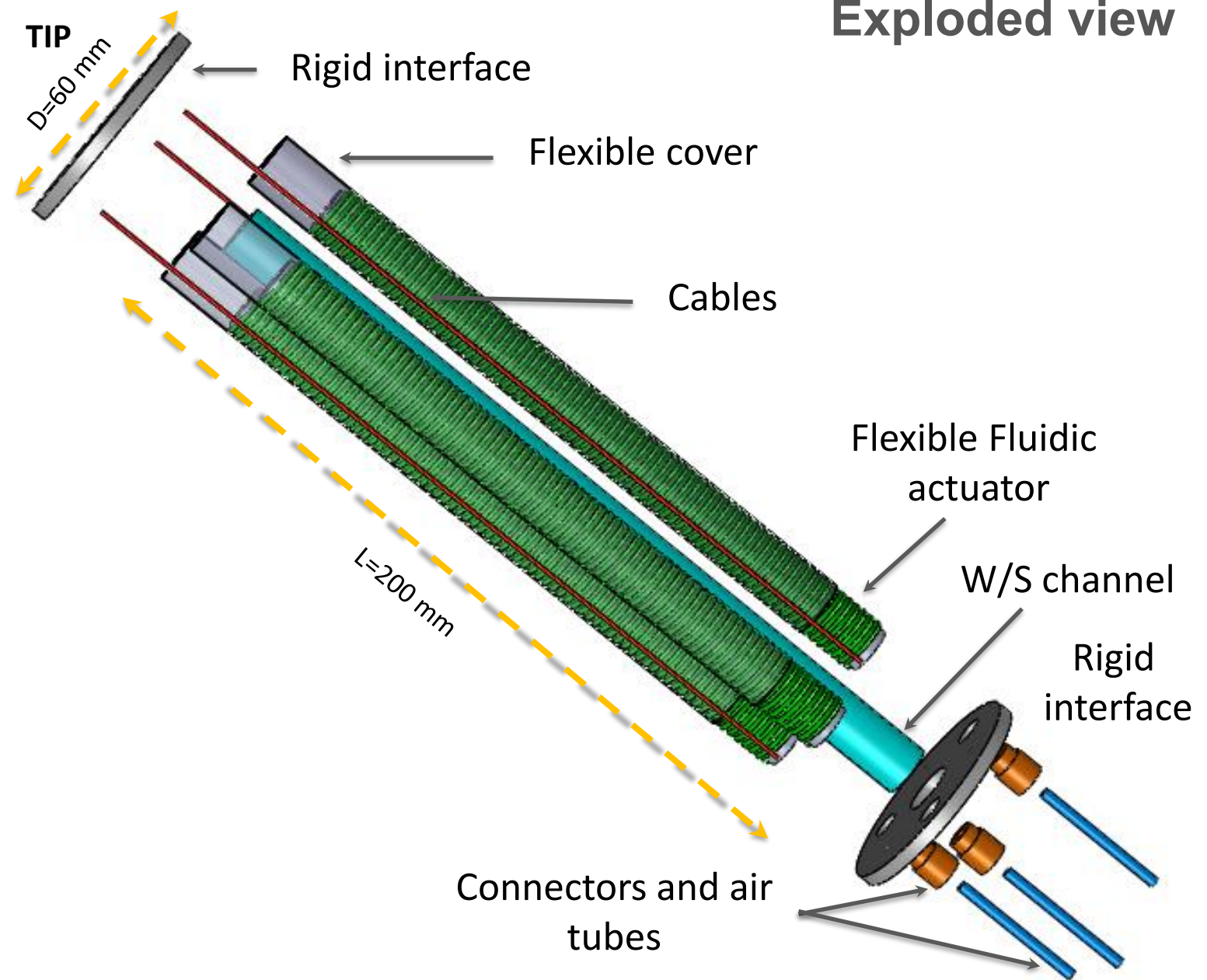
Flexible Fluidic Actuators @ 120°

W/S
channel



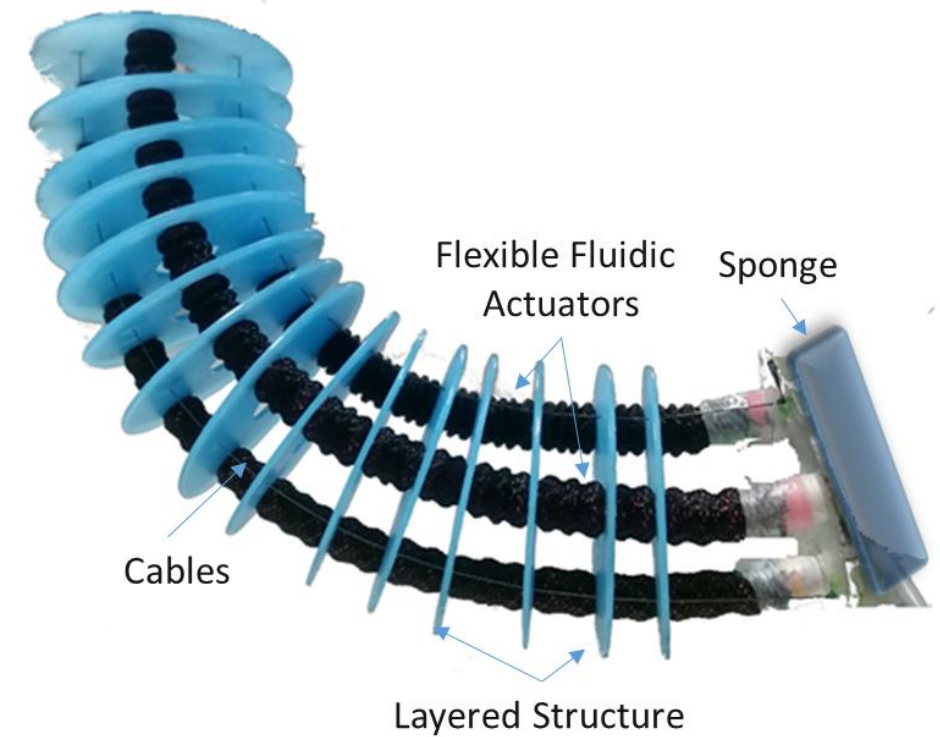
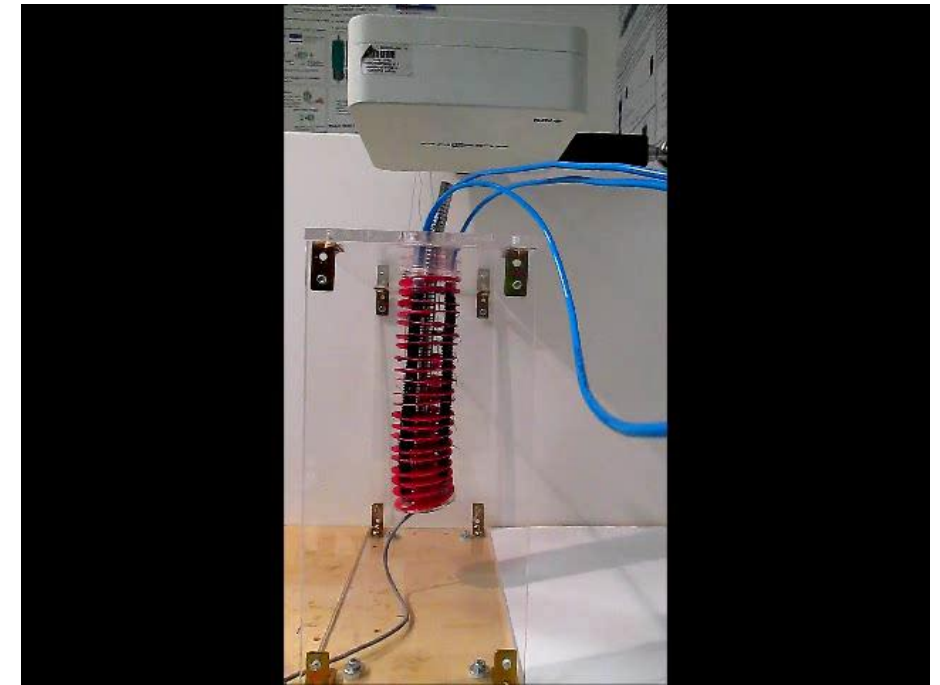
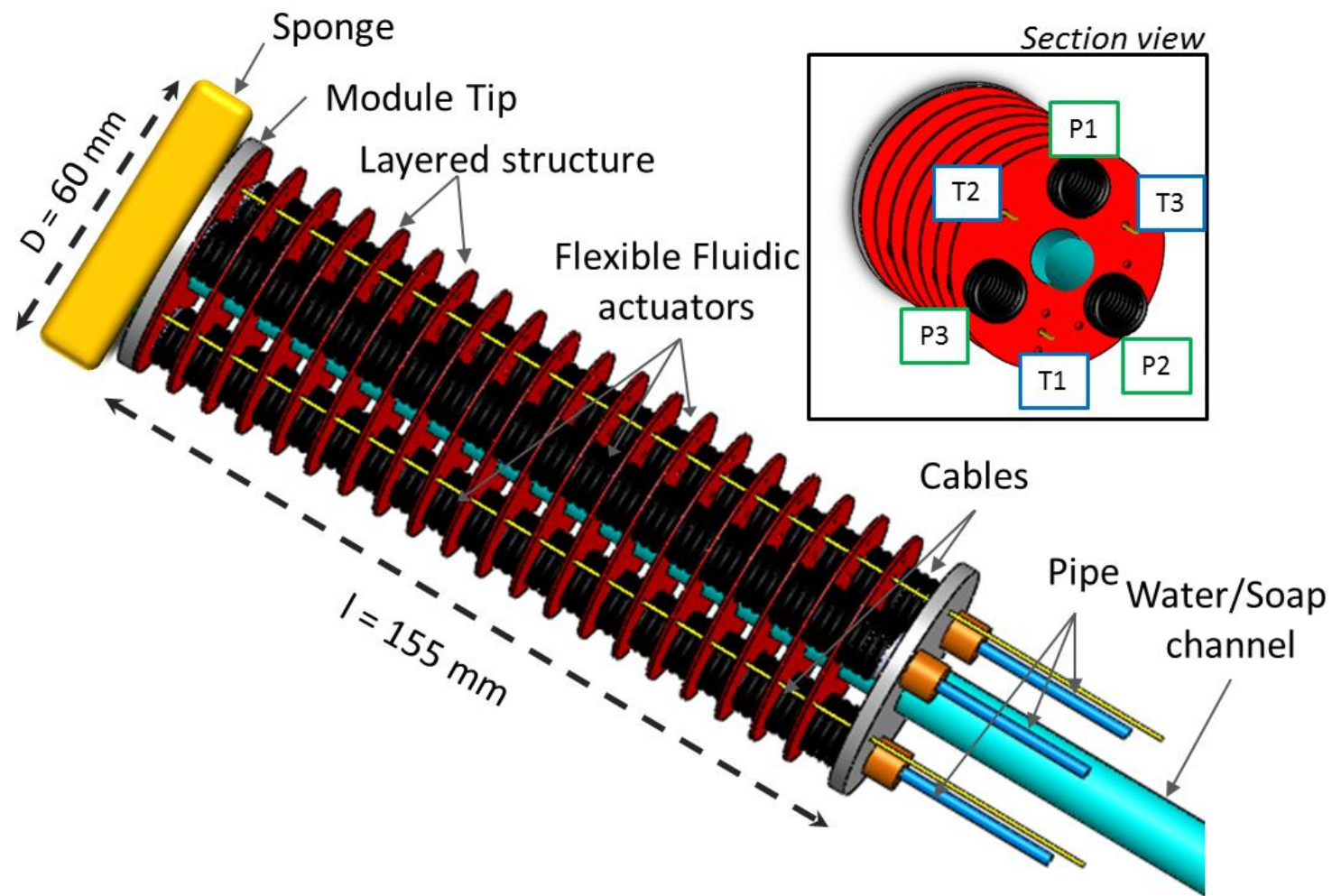
Actuation cables @ 120°

Exploded view



Design of the single module

First version

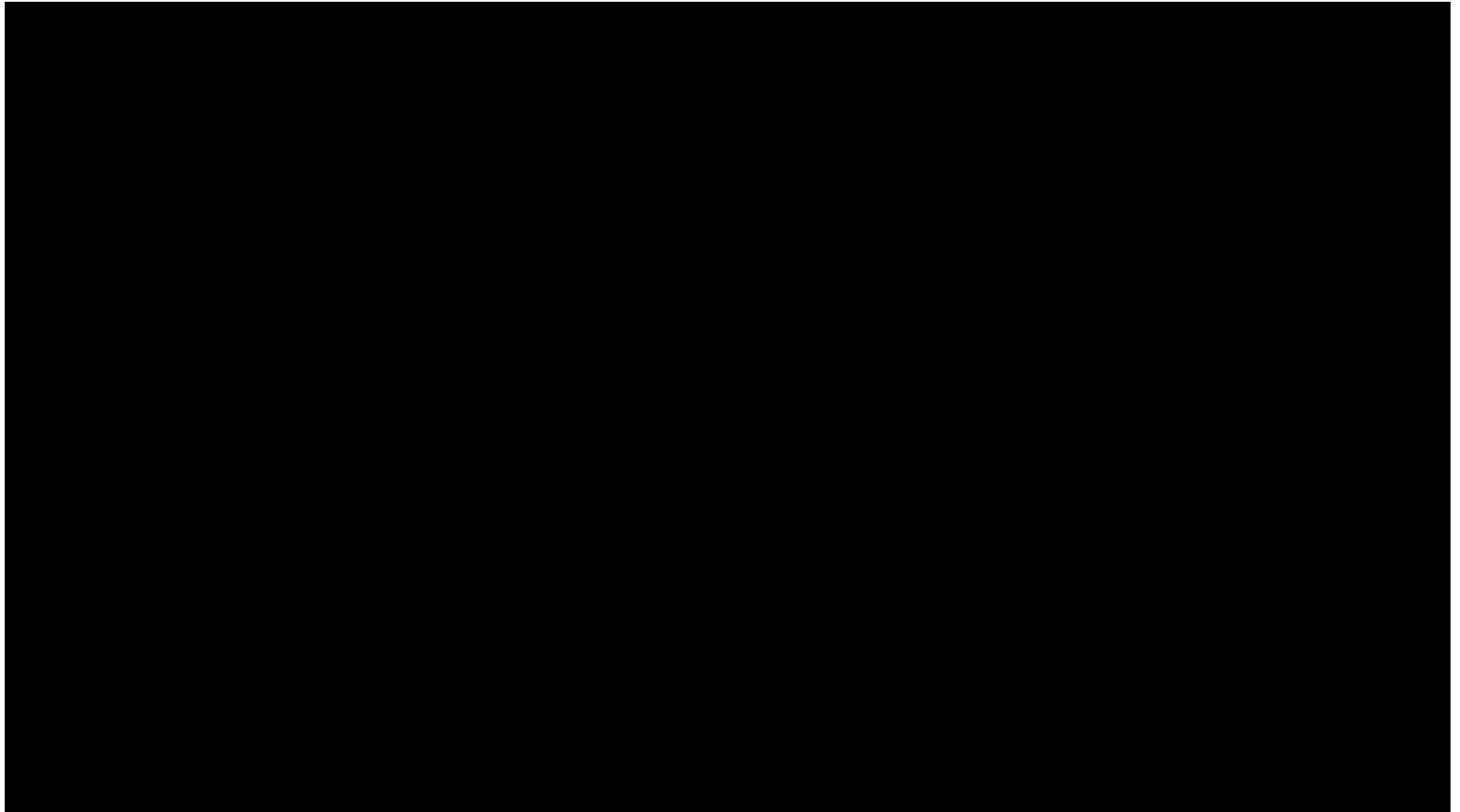


Manti M, Pratesi A, Falotico E, Cianchetti M, Laschi C (2016) "Soft Assistive Robot for Personal Care of Elderly People", Conf Proc IEEE on Biomedical Robotics and Biomechatronics – BioRob 2016, 833-838.



Design of the single module

Current version

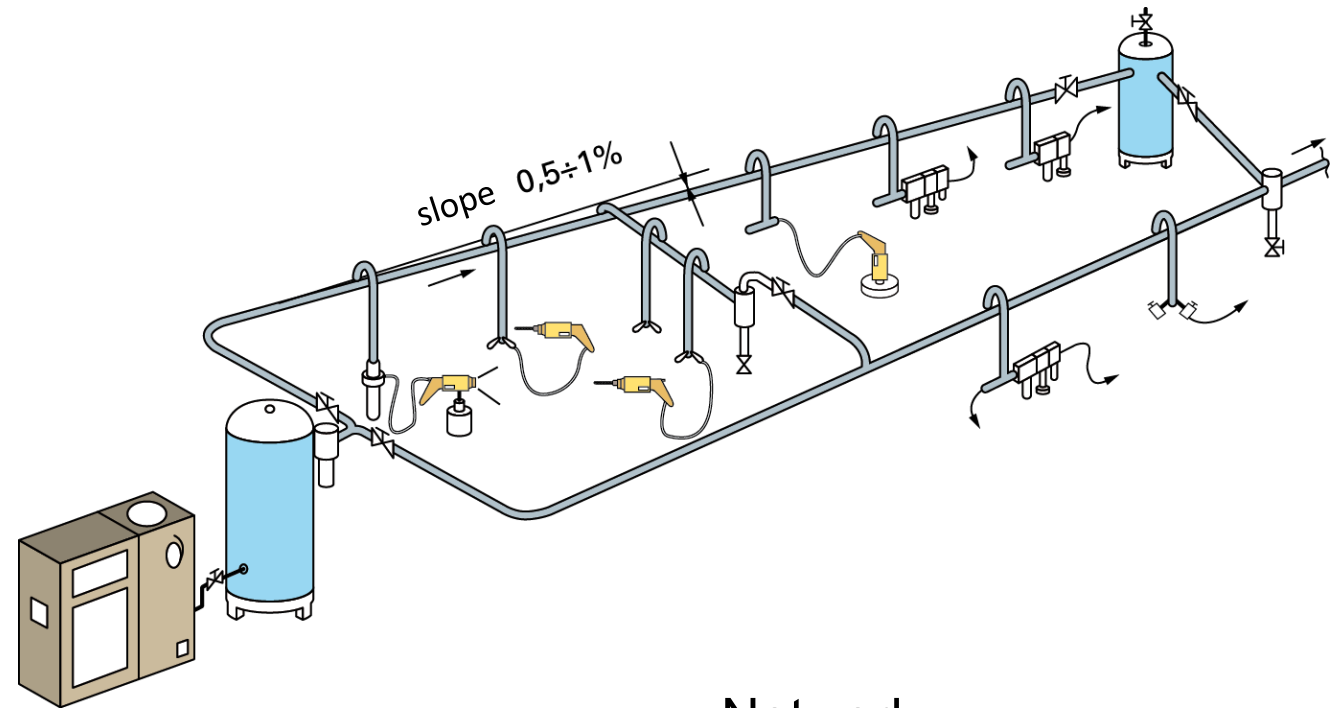


Fluidic energy sources

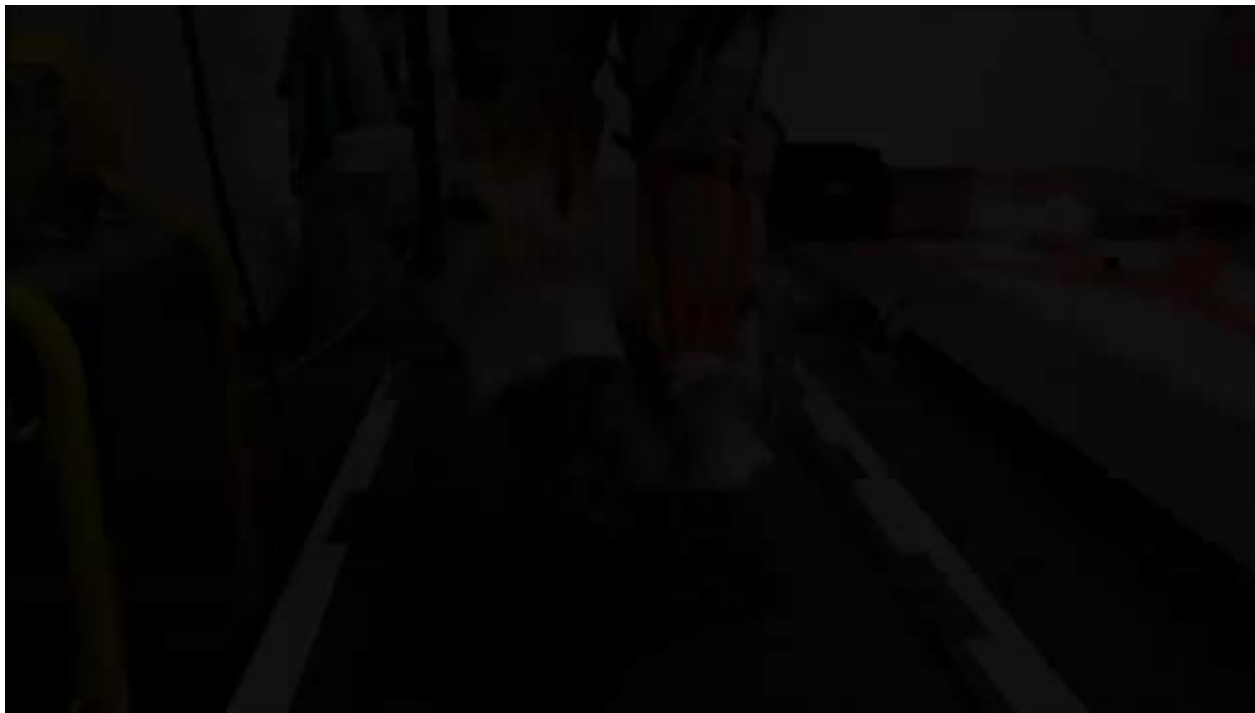
Off-board compressors



Portable



Network



- High pressure
- High volume capacity
- Fast
- Loud
- Bulky
- Heavy
- Exhaust air
- Piping

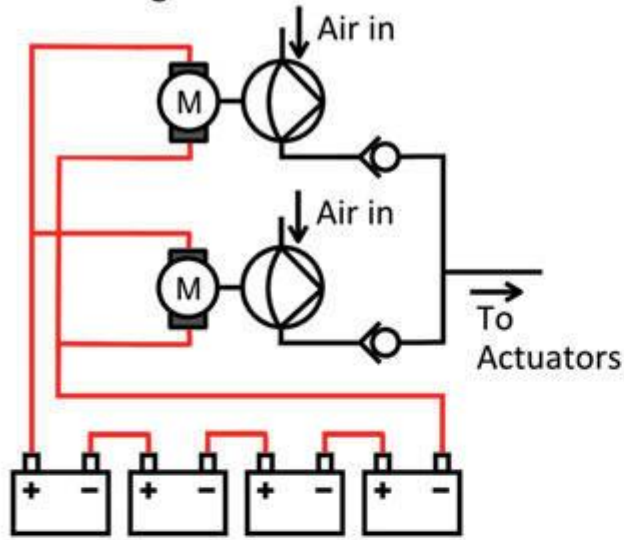


Fluidic energy sources

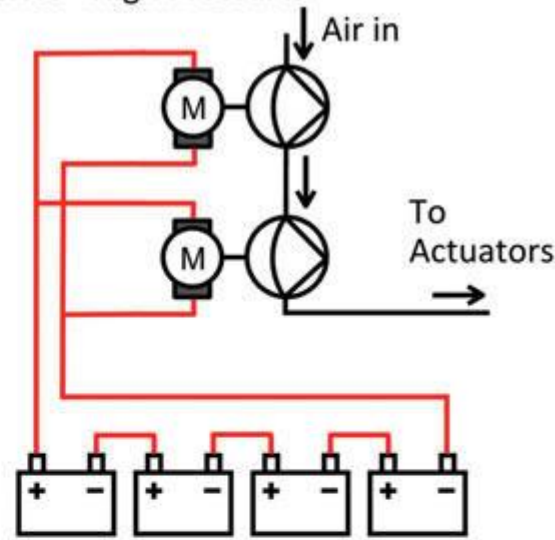
On-board micro-compressors + Lithium batteries

Parker Hargraves microcompressors

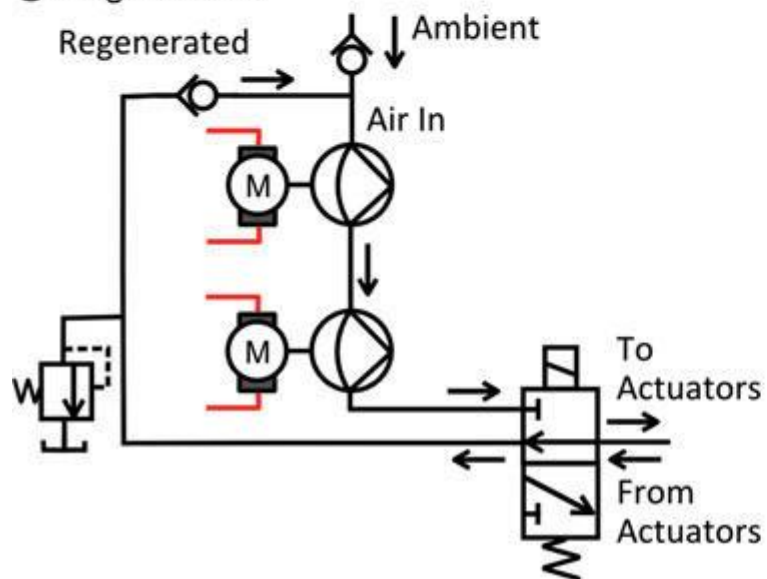
A Parallel - High Flow



B Series - High Pressure

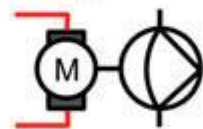


C Regeneration

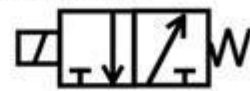


D Legend

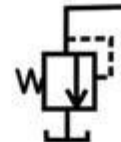
Microcompressor



Solenoid Valve



Relief Valve



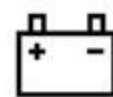
Electric Wire



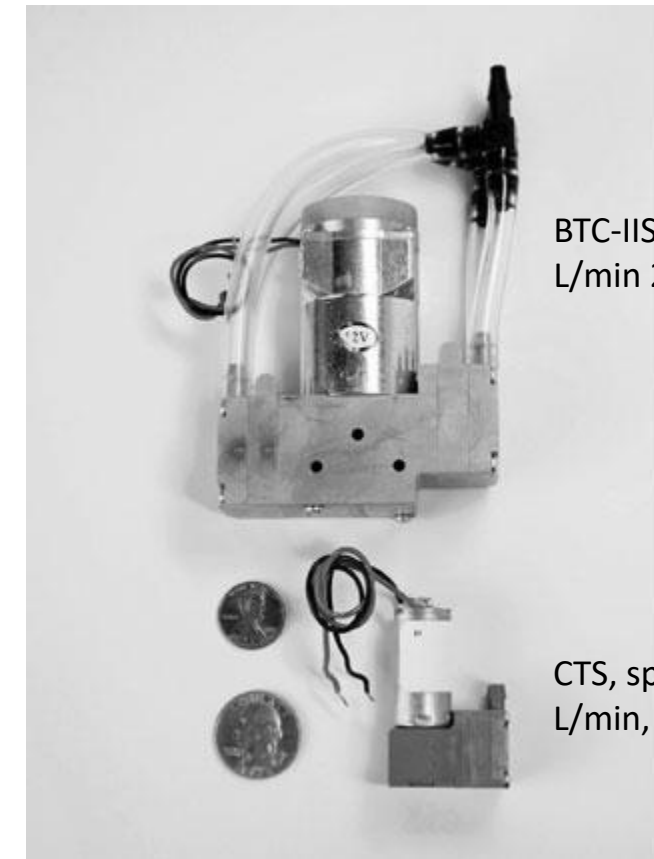
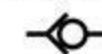
Air Line



Battery



Check Valve



BTC-IIS, specified as 12 V, 11 L/min 28 psig max

CTS, specified as 12 V, 1.8 L/min, 21 psig max

- Light
- Limited dimensions
- No / limited piping
- Loud
- Low pressure → multiple pumps in series
- Slow → multiple pumps in parallel
- Low / no volume capacity

Wehner M, Tolley MT, Mengüç Y, Park YL, Mozeika A, Ding Y, Onal C, Shepherd RF, Whitesides GM, Wood RJ, (2014) "Pneumatic Energy Sources for Autonomous and Wearable Soft Robotics", Soft Robotics, 1(4), 263-274.



Fluidic energy sources

On-board micro-compressors + Lithium batteries

Untethered Soft Quadraped Robot
Ambulatory Gait

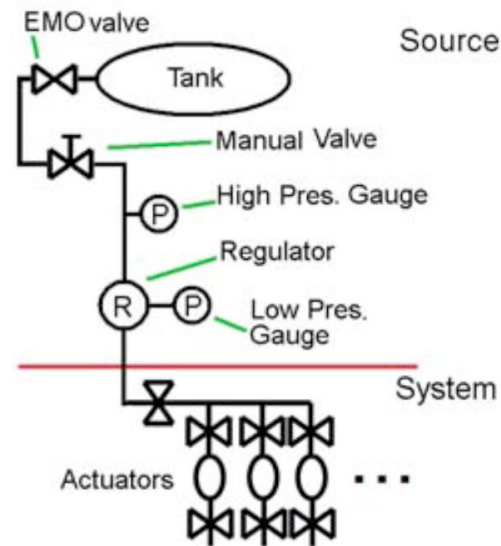


real time



Fluidic energy sources

Compressed fluids



Marchese, Andrew D, Onal, Cagdas D, Rus, Daniela (2014) Autonomous Soft Robotic Fish Capable of Escape Maneuvers Using Fluidic Elastomer Actuators, Soft Robotics 1(1):75--87

TABLE 3. KEY VALUES: COMPRESSED FLUID CYLINDERS

	Mass				Max. flow rate ^a (SLM)	Total flow (std. liter)	Energy ^b		Energy dens. ^c		Flow capacity	
	Net fuel (g)	Gross fuel (g)	Static (g)	Fuel system (g)			Theory (kJ)	0.5 MPa (kJ)	Net fuel (J/g)	Fuel system (J/g)	Net fuel (L/g)	Fuel system (L/g)
CO ₂ 8 g	7.8	33.6	16.3	49.9	18	4.1	1.70	0.66	219.9	34.2	0.51	0.082
CO ₂ 12 g	11.7	41.9	16.3	58.2	18	6.2	2.56	0.99	219.9	44.1	0.51	0.106
CO ₂ 16 g	17.4	57.7	53.5	111	18	8.6	3.82	1.48	219.9	34.4	0.51	0.077
CO ₂ 45 g	46.9	185	227	412	127	25.1	10.3	4.00	219.9	30.4	0.51	0.074
CO ₂ 74 g	70.7	298	227	524	127	38.4	15.5	6.01	219.9	34.4	0.51	0.085
CO ₂ 94 g	86.1	435	227	662	127	46.8	18.9	7.33	219.9	32.1	0.51	0.08
CO ₂ 20 oz	416	1281	291	1573	300	212.1	91.4	35.5	219.9	58.1	0.51	0.135
Air 13 in ³	56.0	513	395	909	350	43.1	24.6	12.1	440.1	27.1	0.77	0.047
20 MPa												
Air 48 in ³	196	1579	395	1975	350	151.1	86.3	42.5	440.1	43.7	0.77	0.077
20 MPa												
Air 68 in ³	333	1490	395	1885	650	256.3	157.6	49.5	473.7	83.6	0.77	0.136
30 MPa												

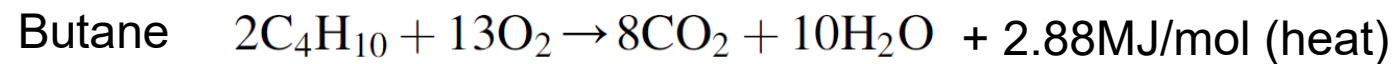
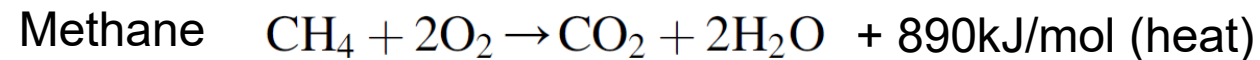
- High pressure
- Fast
- Light
- Limited volume capacity



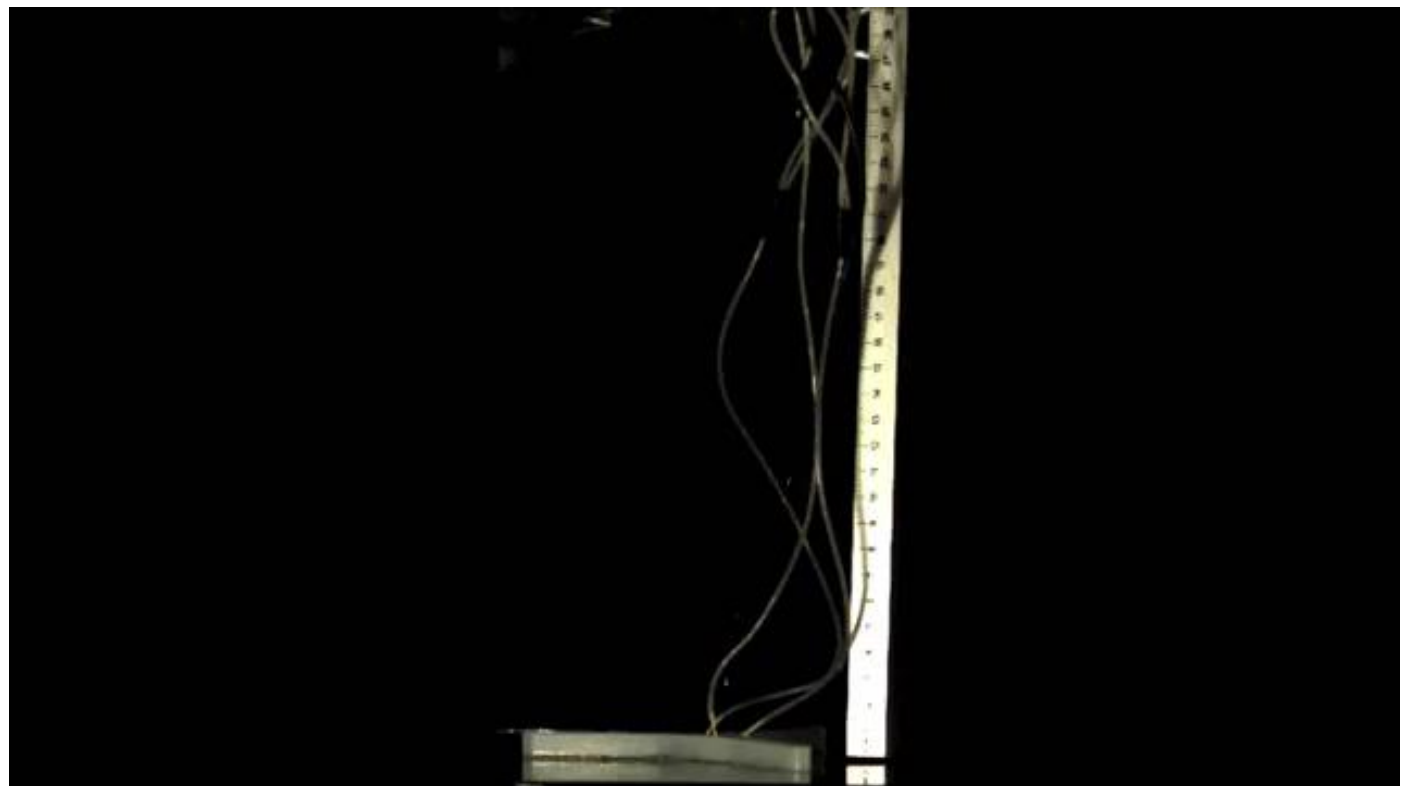
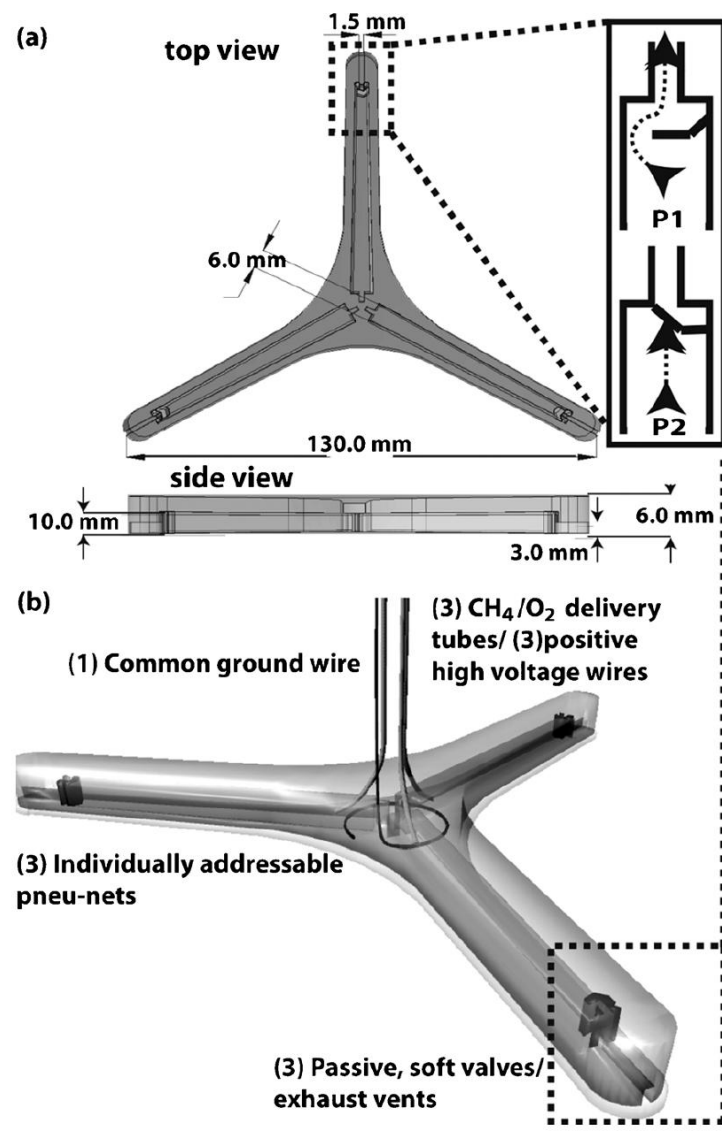
Fluidic energy sources

Combustion of hydrocarbons

Fuel and oxidizer reacting in the presence of a spark/flame



- High energy density
- Highest speed
- Very loud
- High local temperature
- On-board fuel, oxygen and ignition required
- Valves and control hardware (for stoichiometry)
- No / limited control (ok for jumping)
- Secondary products
- Difficult to model



Shepherd R, Stokes A, Freake J, Barber J, Snyder PW, Mazzeo ad, Cademartiri L, Morin SA, Whitesides GM. Using explosions to power a soft robot. Angew Chem 2013;125: 2964–2968.

– Flexible Fluidic Actuators



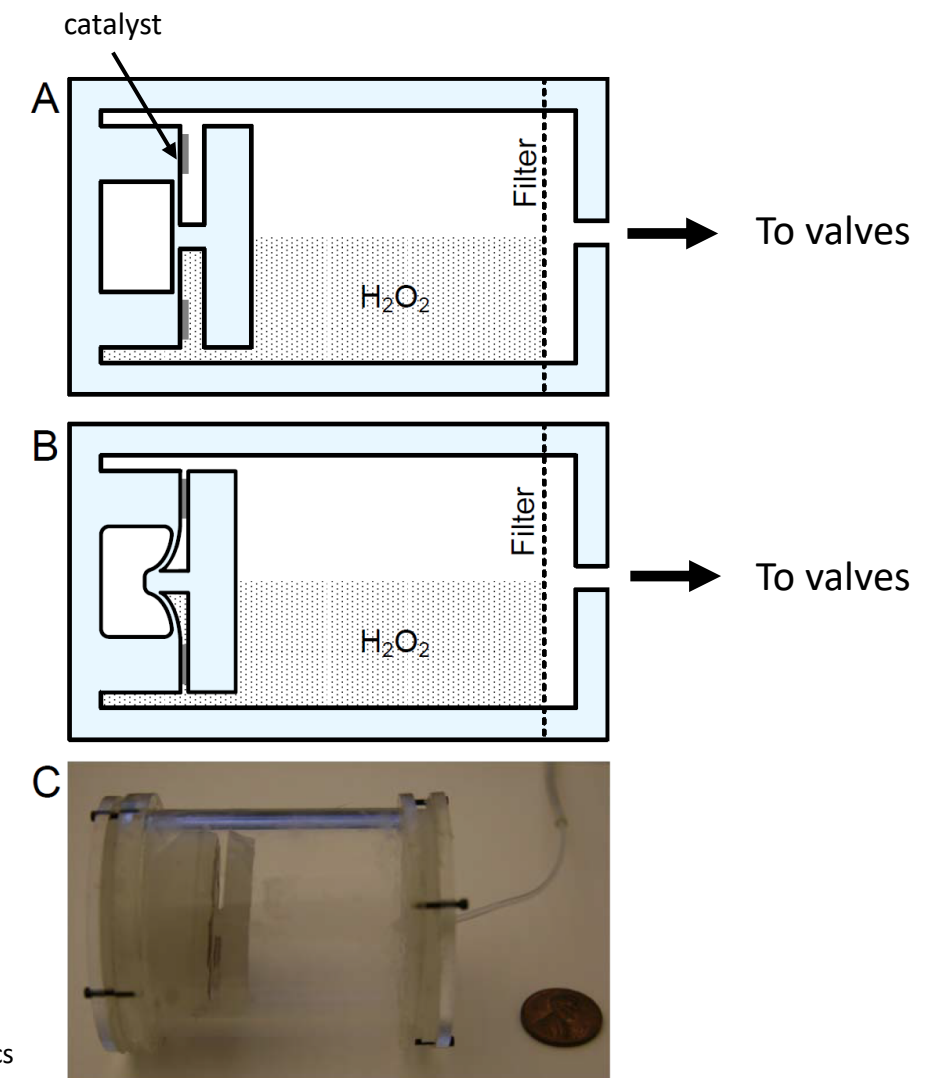
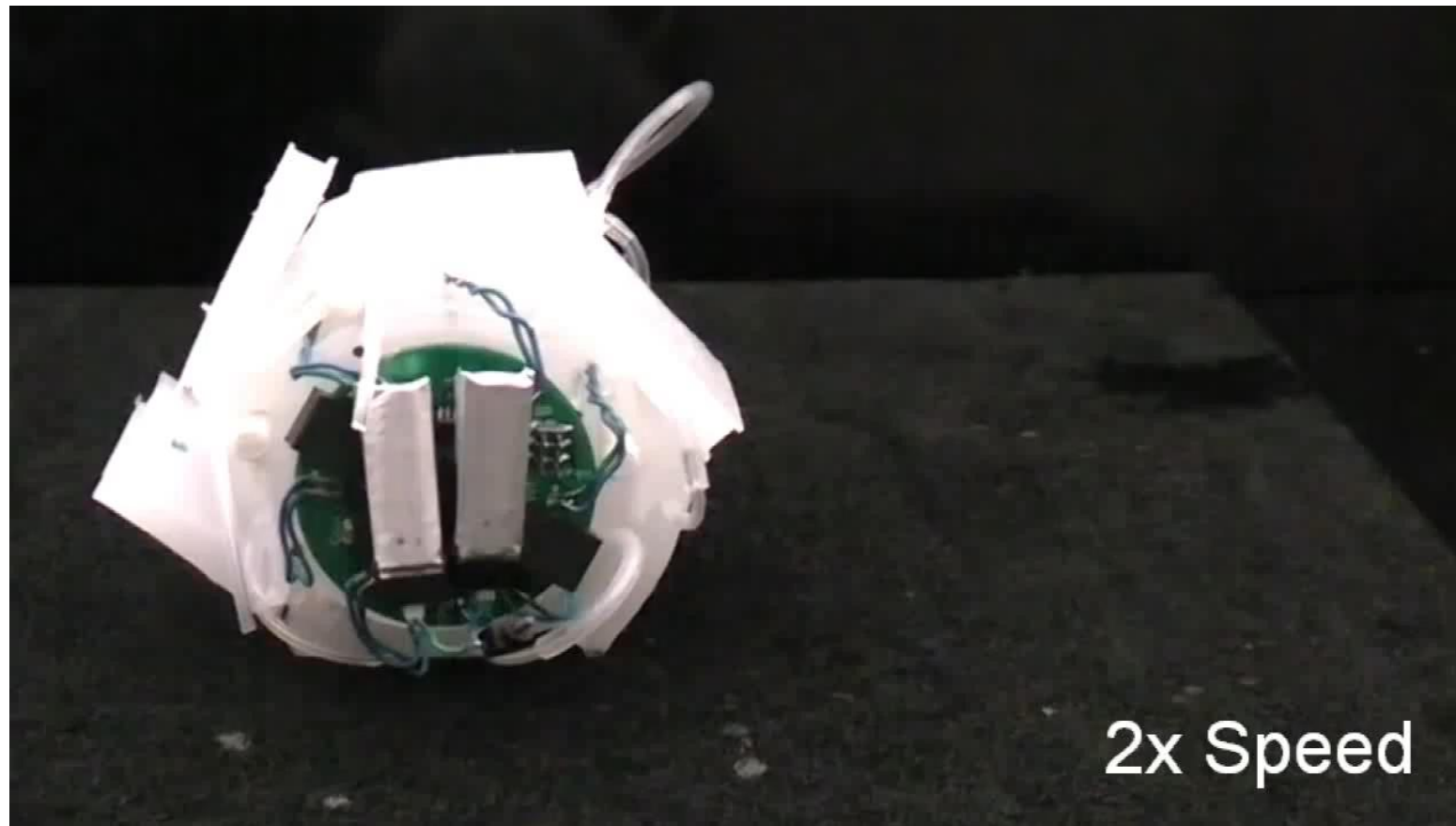
Fluidic energy sources

Monopropellant decomposition

Chemical breaking down exothermically in the presence of a catalyst, not requiring a separate oxidizer

Hydrogen peroxide $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 98.2\text{kJ/mol (heat)}$

- Acceptable energy density (dilution needed to avoid explosion)
- Low speed
- High local temperature
- On-board fuel
- Difficult to model



Onal C, Chen X, Whitesides G, Rus D. Soft mobile robots with on-board chemical pressure generation. International Symposium on Robotics Research, 2011



Fluidic energy sources

TABLE 5. KEY VALUES: FUEL SYSTEM-LEVEL SUMMARY

	<i>Fuel system mass as evaluated (g)</i>	<i>Maximum pressure (MPa)</i>	<i>Maximum flow rate (SLM)</i>	<i>Fuel system energy density (J/g)</i>	<i>Efficiency as evaluated %</i>	<i>Fuel system flow capacity (L/g)</i>
Microcompressor small	130–171	0.2–0.34	1.7–3.5	72.6–99.3	36.7–39.9%	0.78–1.38
Microcompressor large	422–756	0.26–0.55	5.7–10.7	14.8–33.6	31.2–40.3%	0.17–0.37
Liquid CO ₂ small	49.9–662	5.6	18–127	30.4–44.1	< 49% ^a	0.074–0.11
Liquid CO ₂ (refillable)	1573	5.6	300	58.1	< 49% ^a	0.14
Air cylinder (refillable)	909–1975	20 and 30	350–650	27.1–83.6	< 39% ^a	0.05–0.14
Butane combustion ³²	62.7	n.a.	n.a.	2.37	0.5%	0.094
Hydrogen peroxide decomposition	357 ¹²	0.0507 ¹²	n.a.	10–28 ^b	3.3–9%, 45% ^{37–40}	0.04–0.05 ^{12,c}

Microcompressor values are for 800 mAh batteries at pressure of maximum efficiency. Air and CO₂ are for stored pressure, not reduced working pressure. Butane and hydrogen peroxide values are from the literature based on custom architecture and may not represent future designs. Methane was supplied via tether, and so fuel system mass not applicable.⁶ No maximum pressure and rate values are available for butane.³²

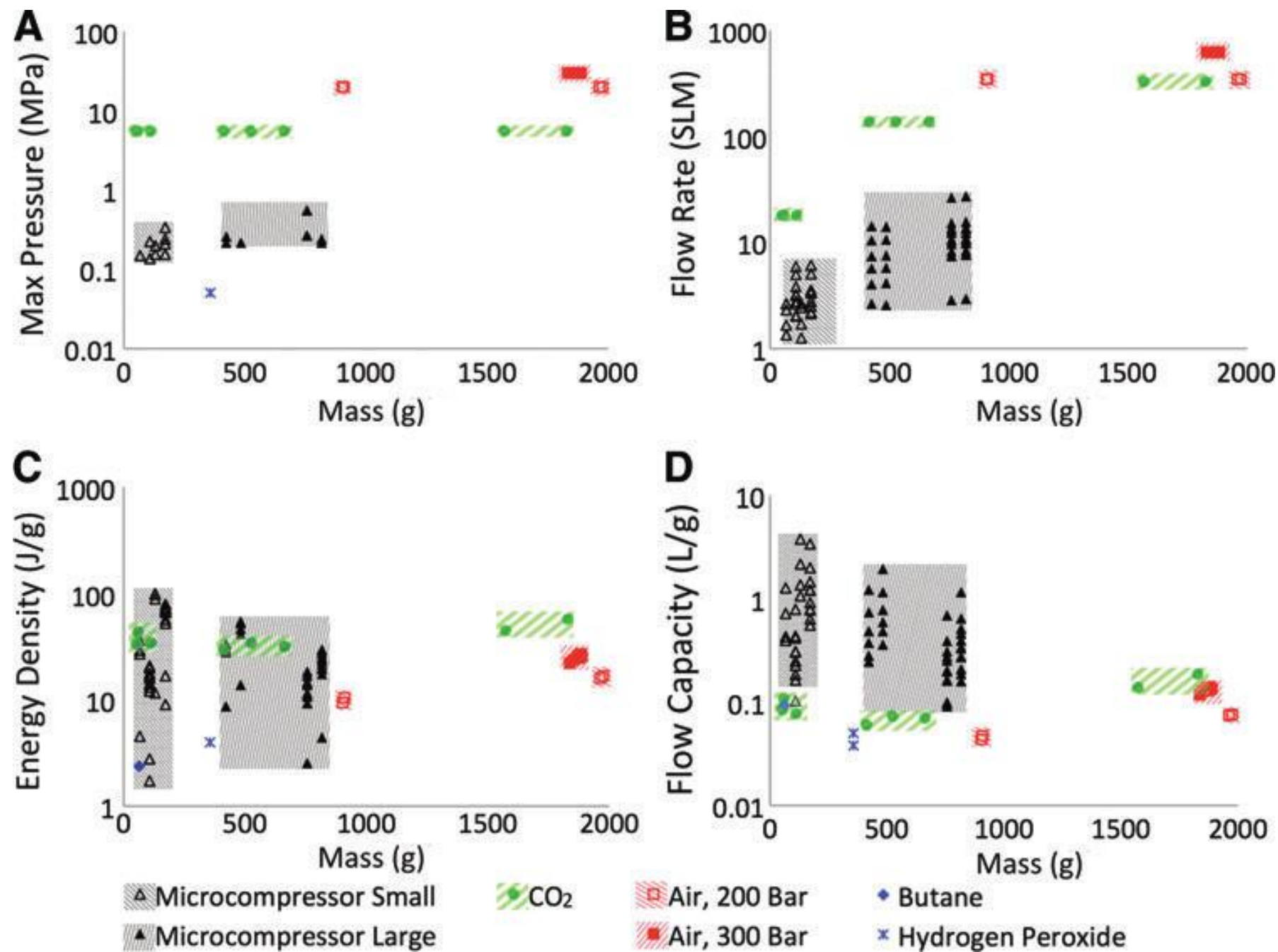
^aAssumes working pressures of 0.5 MPa and isothermal conditions. Polytropic systems at lower operating pressures will have lower values.

^bBased on mass of pneumatic battery¹² and efficiency of 3.3–9% of 1.44 kJ/g net fuel.^{37–40}

^cRange assumes that all reaction products are liquid between 20°C and 100°C.



Fluidic energy sources



Wehner M, Tolley MT, Mengüç Y, Park YL, Mozeika A, Ding Y, Onal C, Shepherd RF, Whitesides GM, Wood RJ, (2014) "Pneumatic Energy Sources for Autonomous and Wearable Soft Robotics", *Soft Robotics*, 1(4), 263-274.

